

Computer Science

Chapter 3: Hardware

3.1 Computer architecture

3.2 Input and output devices

3.3 Data storage

3.4 Network hardware

By: Nouredine Tadjerout

Learning Objectives

3.1 Computer architecture

Candidates should be able to:

- 1 (a) Understand the role of the central processing unit (CPU) in a computer
- (b) Understand what is meant by a microprocessor
- 2 (a) Understand the purpose of the components in a CPU, in a computer that has a Von Neumann architecture
- (b) Describe the process of the fetch–decode–execute cycle including the role of each component in the process

Notes and guidance

- The CPU processes instructions and data that are input into the computer so that the result can be output
- A microprocessor is a type of integrated circuit on a single chip
- Including:
 - units: arithmetic logic unit (ALU) and control unit (CU)
 - registers: program counter (PC), memory address register (MAR), memory data register (MDR), current instruction register (CIR) and accumulator (ACC)
 - buses: address bus, data bus and control bus
- How instructions and data are fetched from random access memory (RAM) into the CPU, how they are processed using each component and how they are then executed
- Storing data and addresses into specific registers
- Using buses to transmit data, addresses and signals
- Using units to fetch, decode and execute data and instructions

Learning Objectives

3.1 Computer architecture

- 3 Understand what is meant by a core, cache and clock in a CPU and explain how they can affect the performance of a CPU
 - 4 Understand the purpose and use of an instruction set for a CPU
 - 5 Describe the purpose and characteristics of an embedded system and identify devices in which they are commonly used
- The number of cores, size of the cache and speed of the clock can affect the performance of a CPU
 - An instruction set is a list of all the commands that can be processed by a CPU and the commands are machine code
 - An embedded system is used to perform a dedicated function, e.g. domestic appliances, cars, security systems, lighting systems or vending machines. This is different to a general purpose computer that is used to perform many different functions, e.g. a personal computer (PC) or a laptop

Learning Objectives

3.2 Input and output devices

Candidates should be able to:

- 1 Understand what is meant by an input device and why it is required
- 2 Understand what is meant by an output device and why it is required

Notes and guidance

- Including:
 - barcode scanner
 - digital camera
 - keyboard
 - microphone
 - optical mouse
 - QR code scanner
 - touch screen (resistive, capacitive and infra-red)
 - two-dimensional (2D) and three-dimensional (3D) scanners
- Including:
 - actuator
 - digital light processing (DLP) projector
 - inkjet printer
 - laser printer
 - light emitting diode (LED) screen
 - liquid crystal display (LCD) projector
 - liquid crystal display (LCD) screen
 - speaker
 - 3D printer

Learning Objectives

3.2 Input and output devices

- 3 (a) Understand what is meant by a sensor and the purposes of sensors
- Limited to:
 - acoustic
 - accelerometer
 - flow
 - gas
 - humidity
 - infra-red
 - level
 - light
 - magnetic field
 - moisture
 - pH
 - pressure
 - proximity
 - temperature
- (b) Identify the type of data captured by each sensor and understand when each sensor would be used, including selecting the most suitable sensor for a given context

Learning Objectives

3.3 Data storage

Candidates should be able to:

- 1 Understand what is meant by primary storage
- 2 Understand what is meant by secondary storage
- 3 Describe the operation of magnetic, optical and solid-state (flash memory) storage and give examples of each

Notes and guidance

- Primary storage is directly accessed by the CPU
- Including the role of:
 - random access memory (RAM)
 - read only memory (ROM)
- Including why a computer needs both RAM and ROM, and the difference between them
- Secondary storage is not directly accessed by the CPU and is necessary for more permanent storage of data
- Magnetic storage uses platters which are divided into tracks and sectors. Data is read and written using electromagnets
- Optical storage uses lasers to create and read pits and lands
- Solid-state (flash memory) uses NAND or NOR technology. Transistors are used as control gates and floating gates

Learning Objectives

3.3 Data storage

- 4 Describe what is meant by virtual memory, how it is created and used and why it is necessary
- 5 Understand what is meant by cloud storage
 - Pages of data are transferred between RAM and virtual memory when needed
 - Cloud storage can be accessed remotely in comparison to storing data locally
 - Physical servers and storage are needed to store data in cloud storage
- 6 Explain the advantages and disadvantages of storing data on the cloud in comparison to storing it locally

Learning Objectives

3.4 Network hardware

Candidates should be able to:

- 1 Understand that a computer needs a network interface card (NIC) to access a network
- 2 Understand what is meant by and the purpose of a media access control (MAC) address, including its structure
- 3 (a) Understand what is meant by and the purpose of an internet protocol (IP) address
(b) Understand that there are different types of IP address
- 4 Describe the role of a router in a network

Notes and guidance

- A network interface card is given a MAC address at the point of manufacture
- MAC addresses are usually written as hexadecimal
- MAC addresses are created using the manufacturer code and the serial code
- An IP address is allocated by the network and they can be static or dynamic
- Including the characteristics of and differences between IPv4 and IPv6
- A router sends data to a specific destination on a network
- A router can assign IP addresses
- A router can connect a local network to the internet

Resources



Craig 'n' Dave For Students

CRAIG 'N' DAVE

To explore the entire series,
visit <https://craigndave.org/free-videos/>

Resources



To explore the entire series,
visit www.hoddereducation.com/cambridge-igcse-computerscience

Recap last two week lesson: IGCSE Exam Question 1

7 Six internet terms and six definitions are listed.

Draw a line to connect each term to a correct definition.

Internet term	Definition
Browser	A program that allows a user to view webpages
Internet Service Provider (ISP)	The main protocol that governs the transmission of data using the Internet
Hyper Text Transfer Protocol (HTTP)	The website address that is typed into the address bar
Uniform Resource Locator (URL)	An address given to each device on a network. It is provided by the network
MAC address	A unique address given to a device on a network. It is provided by the manufacturer
IP address	A company that provides a connection to access the Internet

[5]

Recap last two week lesson: IGCSE Exam Question 1 with Answer

Question	Answer	Marks												
7	1 mark for each correct line (to a maximum of 5) <table><tr><td>Browser</td><td>A program that allows a user to view webpages</td></tr><tr><td>Internet Service Provider (ISP)</td><td>The main protocol that governs the transmission of data using the Internet</td></tr><tr><td>Hypertext Transfer Protocol (HTTP)</td><td>The website address that is typed into the address bar</td></tr><tr><td>Uniform Resource Locator (URL)</td><td>An address given to each device on a network. It is provided by the network</td></tr><tr><td>MAC address</td><td>A unique address given to a device on a network. It is provided by the manufacturer</td></tr><tr><td>IP address</td><td>A company that provides a connection to access the Internet</td></tr></table>	Browser	A program that allows a user to view webpages	Internet Service Provider (ISP)	The main protocol that governs the transmission of data using the Internet	Hypertext Transfer Protocol (HTTP)	The website address that is typed into the address bar	Uniform Resource Locator (URL)	An address given to each device on a network. It is provided by the network	MAC address	A unique address given to a device on a network. It is provided by the manufacturer	IP address	A company that provides a connection to access the Internet	5
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Recap last two week lesson: IGCSE Exam Question 2

(a) Computers can transmit data using different methods.

Describe the **three** data transmission methods given.

(i) Serial data transmission

.....
.....
.....
.....[2]

(ii) Parallel data transmission

.....
.....
.....
.....[2]

(iii) Duplex data transmission

.....
.....
.....
.....[2]

Recap last two week lesson: IGCSE Exam Question 2 with Answer

Question	Answer	Marks
9(a)(i)	Two from: ☐ Data is transmitted one bit at a time ☐ Data is transmitted using a single wire ☐ Bits arrive in order/sequence	2
9(a)(ii)	Two from: ☐ Data is transmitted multiple bits at a time/simultaneously ☐ Data is transmitted using multiple wires ☐ Bits may arrive out of sequence/skewed (and are reordered)	2
9(a)(iii)	1 mark for each: ☐ Data is transmitted in both directions ☐ ... at the same time/simultaneously	2

Recap last month lesson: IGCSE Exam Question 3

(a) Six binary or hexadecimal numbers and six denary conversions are given.

Draw a line to connect each binary or hexadecimal number to the correct denary conversion.

Binary or hexadecimal

01001011

4E

11011010

10011101

A7

19

Denary

75

78

157

167

25

218

[5]

Recap last month lesson: IGCSE Exam Question 3 with Answer

Question	Answer	Marks																					
2(a)	1 mark for each correct line (to a maximum of 5) <table><thead><tr><th>Binary or hexadecimal</th><th></th><th>Denary</th></tr></thead><tbody><tr><td>01001011</td><td>—</td><td>75</td></tr><tr><td>4E</td><td>—</td><td>78</td></tr><tr><td>11011010</td><td>—</td><td>157</td></tr><tr><td>10011101</td><td>—</td><td>167</td></tr><tr><td>A7</td><td>—</td><td>25</td></tr><tr><td>19</td><td>—</td><td>218</td></tr></tbody></table>	Binary or hexadecimal		Denary	01001011	—	75	4E	—	78	11011010	—	157	10011101	—	167	A7	—	25	19	—	218	5
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Learning Objectives

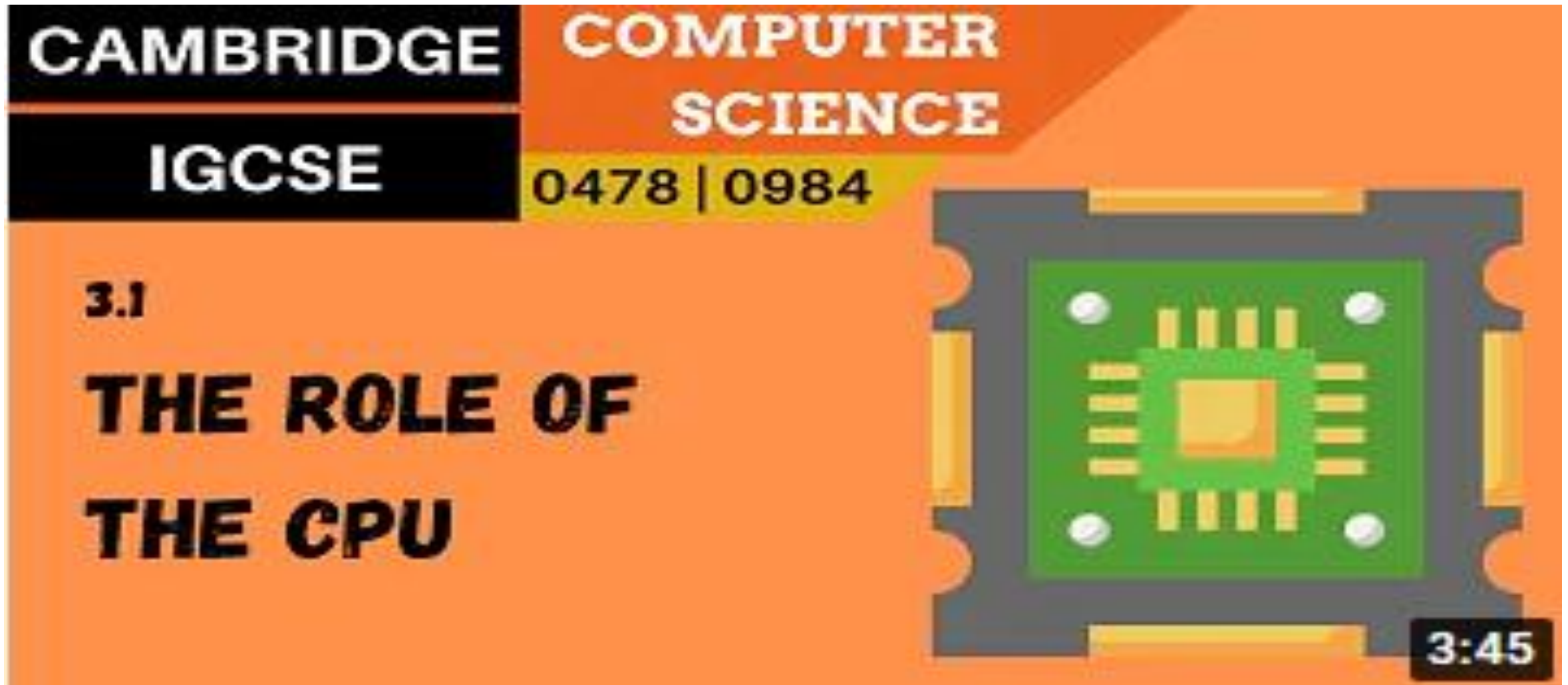
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3.1:Computer Architecture

Watch the video for The role of the central processing unit

<https://www.youtube.com/watch?v=cLL1AuQNHmQ>



3.1:Computer Architecture

The role of the central processing unit



The role of the CPU

The **central processing unit (CPU)**:

- takes in data and instructions;
- processes them; and
- produces output.

The terms **CPU** and **microprocessor** essentially mean the same thing.

A **CPU/microprocessor** is typically installed as an integrated circuit on a single microchip.

3.1:Computer Architecture

Watch the video for Purpose of the components in a CPU

<https://www.youtube.com/watch?v=K9EuHc5kzJc>



3.1:Computer Architecture

Components in a CPU



Components of the CPU

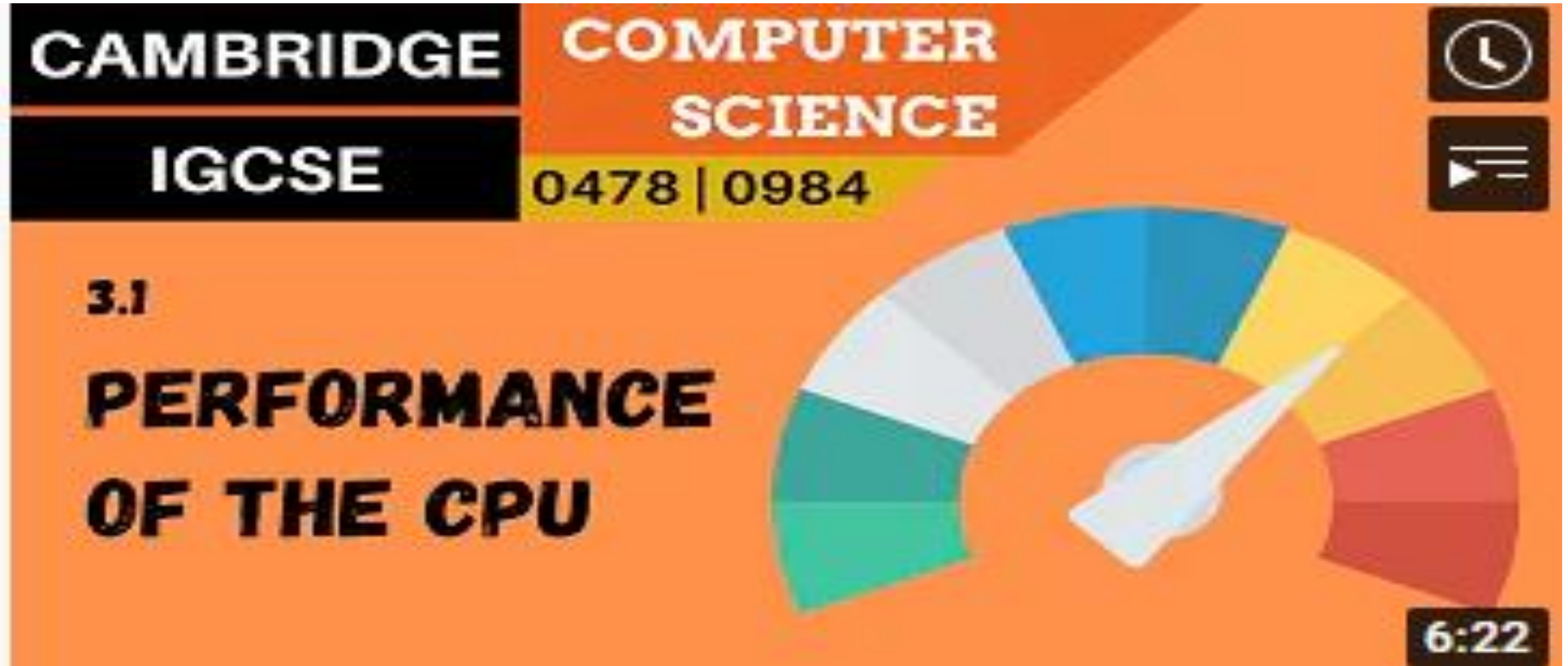
The CPU consists of the following components:

- **Arithmetic logic unit (ALU):** Performs calculations and logical decisions.
- **Control unit (CU):** Sends signals to control how data moves around the CPU.
- **Buses:** A collection of wires through which data and instructions are transmitted from one component to another.
 - **Address bus:** Unidirectional – carries the addresses that data needs to be written to or read from.
 - **Data bus:** Bidirectional – carries actual data or instructions.
 - **Control bus:** Bidirectional – carries command and control signals telling components when they should be receiving reads, writes, etc.
- **Registers:** Tiny, super-fast pieces of onboard memory inside the CPU, each with a very specific purpose.
 - **Program counter (PC):** Holds the address of the next instruction to be executed.
 - **Memory address register (MAR):** Holds the address of the memory location that data or an instruction is to be fetched from or written to.
 - **Memory data register (MDR):** Used to temporarily store data that is read from or written to memory.
 - **Current instruction register (CIR):** Holds the current instruction being executed.
 - **Accumulator (ACC):** One of several general-purpose registers. The results of calculations carried out by the ALU can be temporarily stored here.

3.1:Computer Architecture

Watch the video for The common characteristics of CPUs or Performance of the CPU

<https://www.youtube.com/watch?v=F0ZQQLPCWH8>



3.1:Computer Architecture

The common characteristics of CPUs or Performance of the CPU



Common characteristics of the CPU

A CPU's overall performance can be affected by several factors. The three most important factors are:

Clock speed

- Measured in number of cycles per second or hertz (Hz).
- 3.2GHz clock speed = 3.2 billion instructions fetched per second – in theory.

Cache size

- Temporary storage of data and instructions being read from and written to main memory (RAM).
- Stores copies of recent data and instructions.
- Much quicker to read from than main memory (RAM).
- Reduces the need to fetch instructions and data from memory, saving time.

Number of cores

- A core is a complete copy of a CPU.
- A quad-core processor has four separate processing units, each with its own registers, ALU, accumulator and CU.
- Doubling the number of cores won't double processing speed. Cores need to communicate with each other, which takes time. Many programs are not designed to make use of multiple cores.

3.1:Computer Architecture

Watch the video for CPU instruction sets

https://www.youtube.com/watch?v=JS_cey2vHwY



3.1:Computer Architecture

CPU instruction sets



CPU instruction sets

Processors execute **machine code**.

Every type of processor has its own specific lists of commands that it can understand, called an **instruction set**.

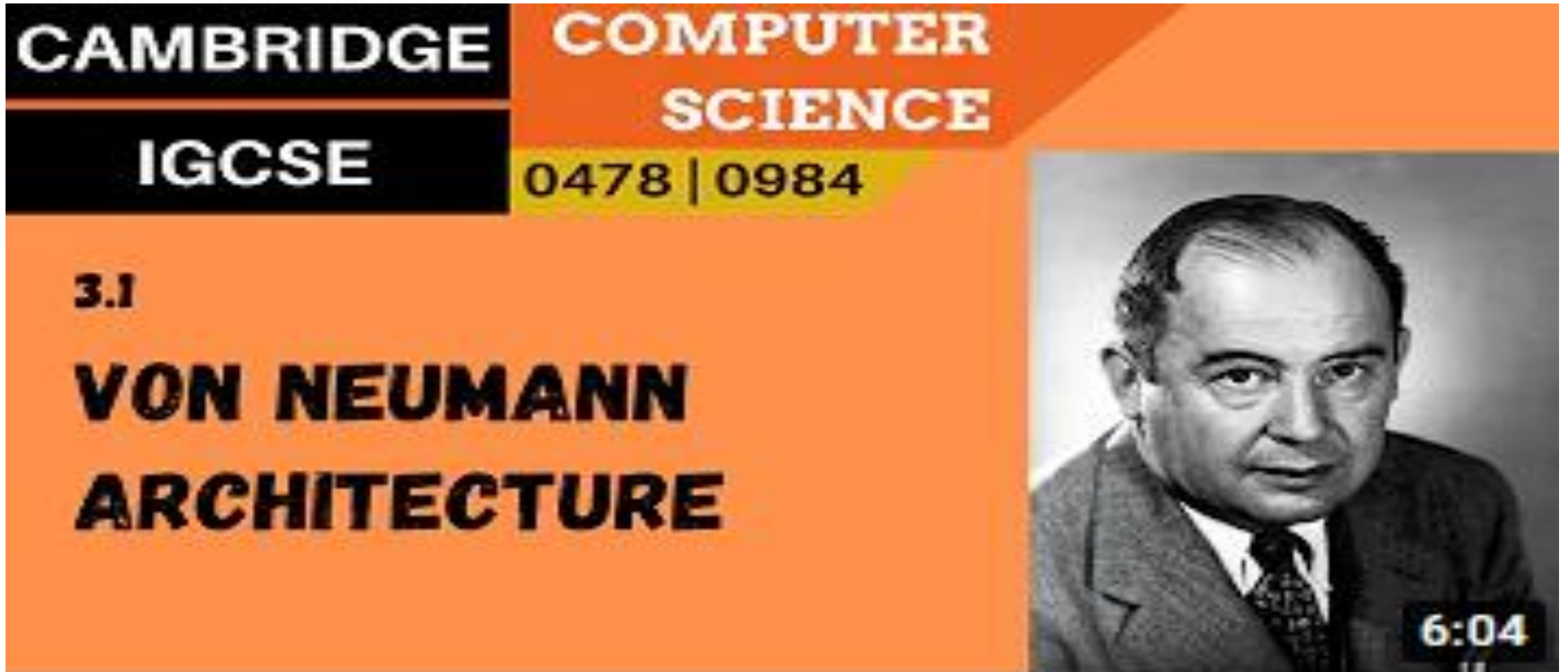
Machine code produced for one type of computer will not work on another – it is machine-specific.

These specific **machine codes** can be represented using **mnemonics**, short codes like LDA.

Mnemonics are easier for humans to understand and form the basis of a **low-level programming** language called **assembly**.

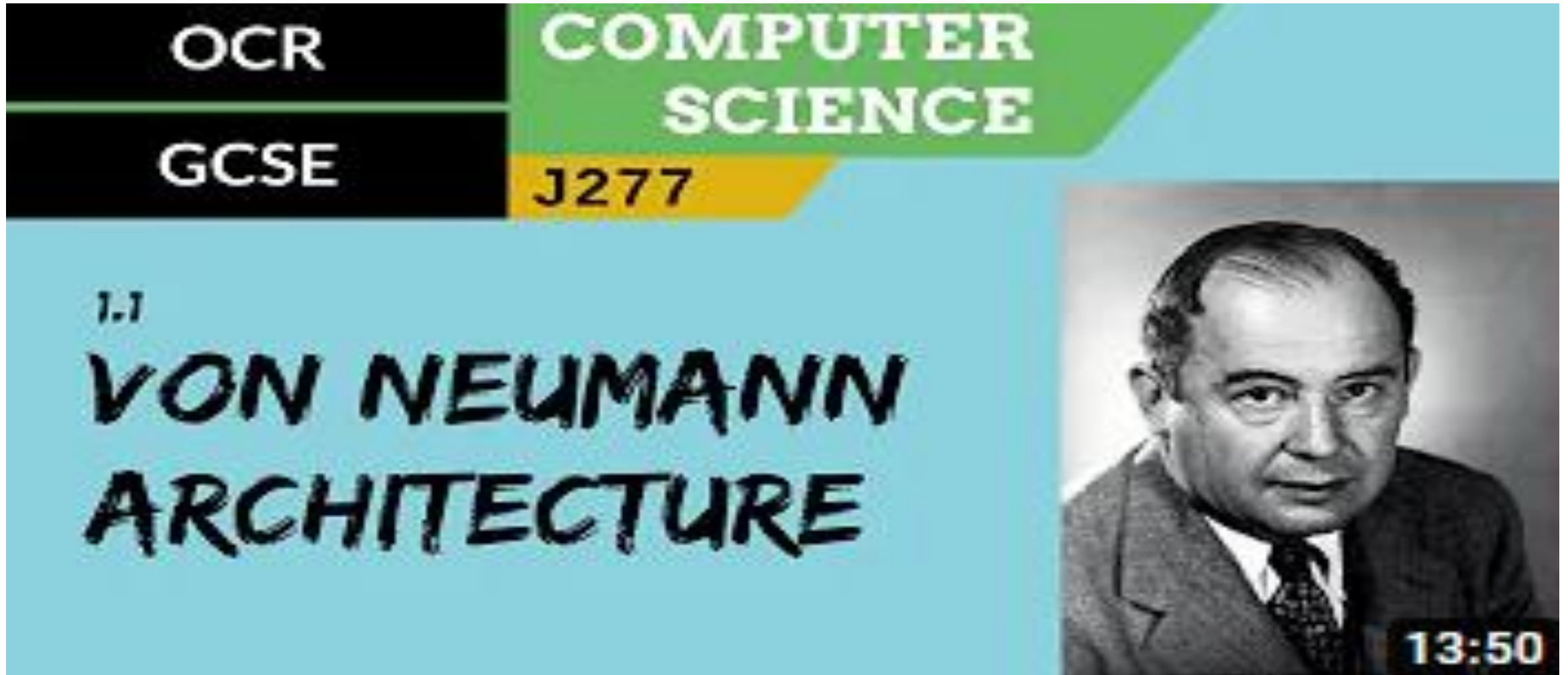
3.1:Computer Architecture

<https://www.youtube.com/watch?v=C0Pch0zpsT4>



3.1:Computer Architecture

<https://www.youtube.com/watch?v=KBmoqwVt4Qg>



3.1:Computer Architecture



Von Neumann architecture

The von Neumann architecture consists of a:

- Control unit (CU)
- Arithmetic logic unit (ALU)
- Main memory (RAM)
- Inputs and outputs

The von Neumann architecture is based on the stored program concept.

Both instruction data *and* program data are stored in the same memory in binary form. There is no way to know if the pure binary held in memory represents instructions or data simply by looking at it.

A von Neumann CPU contains the following registers:

- Program counter (PC): Holds the address of the next instruction to be executed in memory.
- Memory address register (MAR): Holds the memory address where data is to be fetched from or written to.
- Memory data register (MDR): Holds data fetched from or to be written to memory.
- Accumulator: Holds the results of calculations.

3.1:Computer Architecture

In the early days of computing (1940s), computers were built to carry out a very specific task, for example breaking secret wartime codes. But if the computer then had to do another job, it literally had to be completely re-wired by hand. This could take weeks. There was no such thing as a 'software update' in those days!

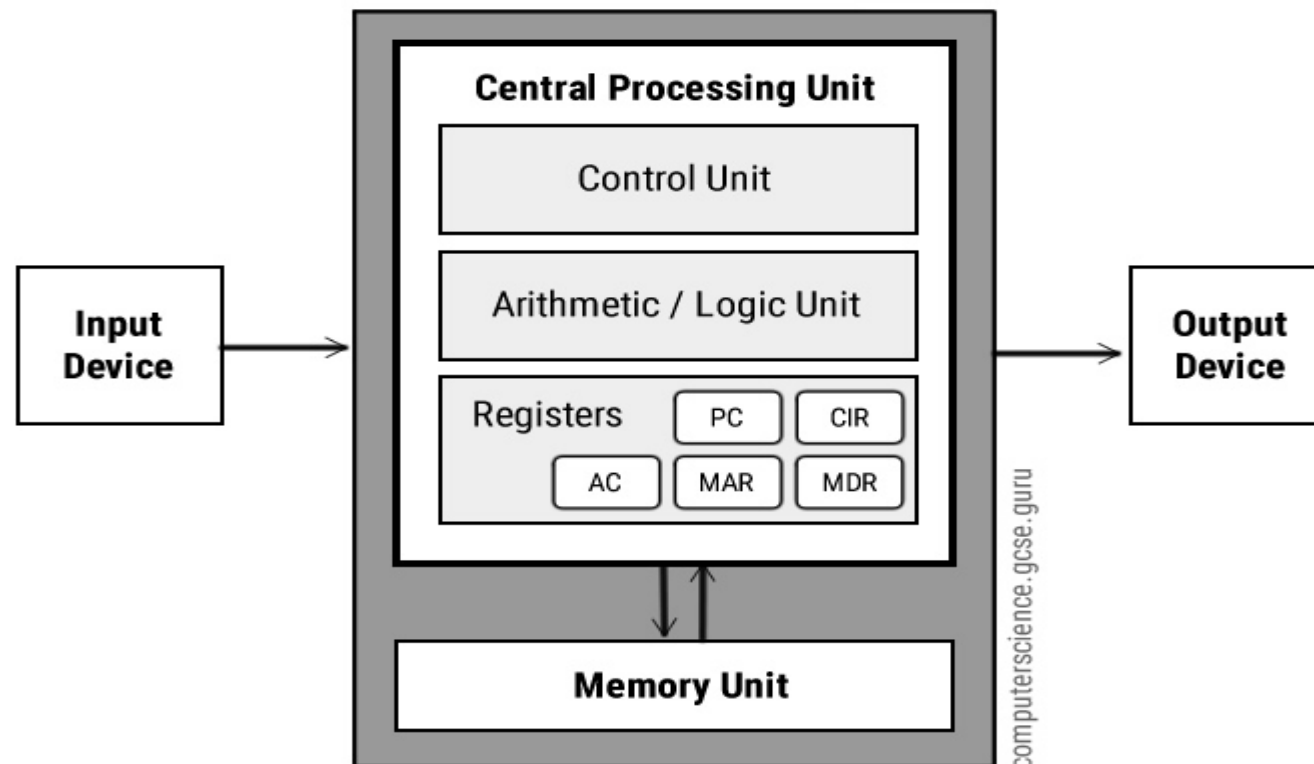
- Mr Von Neumann was a scientist who had an idea of how to create a computer that was far easier to change, this is known as the **Von Neumann architecture**.
- Von Neumann architecture is where both the data and the software that are currently being used are stored in computer memory (RAM).
- It is also known as a '**stored program**' computer, with this architecture, the task a computer is working on can be changed by simply loading a different program into memory



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Noureddine Tadjerout

3.1:Computer Architecture

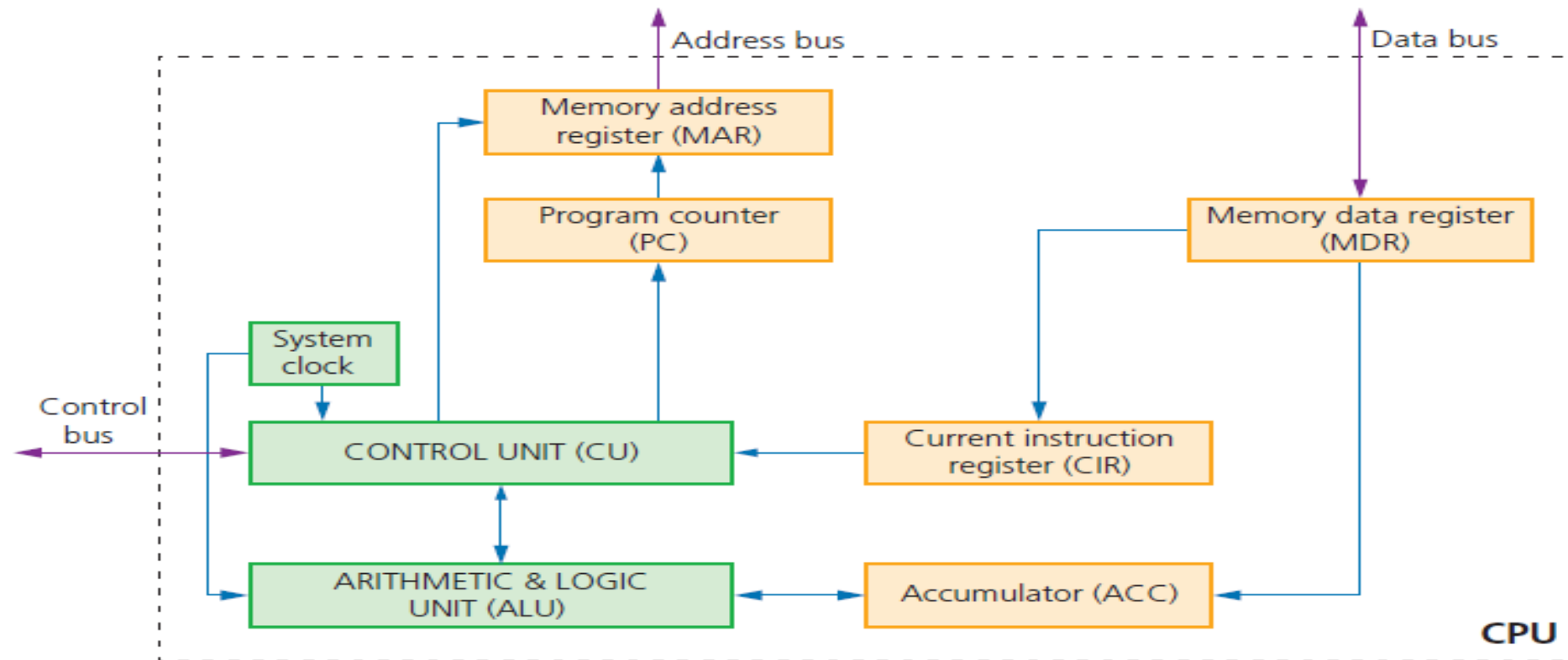
Von Neumann computer architecture design consists of a Control Unit, [Arithmetic and Logic Unit \(ALU\)](#), Memory Unit, [Registers](#) and Inputs/Outputs. Von Neumann architecture is based on the stored-program computer concept, where instruction data and program data are stored in the same memory. This design is still used in most computers produced today.



3.1:Computer Architecture

Von Neumann layout

Figure below shows a slightly more detailed diagram of the von Neumann architecture. It brings to our attention another new concept in this computer model – the idea of **ADDRESSES** and **REGISTERS**. Addresses indicate where the data is stored and registers are needed so that data can be manipulated within the computer..



3.1:Computer architecture

System buses and memory

describes the function of each of the three buses shown in [Figure below](#). Buses essentially move data around the computer and also send out control signals to make sure everything is properly synchronized.

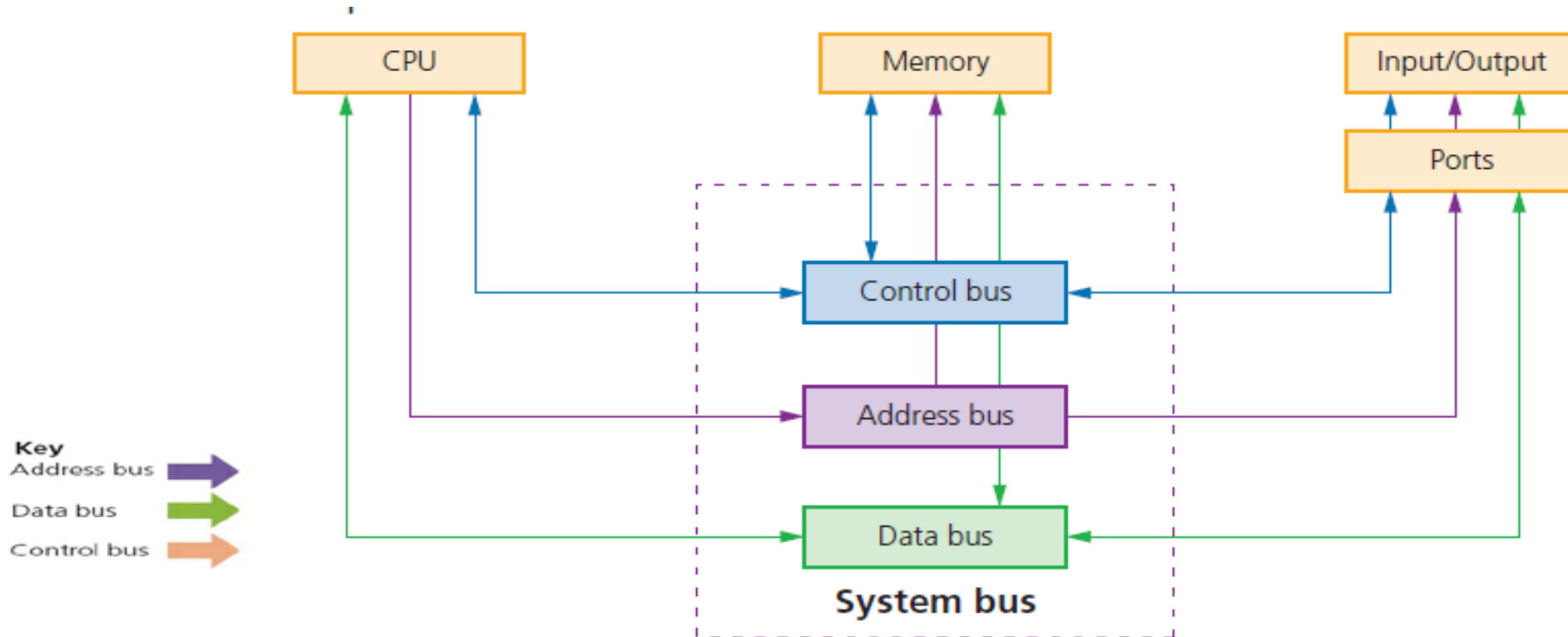
Type of bus	Description of bus	Data/signal direction
address bus	carries signals relating to addresses (see later) between the processor and the memory	unidirectional (signals travel in one direction only)
data bus	sends data between the processor, the memory unit and the input/output devices	bi-directional (data can travel in both directions)
control bus	carries signals relating to the control and coordination of all activities within the computer (examples include: the read and write functions)	unidirectional (signals travel in one direction only)

Address Bus	Carries the addresses of data (but not the data) between the processor and memory
Data Bus	Carries data between the processor, the memory unit and the input/output devices
Control Bus	Carries control signals/commands from the CPU (and status signals from other devices) in order to control and coordinate all the activities within the computer

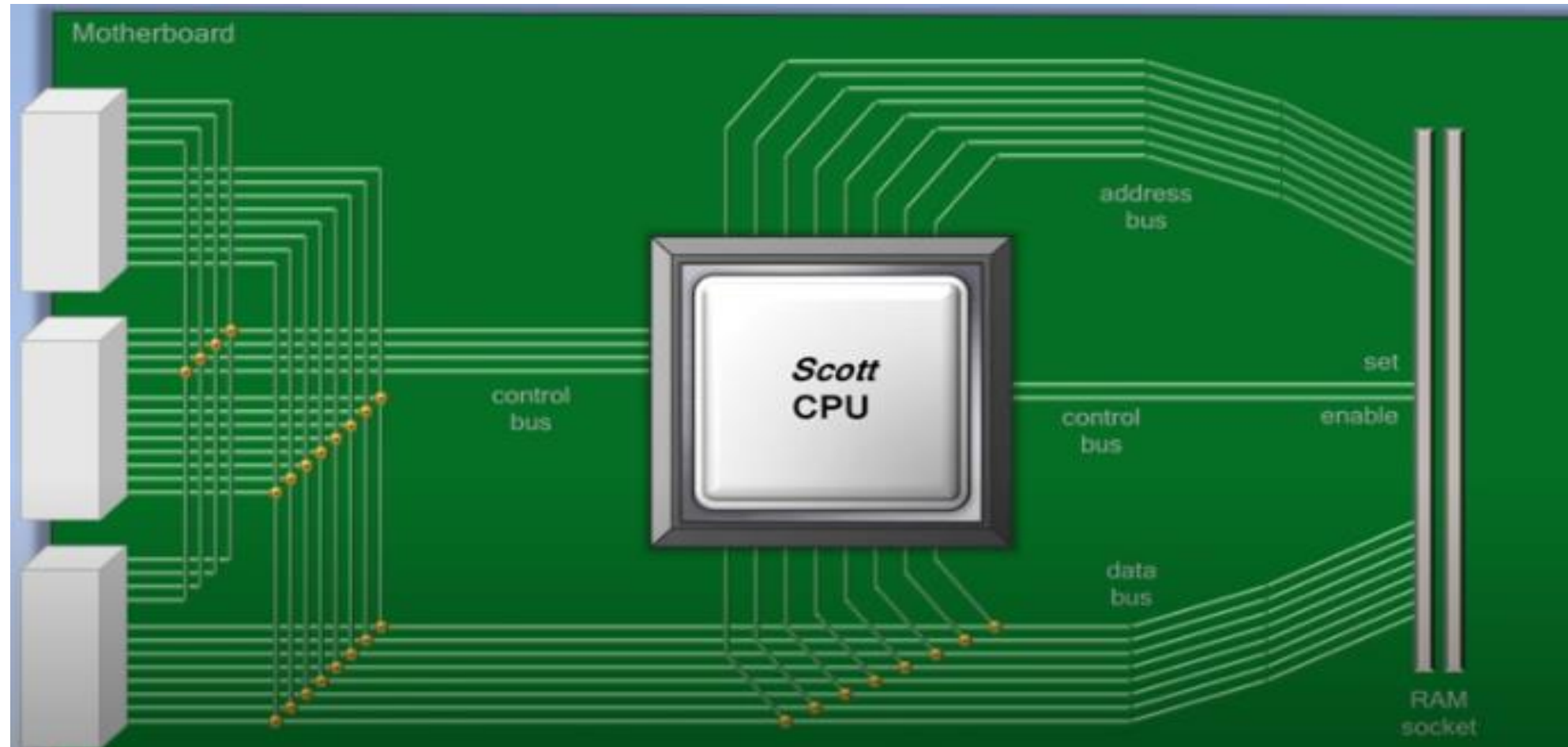
3.1:Computer architecture

System buses and memory

Figure below referred to some components labelled as buses. Figure below shows how these buses are used to connect the CPU to the memory and to input/output devices.



System buses and memory



3.1:Computer Architecture

Registers

One of the most fundamental components of the von Neumann system are the **registers**. Registers can be general or special purpose. We will only consider the special purpose registers. The use of these registers is explained more fully in the Fetch–Decode–Execute cycle

▼ **Table 3.1** Specific purpose registers

Register	Abbreviation used	Function/purpose of register
current instruction register	CIR	this register stores the current instruction being decoded and executed
accumulator	ACC	this register is used when carrying out ALU calculations; it stores data temporarily during the calculations
memory address register	MAR	this register stores the address of the memory location currently being read from or written to
memory data/ buffer register	MDR	this register stores data which has just been read from memory or data which is about to be written to memory
program counter	PC	this register stores the address where the next instruction to be read can be found

3.1:Computer architecture

System Clock

system clock is used to produce timing signals on the control bus to ensure this vital synchronisation takes place – without the clock the computer would simply crash! The RAM holds all the data and programs needed to be accessed by the CPU. The RAM is often referred to as the **Immediate Access Store (IAS)**. The CPU takes data and programs held in **backing store** (e.g. a hard disk drive) and puts them into RAM temporarily. This is done because read/write operations carried out using the RAM are considerably faster than read/write operations to backing store; consequently, any key data needed by an application will be stored temporarily in RAM to considerably speed up operations..

3.1:Computer Architecture

Watch the video for Fetch-decode-execute cycle

https://www.youtube.com/watch?v=8Qx6Dr_aKXs



3.1:Computer Architecture

Fetch-decode-execute cycle

Fetch-decode-execute cycle

The purpose of the **CPU** is to **fetch**, **decode** and **execute** instructions – it does this billions of times a second.



Fetch

1. The **program counter** is checked for the address of the next instruction to be executed.
2. The contents of the **program counter** are copied into the **memory address register**.
3. The **program counter** is incremented.
4. The **control unit** sends a memory read request to **main memory** down the **control bus**.
5. The address is sent down the **address bus** from the **memory address register** to **main memory**.
6. The contents of **main memory** are sent down the **data bus** into the **memory data register**.
7. The contents of the **memory data register** are copied into the **current instruction register**.

Decode

8. The instruction in the **current instruction register** is inspected and **decoded** to work out what needs to be done.

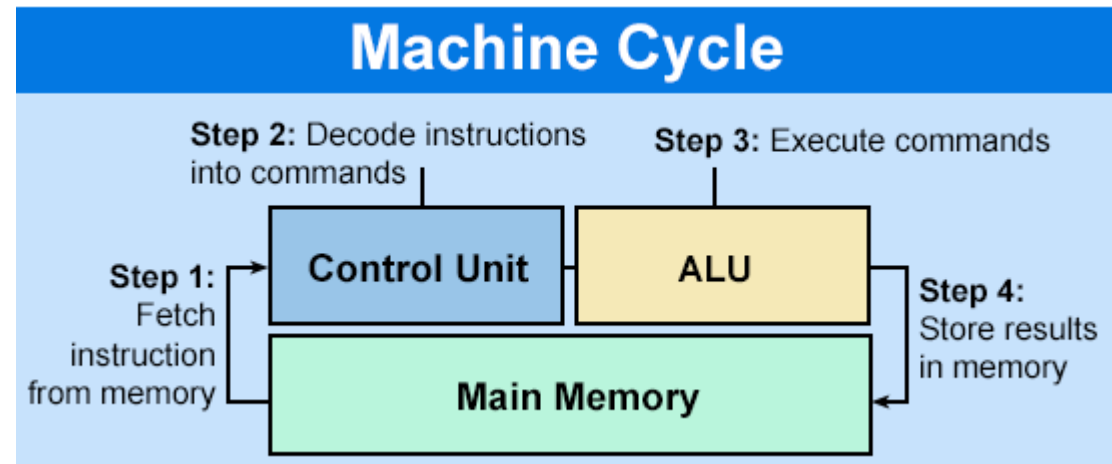
Execute

9. The instruction is carried out – this could be:
 - a) Going back to **main memory** and **fetching** data.
 - b) Performing a calculation.
 - c) Storing information back in **main memory**.
10. Depending on the instruction, data may end up being copied into the **ALU** and/or the **accumulator**.

The fetch–execute cycle

- **Fetch** - Retrieve an [instruction](#) from [memory](#).
- **Decode** - Translate the retrieved instruction into a series of computer commands.
- **Execute** - Execute the computer commands.
- **Store** - Send and write the results back in memory.

Computer processors use [pipelining](#) (**micro processing technique that handles multiple stages of different computer instructions at once**), which allows the processor to begin fetching a second instruction before it has completed the machine cycle for another instruction



Fetch mean: to go to another place to get something

The fetch–execute cycle

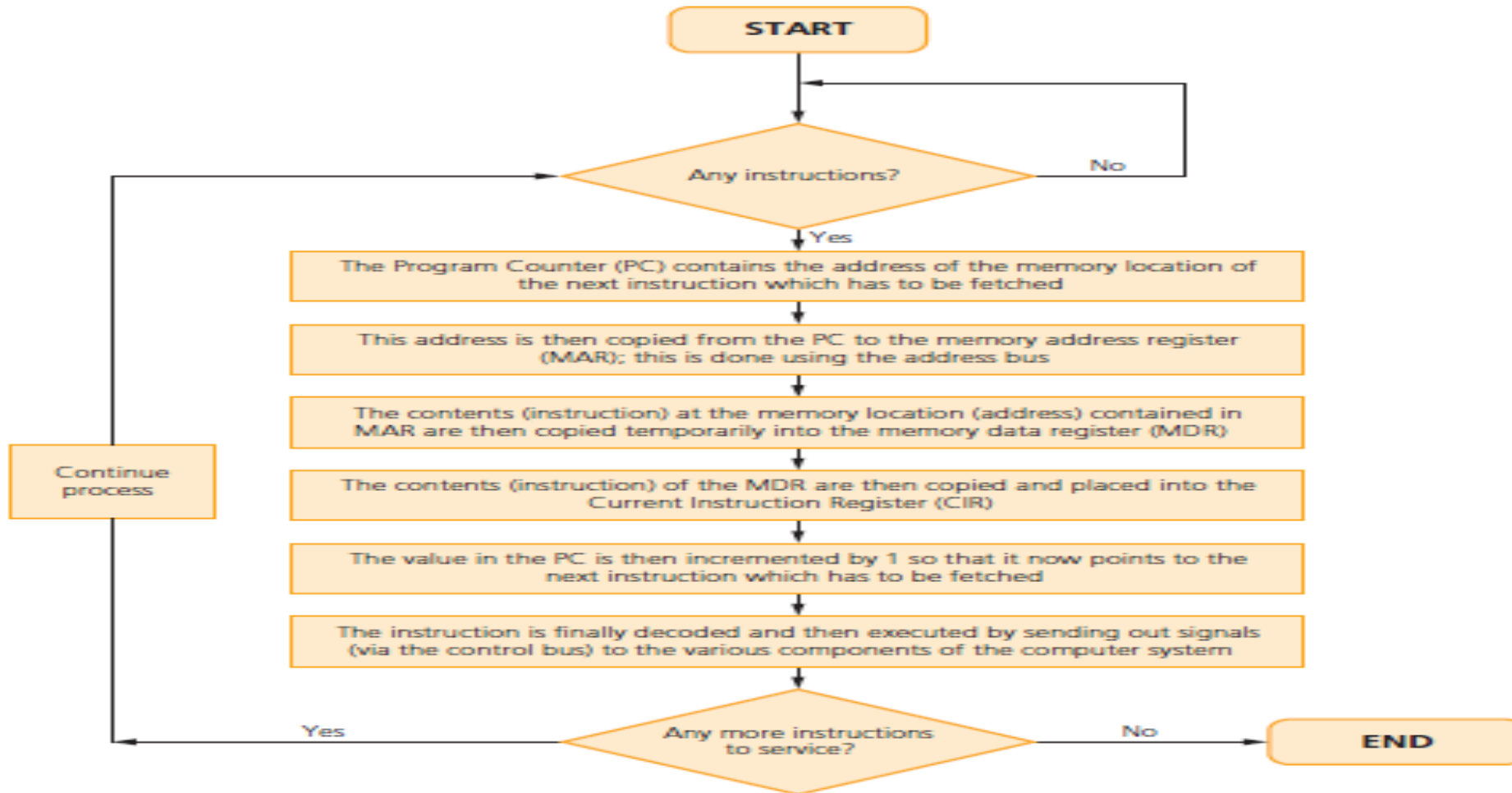
Example of a machine cycle

Below is an example of a machine cycle performing the steps mentioned above for a math problem.

- The computer user enters a math problem that's stored in memory. The computer fetches that instruction from memory.
- The control unit decodes that math problem into instructions the computer understands.
- The ALU executes the instructions to get the answer to the math problem.
- The problem's answer is sent to and stored in memory where it can be accessed and displayed to the user.

The fetch–execute cycle

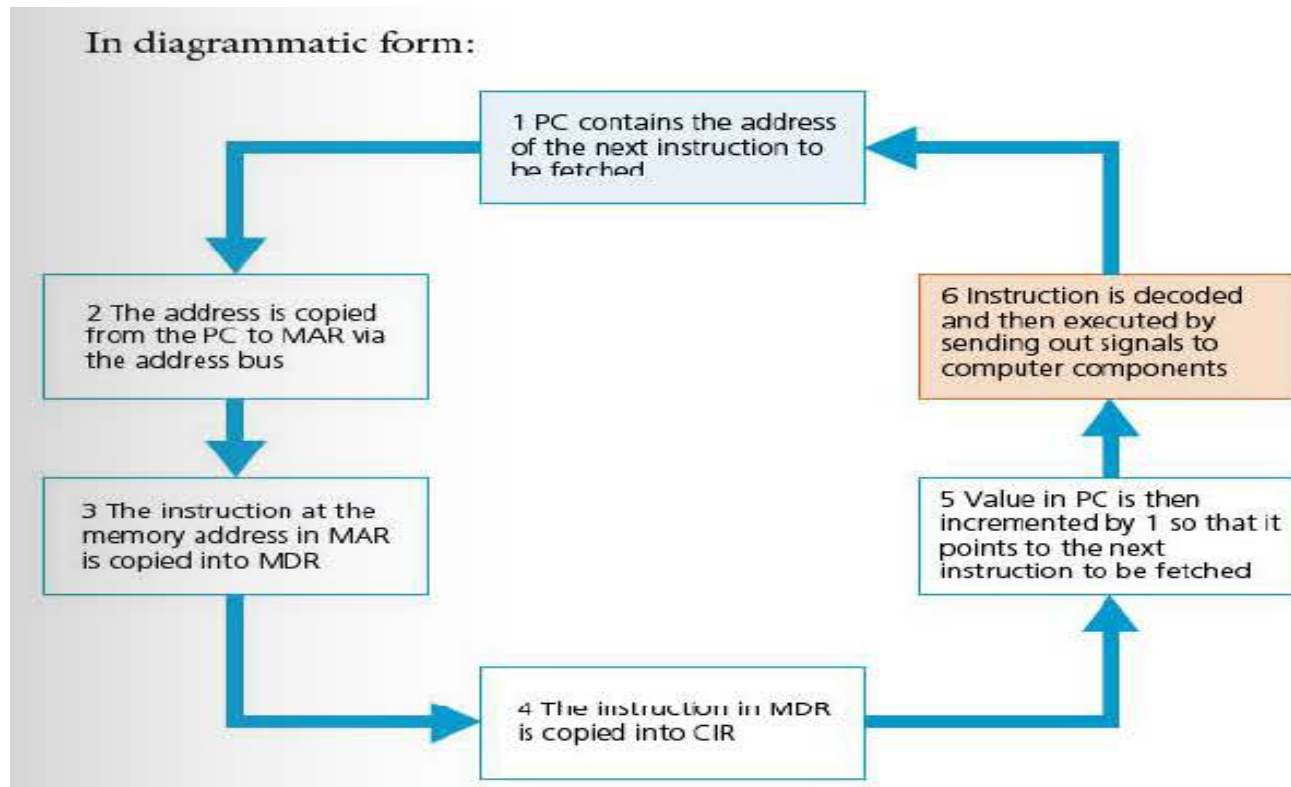
Figure 3.3 shows how the Fetch–Decode–Execute cycle is carried out in the von Neumann computer model.

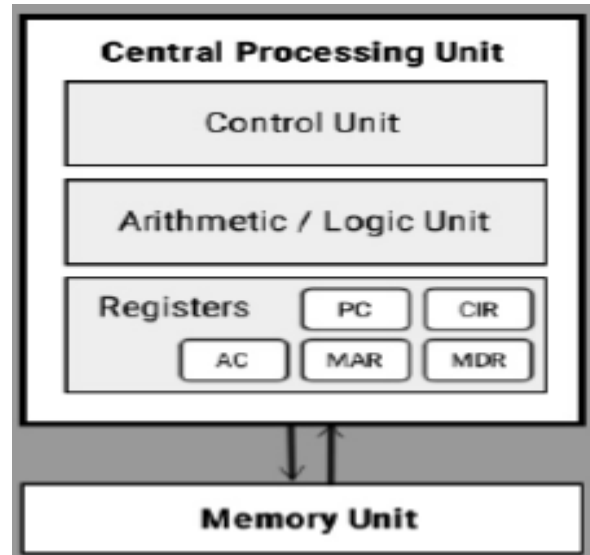


▲ **Figure 3.3** Fetch–Decode–Execute cycle flowchart

The fetch–execute cycle

- Figure below shows how the fetch–execute cycle is carried out in the von Neumann computer model.

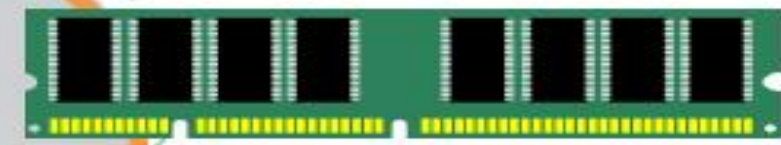




Address Bus - asks for the instruction from RAM

Control Unit

- Responsible for decoding the instructions



Main Memory (RAM)
Random Access Memory
Volatile as loses information when switched OFF.

Data Bus - brings the instruction back



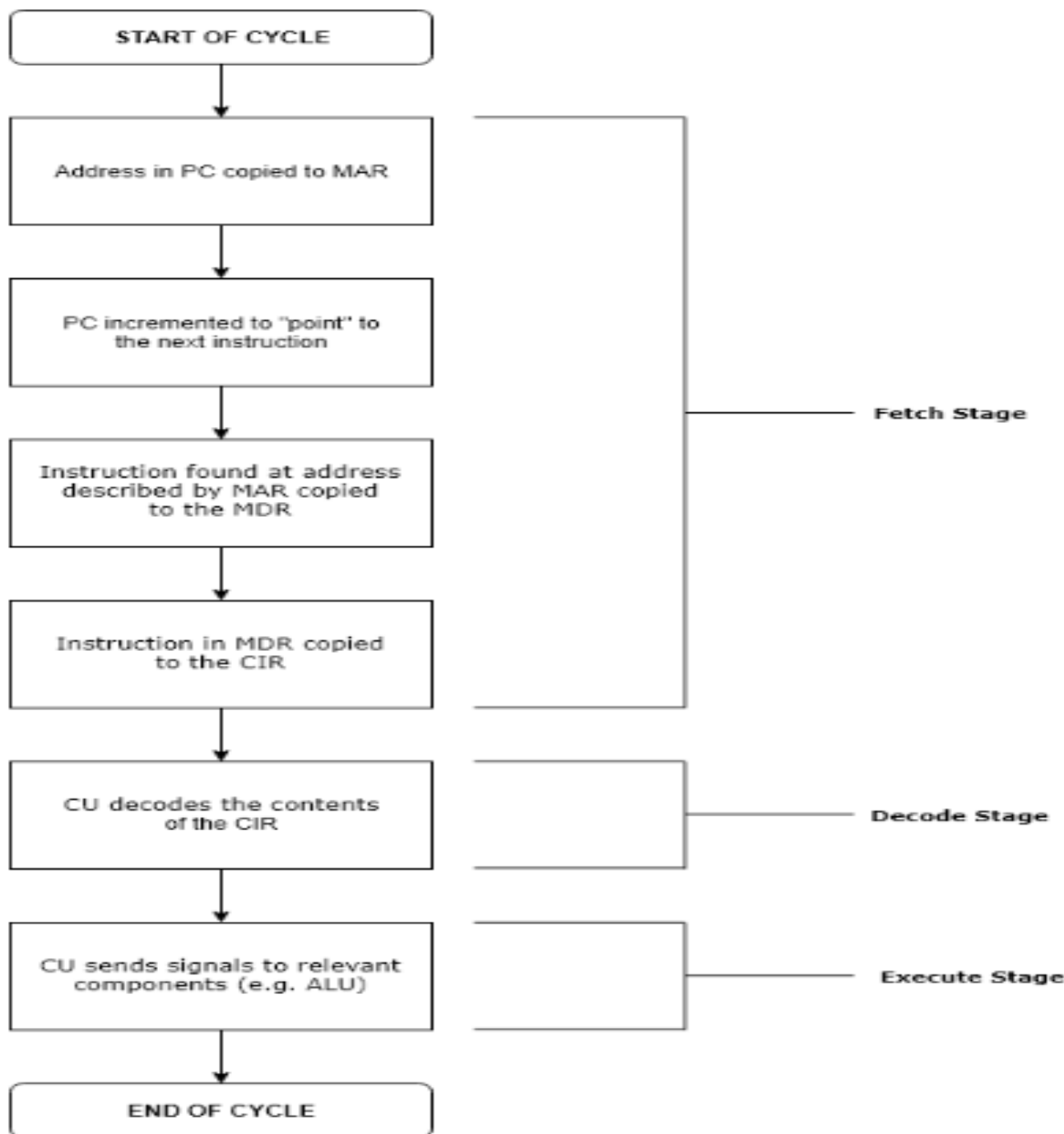
Stage	Order
address is copied from PC to MAR; this is done using the address bus	2
contents at memory location contained in MAR are copied temporarily into MDR	3
contents of MDR are copied and placed in CIR	4
instruction is finally decoded and then executed by sending out signals (via the control bus) to various components of the computer system	6
program counter (PC) contains address of memory location of next instruction to be fetched	1
value in PC is incremented by 1 so that it now points to the next instruction to be fetched	5

This cycle runs inside the CPU continuously

(CPU) - Central Pro



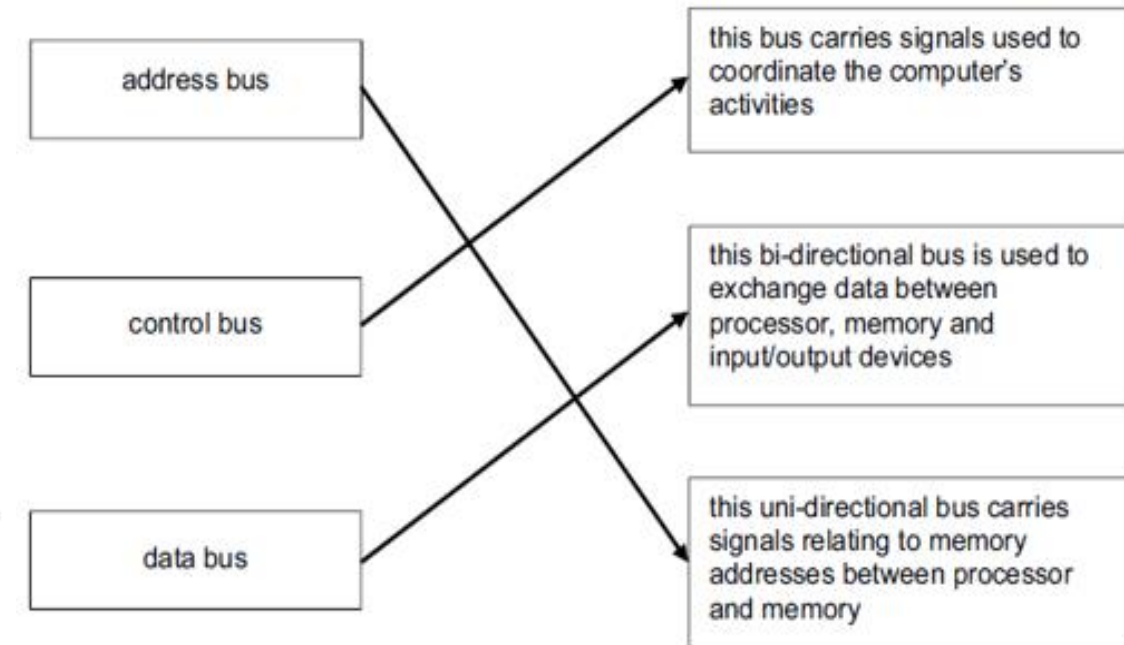
Accumulator: Stores the results of calculations made by the ALU.



Registers

Registers are high speed storage areas in the CPU. All data must be stored in a register before it can be processed.

MAR	Memory Address Register	Holds the memory location of data that needs to be accessed
MDR	Memory Data Register	Holds data that is being transferred to or from memory
AC	Accumulator	Where intermediate arithmetic and logic results are stored
PC	Program Counter	Contains the address of the next instruction to be executed
CIR	Current Instruction Register	Contains the current instruction during processing



Exam style IGCSE Exam Question 1

4 a (i) Describe the function of the ALU.

(ii) Describe the function of the Control Unit. [4 marks]

b The following table shows the stages in the fetch–execute cycle. They are not written in the correct order. Place the numbers 1 to 6 in the right-hand column to show each stage in its correct order.

The first one has been done for you. [5 marks]

Stage	Order
address is copied from PC to MAR; this is done using the address bus	
contents at memory location contained in MAR are copied temporarily into MDR	
contents of MDR are copied and placed in CIR	
instruction is finally decoded and then executed by sending out signals (via the control bus) to various components of the computer system	
program counter (PC) contains address of memory location of next instruction to be fetched	1
value in PC is incremented by 1 so that it now points to the next instruction to be fetched	

Exam style IGCSE Exam Answer 1

- 4 a (i) • carries out arithmetic operations
• carries out logic operations
- (ii) • controls operation of memory, processor and I/O devices
• sends signals to coordinate all computer components

b

Stage	Order
address is copied from PC to MAR; this is done using the address bus	2
contents at memory location contained in MAR are copied temporarily into MDR	3
contents of MDR are copied and placed in CIR	4
instruction is finally decoded and then executed by sending out signals (via the control bus) to various components of the computer system	6
program counter (PC) contains address of memory location of next instruction to be fetched	1
value in PC is incremented by 1 so that it now points to the next instruction to be fetched	5

Memory unit

The computer memory unit is made up of a number of partitions. Each partition consists of an **ADDRESS** and its **CONTENTS**. The address will uniquely identify every **LOCATION** in the memory and the contents will be the binary. Let us now consider how the two registers (MAR and MDR) shown in the memory unit are used.

- MAR memory address register
- MDR memory data register

Memory unit

● Sample question

- 3 a** The location in a memory unit contains the address 1 1 0 1 1 0 0 0 and the contents 0 0 0 1 1 0 0 0.

Describe a READ operation on location 1 1 0 1 1 0 0 0 indicating the role of the MAR, MDR and buses.

- b** If the value 1 1 1 0 1 1 1 0 is to be stored in location 1 1 1 1 0 1 1 1, describe the WRITE operation indicating the role of the MAR, MDR and buses.

Student answer

- 3 a** MAR will contain 11011000
MDR will contain 00011000
A read signal is sent to the computer memory along the control bus.
- b** MAR will contain 11110111
MDR will contain 11101110
A write signal is sent to the computer memory along the control bus.

Examiner's comments

Whilst the student has answered the question correctly in both parts, it is often a good idea to show a memory map to aid in the description.

Address	Contents
1101 1000	0001 1000
↓	↓
1111 0111	1110 1110

This will help greatly in the description of the READ and WRITE operation. The location contains the address and contents; it is important to distinguish between location and address.

3.1:Computer Architecture

Watch the video for Embedded systems

<https://www.youtube.com/watch?v=NwGQLfEBR5E>



3.1:Computer Architecture

Embedded systems

Embedded systems

Properties of embedded systems compared to general-purpose computers include:

- Low power consumption
- Small size
- Rugged operating systems
- Low cost per unit

These benefits come at the cost of limited processing resources, making them more difficult to program and interact with.



3.1:Computer Architecture

Embedded systems

Embedded systems

Embedded systems are dedicated to a specific task – this means design engineers can optimise the system to:

- Reduce the size and cost of the product.
- Increase its reliability for the given task.

Many embedded systems are mass-produced, meaning further benefits can be gained from economies of scale.



3.1:Computer Architecture

Embedded systems



Embedded systems

Embedded system: A computer system with a dedicated function within a larger mechanical system.

Typical properties of embedded systems:

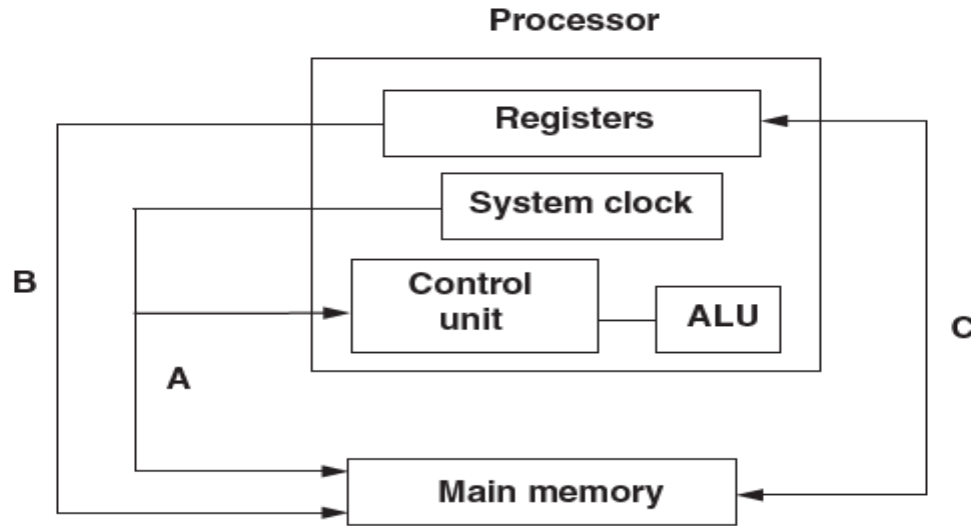
- Low power consumption
- Small size
- Rugged operating ranges
- Low cost per unit

Examples of embedded systems:

- Traffic lights
- Domestic appliances
- Factory equipment
- Engine management systems
- Hospital equipment

IGCSE Exam Question 1

2 (a)



The diagram above shows a simplified form of processor architecture.

Name the three buses labelled A, B and C.

A

B

C

[3]

IGCSE Exam Question 1

(b) State the role of each of the following special purpose registers used in a typical processor.

Program Counter

.....

Memory Data Register

.....

Current Instruction Register

.....

Memory Address Register

.....

[4]

IGCSE Exam Answer 1

2 (a) A = control bus

B = address bus

C = data bus

[3]

(b) Program Counter – stores the address of next instruction to be executed

Memory Data Register – stores the data in transit between memory and other registers //
holds the instruction before it is passed to the CIR

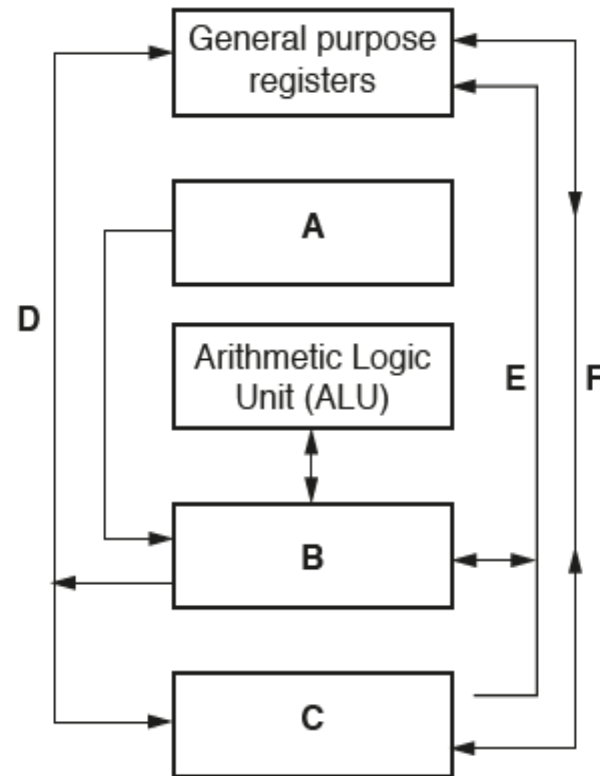
Current Instruction Register – stores the current instruction being executed

Memory Address Register – stores the address of the memory location which is about to be
accessed

[4]

IGCSE Exam Question 2

- 4 (a) The diagram shows the components and buses found inside a typical Personal Computer (PC).



Some components and buses only have labels **A** to **F** to identify them.

IGCSE Exam Question 2

For each label, choose the appropriate title from the following list. The title for label **D** is already given.

- Control bus
- System clock
- Data bus
- Control unit
- Main memory
- Secondary storage

A

B

C

D Address bus

E

F

[5]

IGCSE Exam Answer 2

Question	Answer	Marks
4(a)	A – System clock B – Control unit C – Main memory E – Control bus F – Data bus	5

IGCSE Exam Question 3

Six components of the Von Neumann model for a computer system and **six** descriptions are given.

Draw a line to match each component to the most suitable description.

Component	Description
Immediate access store (IAS)	Holds data and instructions when they are loaded from main memory and are waiting to be processed.
Register	Holds data temporarily that is currently being used in a calculation.
Control unit (CU)	Holds data or instructions temporarily when they are being processed.
Accumulator (ACC)	Manages the flow of data and interaction between the components of the processor.
Arithmetic logic unit (ALU)	Carries out the calculations on data.
Bus	Pathway for transmitting data and instructions.

IGCSE Exam Answer 3

Question	Answer	Marks														
5	1 mark for each correct line, up to a maximum of 5 marks: <table><thead><tr><th>Component</th><th>Description</th></tr></thead><tbody><tr><td>Immediate access store (IAS)</td><td>Holds data and instructions when they are loaded from main memory and are waiting to be processed.</td></tr><tr><td>Register</td><td>Holds data temporarily that is currently being used in a calculation.</td></tr><tr><td>Control unit (CU)</td><td>Holds data or instructions temporarily when they are being processed.</td></tr><tr><td>Accumulator (ACC)</td><td>Manages the flow of data and interaction between the components of the processor.</td></tr><tr><td>Arithmetic logic unit (ALU)</td><td>Carries out the calculations on data.</td></tr><tr><td>Bus</td><td>Pathway for transmitting data and instructions.</td></tr></tbody></table>	Component	Description	Immediate access store (IAS)	Holds data and instructions when they are loaded from main memory and are waiting to be processed.	Register	Holds data temporarily that is currently being used in a calculation.	Control unit (CU)	Holds data or instructions temporarily when they are being processed.	Accumulator (ACC)	Manages the flow of data and interaction between the components of the processor.	Arithmetic logic unit (ALU)	Carries out the calculations on data.	Bus	Pathway for transmitting data and instructions.	5
Component	Description															
Immediate access store (IAS)	Holds data and instructions when they are loaded from main memory and are waiting to be processed.															
Register	Holds data temporarily that is currently being used in a calculation.															
Control unit (CU)	Holds data or instructions temporarily when they are being processed.															
Accumulator (ACC)	Manages the flow of data and interaction between the components of the processor.															
Arithmetic logic unit (ALU)	Carries out the calculations on data.															
Bus	Pathway for transmitting data and instructions.															

IGCSE Exam Question 4

Six components of a computer system and six descriptions are shown.

Draw a line to match each component with the most suitable description.

Component	Description
Arithmetic Logic Unit (ALU)	Used to connect together the internal components of the CPU.
Buses	Used to carry out calculations on data.
Control Unit (CU)	Used to temporarily hold data and instructions during processing.
Immediate Access Store (IAS)	Used to allow interaction with the computer.
Input/Output	Used to hold data and instructions before they are processed.
Registers	Used to manage the flow of data through the CPU.

IGCSE Exam Answer 4

Question	Answer														
4	1 mark for each correct line up to a total of 5 marks <table border="1"> <thead> <tr> <th data-bbox="537 311 823 339">Component</th><th data-bbox="1090 311 1352 339">Description</th></tr> </thead> <tbody> <tr> <td data-bbox="537 361 823 461">Arithmetic Logic Unit (ALU)</td><td data-bbox="1090 361 1447 461">Used to connect together the internal components of the CPU.</td></tr> <tr> <td data-bbox="537 518 823 618">Buses</td><td data-bbox="1090 518 1447 618">Used to carry out calculations on data.</td></tr> <tr> <td data-bbox="537 675 823 775">Control Unit (CU)</td><td data-bbox="1090 675 1447 775">Used to temporarily hold data and instructions during processing.</td></tr> <tr> <td data-bbox="537 832 823 932">Immediate Access Store (IAS)</td><td data-bbox="1090 832 1447 932">Used to allow interaction with the computer.</td></tr> <tr> <td data-bbox="537 989 823 1089">Input/Output</td><td data-bbox="1090 989 1447 1089">Used to hold data and instructions before they are processed.</td></tr> <tr> <td data-bbox="537 1146 823 1246">Registers</td><td data-bbox="1090 1146 1447 1246">Used to manage the flow of data through the CPU.</td></tr> </tbody> </table>	Component	Description	Arithmetic Logic Unit (ALU)	Used to connect together the internal components of the CPU.	Buses	Used to carry out calculations on data.	Control Unit (CU)	Used to temporarily hold data and instructions during processing.	Immediate Access Store (IAS)	Used to allow interaction with the computer.	Input/Output	Used to hold data and instructions before they are processed.	Registers	Used to manage the flow of data through the CPU.
Component	Description														
Arithmetic Logic Unit (ALU)	Used to connect together the internal components of the CPU.														
Buses	Used to carry out calculations on data.														
Control Unit (CU)	Used to temporarily hold data and instructions during processing.														
Immediate Access Store (IAS)	Used to allow interaction with the computer.														
Input/Output	Used to hold data and instructions before they are processed.														
Registers	Used to manage the flow of data through the CPU.														

IGCSE Exam Question 5

(a) One of the key features of von Neumann computer architecture is the use of buses.

Three buses and three descriptions are shown below.

Draw a line to connect each bus to its correct description.

Bus

address bus

control bus

data bus

Description

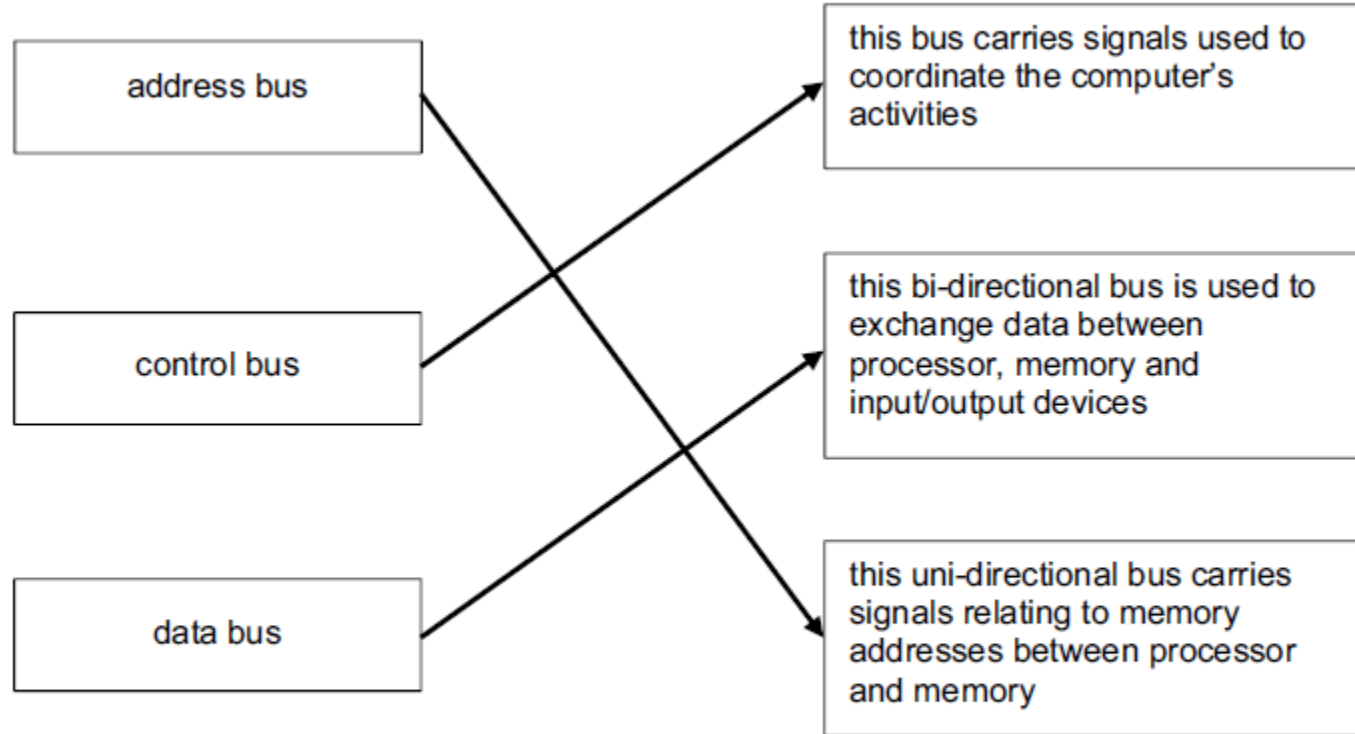
this bus carries signals used to coordinate the computer's activities

this bi-directional bus is used to exchange data between processor, memory and input/output devices

this uni-directional bus carries signals relating to memory addresses between processor and memory

IGCSE Exam Answer 5

(a)



2/3 matches – 2 marks
1 match – 1 mark

[2]

IGCSE Exam Question 6

4 A computer has a Von Neumann architechure.

(a) Circle **three** components that are part of the central processing unit (CPU) in this computer.

accumulator (ACC)

hard disk drive (HDD)

memory address register (MAR)

program counter (PC)

random access memory (RAM)

read only memory (ROM)

sensor

solid state drive (SSD)

[3]

(b) Describe the purpose of the control unit (CU) within this computer.

.....

.....

.....

..... [2]

IGCSE Exam Question 6

(c) The computer has a single core CPU.

(i) State one purpose of a core in a CPU.

.....
..... [1]

(ii) The computer is upgraded to a dual core CPU.

Explain how the upgrade can affect the performance of the computer.

.....
.....
.....
..... [2]

IGCSE Exam Answer 6

Question	Answer	Marks
4(a)	One mark for each correctly circled component: • accumulator (ACC) • program counter (PC) • memory address register (MAR)	3
4(b)	Any two from: • It sends control signals • ... that manage the transfer of data and instructions within the CPU • It decodes an instruction • ... using an instruction set	2
4(c)(i)	Any one from: • to process an instruction • to carry out a fetch-execute cycle.	1

IGCSE Exam Question 7

A Von Neumann model for a computer system has a central processing unit (CPU) that makes use of registers.

- (a) Identify **three** registers that may be used.

Register 1

Register 2

Register 3

[3]

- (b) The CPU is responsible for processing instructions.

One stage of processing instructions is the decode stage.

- (i) Identify the **two other** stages of processing instructions.

Stage 1

Stage 2

[2]

- (ii) Identify the component of the CPU that is responsible for decoding instructions.

..... [1]

IGCSE Exam Answer 7

Question	Answer	Marks
1(a)	Any three from: – MAR – MDR // MBR – PC // IAR // NIR // SCR – ACC – CIR // IR – IAS	3
1(b)(i)	– Fetch – Execute	2
1(b)(ii)	– Control unit	1

IGCSE Exam Question 8

(b) A computer uses the Von Neumann model and the stored program concept.

(i) Explain what is meant by the stored program concept.

.....

.....

.....

..... [2]

(ii) The Von Neumann model has several components that are used in the fetch-execute cycle.

One component is the Arithmetic Logic Unit (ALU).

Describe the role of the ALU.

.....

.....

IGCSE Exam Answer 8

2(b)(i)	Two from: - Programs / instructions are stored in memory - Data is stored in memory - Instructions are fetched and executed one after another	2
2(b)(ii)	- Carries out calculations - Carries out logical operations - Holds temporary / interim values during calculations - ... in a register called the accumulator (ACC)	4

IGCSE Exam Question 9

Kelvin correctly answers an examination question about the Von Neumann model.

Eight different terms have been removed from his answer.

Complete the sentences in Kelvin's answer, using the list given.

Not all items in the list need to be used.

- accumulator (ACC)
- address bus
- arithmetic logic unit (ALU)
- control unit (CU)
- data bus
- executed
- fetches
- immediate access store (IAS)
- memory address register (MAR)
- memory data register (MDR)
- program counter (PC)
- saved
- transmits

The central processing unit (CPU)

the data and instructions needed and stores them in the

..... to wait to be processed.

The holds the address of the next

instruction. This address is sent to the

The data from this address is sent to the

The instruction can then be decoded and

Any calculations that are carried out on the data are done by the

..... . During calculations, the data is temporarily

held in a register called the

IGCSE Exam Answer 9

Question	Answer	Marks
6	1 mark for each correct missing word, in the given order: - fetches - immediate access store // IAS - program counter // PC - memory address register // MAR - memory data register // MDR - executed - arithmetic logic unit // ALU - accumulator // ACC	8

IGCSE Exam Question 10

Signals are sent to and from the components of a processor using buses.

Identify and describe the purpose of **two** different buses.

Bus 1

Purpose

.....

.....

.....

.....

.....

Bus 2

Purpose

.....

.....

IGCSE Exam Answer 10

Question	Answer	Marks
6	1 mark for correct bus name and up to 2 further marks for appropriate purpose. Address (bus) Two from: ☐ Carries / transports an address / location ... ☐ ... of the next item to be fetched ☐ Data travels one way (unidirectional) Data (bus) Two from: ☐ Carries / transports data / example of data ... ☐ ... that is currently being processed // that will be / has been processed ☐ Data can travel in both directions (bidirectional) Control (bus) Two from: ☐ Carries / transports signals ☐ Control / directs the actions of the CPU / processor ☐ Can be either Unidirectional or Bidirectional	6

IGCSE Exam question with Question 11

Name **three** different buses that are used in the fetch-execute cycle.

Bus 1

Bus 2

Bus 3

[3]

IGCSE Exam question with Answer 11

Question	Answer	Marks
1	☐ address (bus) ☐ control (bus) ☐ data (bus)	3

IGCSE Exam Question 12

The fetch-execute cycle make use of registers.

(a) Describe the role of the Program Counter (PC).

.....

.....

.....

.....[2]

(b) Describe the role of the Memory Data Register (MDR).

.....

.....

.....

.....[2]

IGCSE Exam Answer 12

Question	Answer	Marks
11(a)	☐ Holds address of next/current instruction ... ☐ ... to be fetched/processed/executed	2
11(b)	☐ Stores data/instruction that is in use ... ☐ ... from address in MAR	2

IGCSE Exam question 13

- (a) One of the key features of von Neumann computer architecture is the use of buses.

Three buses and three descriptions are shown below.

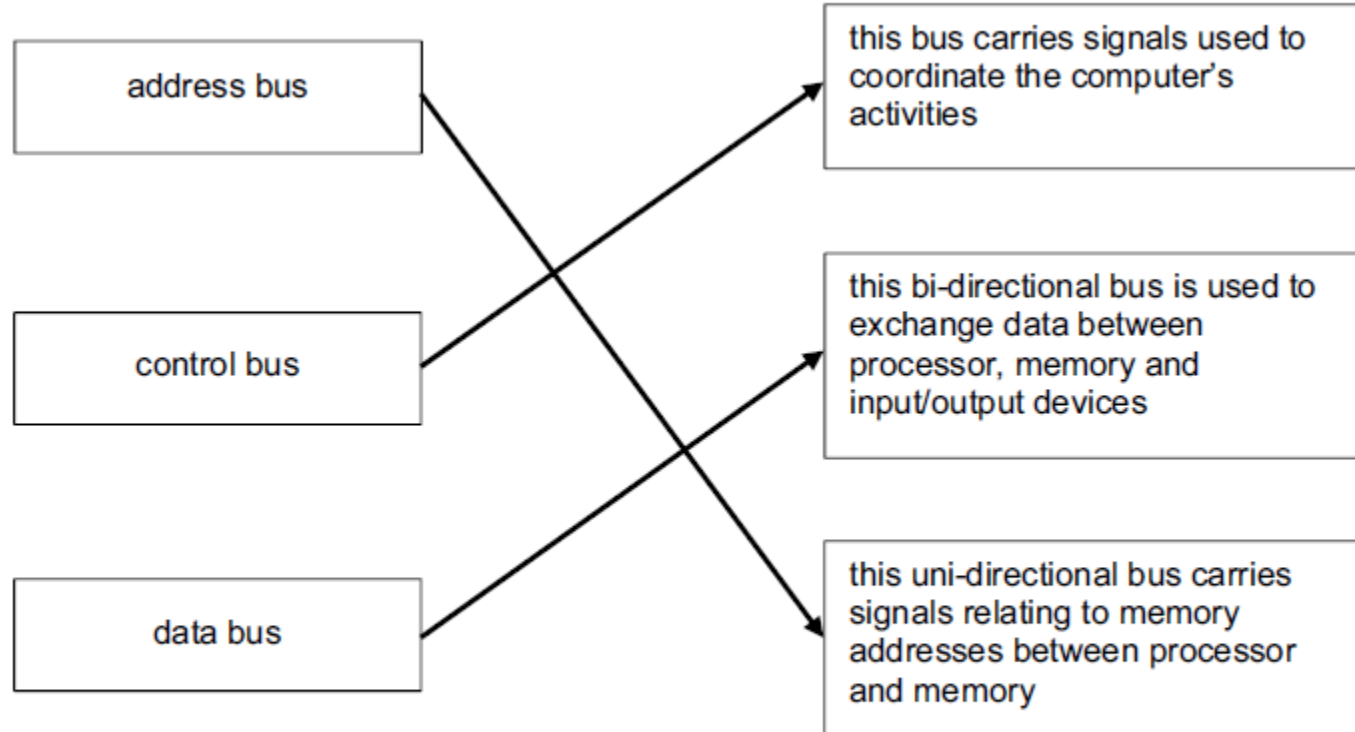
Draw a line to connect each bus to its correct description.

Bus	Description
address bus	this bus carries signals used to coordinate the computer's activities
control bus	this bi-directional bus is used to exchange data between processor, memory and input/output devices
data bus	this uni-directional bus carries signals relating to memory addresses between processor and memory

[2]

IGCSE Exam Answer 13

(a)



2/3 matches – 2 marks
1 match – 1 mark

[2]

IGCSE Exam question 14

11

(b) The seven stages in a von Neumann fetch-execute cycle are shown in the table below.

Put each stage in the correct sequence by writing the numbers 1 to 7 in the right hand column.
The first one has been done for you.

Stage	Sequence number
the instruction is then copied from the memory location contained in the MAR (memory address register) and is placed in the MDR (memory data register)	
the instruction is finally decoded and is then executed	
the PC (program counter) contains the address of the next instruction to be fetched	1
the entire instruction is then copied from the MDR (memory data register) and placed in the CIR (current instruction register)	
the address contained in the PC (program counter) is copied to the MAR (memory address register) via the address bus	
the address part of the instruction, if any, is placed in the MAR (memory address register)	
the value in the PC (program counter) is then incremented so that it points to the next instruction to be fetched	

[6]

IGCSE Exam Answer 14

description of stage	sequence number
the instruction is then copied from the memory location contained in the MAR (memory address register) and is placed in the MDR (memory data register)	3
the instruction is finally decoded and is then executed	7
<i>the PC (program counter) contains the address of the next instruction to be fetched</i>	(1)
the entire instruction is then copied from the MDR (memory data register) and placed in the CIR (current instruction register)	4
the address contained in the PC (program counter) is copied to the MAR (memory address register) via the address bus	2
the address part of the instruction is placed in the MAR (memory address register)	6
the value in the PC (program counter) is then incremented so that it points to the next instruction to be fetched	5*

The incrementation of the program counter can appear at any stage after 2. All other stages must be in the correct given order.

[6]

IGCSE Exam Question 15

A section of computer memory is shown below:

Address	Contents
1000 0000	0110 1110
1000 0001	0101 0001
1000 0010	1000 1101
1000 0011	1000 1100
1000 1100	
1000 1101	
1000 1110	
1000 1111	

- (a) (i) The contents of memory location 1000 0001 are to be read.

Show the contents of the Memory Address Register (MAR) and the Memory Data Register (MDR) during this read operation:

MAR

--	--	--	--	--	--	--	--

MDR

--	--	--	--	--	--	--	--

[2]

- (ii) The value 0111 1001 is to be written into memory location 1000 1110.

Show the contents of the MAR and MDR during this write operation:

MAR

--	--	--	--	--	--	--	--

MDR

--	--	--	--	--	--	--	--

[2]

IGCSE Exam Question 15

- (iii) Show any changes to the computer memory following the read and write operations in part (a)(i) and part (a)(ii).

Address	Contents
1000 0000	0110 1110
1000 0001	0101 0001
1000 0010	1000 1101
1000 0011	1000 1100
	
1000 1100	
1000 1101	
1000 1110	
1000 1111	

[1]

IGCSE Exam Question 15

(b) Name **three** other registers used in computers.

1

2

3

[3]

(c) The control unit is part of a computer system.

What is the function of the control unit?

.....

.....

.....

.....

.....

.....

[3]

IGCSE Exam Answer 15

(a) (i)

MAR

1	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---

MDR

0	1	0	1	0	0	0	1
---	---	---	---	---	---	---	---

[2]

(ii)

MAR

1	0	0	0	1	1	1	0
---	---	---	---	---	---	---	---

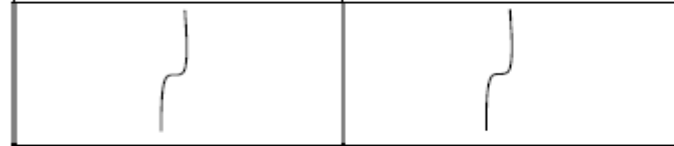
MDR

0	1	1	1	1	0	0	1
---	---	---	---	---	---	---	---

[2]

IGCSE Exam Answer 15

(iii)

Address	Contents
1000 0000	0110 1110
1000 0001	0101 0001
1000 0010	1000 1101
1000 0011	1000 1100
	
1000 1100	
1000 1101	
1000 1110	0111 1001
1000 1111	

[1]

- (b) – CIR (Current Instruction Register)
– PC (Program Counter)
– Acc (Accumulator)

[3]

- (c) – Controls operation of memory, processor and input/output
– Instructions are interpreted
– Sends signals to other components telling them “what to do”

[3]

IGCSE Exam Question 16

- 1 Name **three** different buses that are used in the fetch-execute cycle.

Bus 1

Bus 2

Bus 3

[3]

IGCSE Exam Question 17

1 a What is meant by the two terms:

(i) buffer

(ii) interrupt?

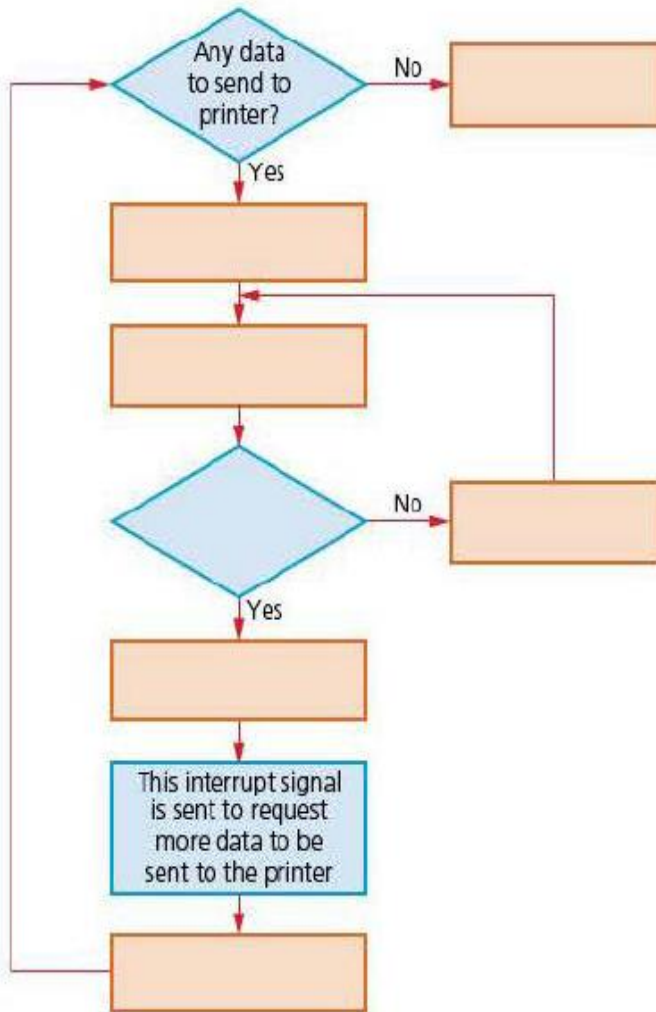
[4 marks]

IGCSE Exam Answer 7

- a (i) • temporary area of memory
• holds data waiting to be printed/sent to the printer
- (ii) • signal sent to processor from hardware or software
• requesting it to be serviced
• processor temporarily stops its current tasks

Exam Question 18

- b** Look at the flowchart which shows what happens when a document is sent to a printer and is printed out.



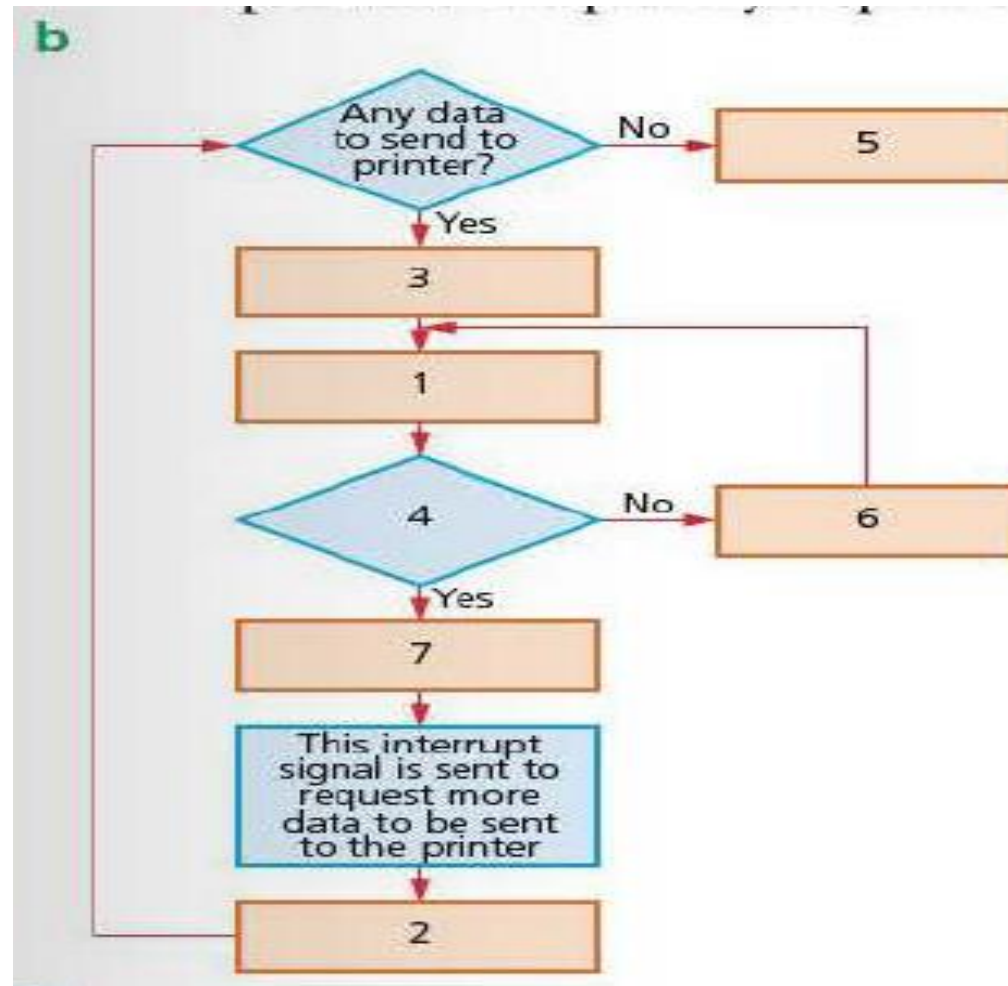
Seven statements are missing from the flowchart.

By writing the statement numbers ONLY, complete the flowchart, using the statements from the following list.

[7 marks]

Statement number	Statement
1	contents of the buffer are emptied to the printer and data from the document printed
2	current processor task is suspended whilst the interrupt is being serviced
3	data from the document to be printed is sent to the buffer from the computer memory
4	has all the data been sent to the printer?
5	processor continues with its tasks
6	processor is able to carry out other tasks while the buffer is being emptied
7	when all the data is printed, the buffer becomes empty; an interrupt signal is then sent to the processor

IGCSE Exam Answer 18



IGCSE Exam Question 19

- 2 a** A student made the following statements. Explain the statements.
- (i)** 'Windows is an example of a single-user multi-tasking operating system.'
 - (ii)** 'Many operating systems support error handling.' [4 marks]
- b** Name **three** other features of an operating system. [3 marks]

IGCSE Exam Answer 19

- a (i)
 - one user can use the computer at a time
 - but it is possible to have many applications open simultaneously
- (ii)
 - when an error occurs the user is warned
 - this prevents the computer from 'hanging up' every time an error is encountered
- b
 - HCI
 - multiprogramming
 - batch processing
 - load/run software
 - manage user accounts
 - file utilities (copy, save, etc.)
 - processor management
 - memory management
 - real-time processing
 - interrupt handling
 - security
 - I/O control

IGCSE Exam Question 20

3 a Name and explain the function of the three main buses used in a typical computer system.

[6 marks]

b Look at this memory map.

- (i)** The contents at address 1 0 0 0 0 0 1 1 are to be read. What values would be stored in the MAR and MDR?
- (ii)** If the value 1 1 1 0 0 1 1 1 is to be written at memory location 1 1 1 1 1 1 1 0, what values would be stored in the MAR and MDR?
- (iii)** Complete the memory map showing the value from part (ii) after the WRITE function has been done. [5 marks]

Address	Contents
1000 0000	1000 1000
1000 0001	0111 0110
1000 0010	1100 1111
1000 0011	0101 0110
1000 0100	1111 1110
1111 1100	
1111 1101	
1111 1110	
1111 1111	

IGCSE Exam Answer 20

3 a address bus

- carries signals relating to addresses between processor and memory

- it is uni-directional

data bus

- sends data between processor, memory unit and I/O devices

- it is bi-directional

control bus

- carries signals relating to control and coordination of all activities within the computer

- can be uni-directional or bi-directional

b (i) MAR: 1 0 0 0 0 0 1 1

MDR: 0 1 0 1 0 1 1 0

(ii) MAR: 1 1 1 1 1 1 1 0

MDR: 1 1 1 0 0 1 1 1

(iii)

Address	Contents
1111 1100	
1111 1101	
1111 1110	1110 0111
1111 1111	

IGCSE Exam Question 21

3 (a) Describe how special purpose registers are used in the fetch stage of the fetch-execute cycle.

.....

.....

.....

.....

.....

.....

.....

.....[4]

IGCSE Exam Question 21

- (b) Use the statements A, B, C and D to complete the description of how the fetch-execute cycle handles an interrupt.

A	the address of the Interrupt Service Routine (ISR) is loaded to the Program Counter (PC).
B	the processor checks if there is an interrupt.
C	when the ISR completes, the processor restores the register contents.
D	the register contents are saved.

At the end of the cycle for the current instruction

If the interrupt flag is set,, and

The interrupted program continues its execution.

[4]

IGCSE Exam Answer 21

3 (a) Four points from:

[4]

- The Program Counter (PC) holds the address of the next instruction to be fetched
- The address in the Program Counter (PC) is copied to the Memory Address Register (MAR)
- The Program Counter (PC) is incremented
- The instruction is copied to the Memory Data Register (MDR)
 - from the address held in the Memory Address Register (MAR)
- The instruction from the Memory Data Register (MDR) is copied to the Current Instruction Register (CIR)

(b) One mark for each statement or letter in the correct place.

[4]

At the end of the cycle for the current instruction **B**
If the interrupt flag is set, **D, A** and **C**
The interrupted program continues its execution

At the end of the cycle for the current instruction the processor checks if there is an interrupt.
If the interrupt flag is set, the register contents are saved, the address of the Interrupt Service Routine (ISR) is loaded to the Program Counter (PC) and when the ISR completes, the processor restores the register contents.

The interrupted program continues its execution.

IGCSE Exam Question 22

(c) The table shows six stages in the von Neumann fetch-execute cycle.

Put the stages into the correct sequence by writing the numbers 1 to 6 in the right hand column.

Description of stage	Sequence number
the instruction is copied from the Memory Data Register (MDR) and placed in the Current Instruction Register (CIR)	
the instruction is executed	
the instruction is decoded	
the address contained in the Program Counter (PC) is copied to the Memory Address Register (MAR)	
the value in the Program Counter (PC) is incremented so that it points to the next instruction to be fetched	
the instruction is copied from the memory location contained in the Memory Address Register (MAR) and is placed in the Memory Data Register (MDR)	

[6]

IGCSE Exam Answer 22

(c)

Description of stage	Sequence number
the instruction is copied from the Memory Data Register (MDR) and placed in the Current Instruction Register (CIR)	3
the instruction is executed	6
the instruction is decoded	5
the address contained in the Program Counter (PC) is copied to the Memory Address Register (MAR)	1
the value in the Program Counter (PC) is incremented so that it points to the next instruction to be fetched	4
the instruction is copied from the memory location contained in the Memory Address Register (MAR) and is placed in the Memory Data Register (MDR)	2

[6]

IGCSE Exam Question 23

2 A computer is designed using the Von Neumann model for a computer system.

The computer has a central processing unit (CPU).

(a) Data is fetched from primary storage into the CPU to be processed.

(i) State the name of the primary storage from where data is fetched.

..... [1]

(ii) The CPU performs a cycle to process data. Fetch is the first stage in this cycle.

State the names of the second and third stages in the cycle.

Second stage

Third stage [2]

(iii) Identify **two** components within the CPU that are used in the fetch stage of the cycle.

Component 1

Component 2 [2]

IGCSE Exam Answer 23

Question	Answer	Marks
2(a)(i)	• Random access memory // RAM	1
2(a)(ii)	One mark for each correct stage Second stage • Decode Third stage • Execute	2
2(a)(iii)	Any two from: • Memory address register // MAR • Memory data register //MDR • Program counter // PC • Control unit // CU • Address bus • Data bus • Control bus	2

IGCSE Exam Question 24

5 Harriet has a computer that is based on a Von Neumann model for a computer system.

The computer has a central processing unit (CPU).

(a) Explain the purpose of the CPU.

Identify the **three** stages of the cycle performed by the CPU in your answer.

[5]

IGCSE Exam Question 24

(b) The memory address register (MAR) is one component built into the CPU.

(i) Identify **four** other components that are built into the CPU.

Component 1

Component 2

Component 3

Component 4

[4]

(ii) Describe the purpose of the MAR in the CPU.

.....

.....

.....

..... [2]

IGCSE Exam Answer 24

Question	Answer	Marks
5(a)	Any two from: • It processes data • It processes/executes instructions • It carries out calculations • It carries out logical operations Any three from: • Fetch • Decode • Execute	5

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Question	Answer	Marks
5(b)(i)	Any four from: • Memory data register / MDR • Control unit / CU • Arithmetic logic unit / ALU • Program counter / PC • Current instruction register / Instruction register / CIR / IR • Accumulator / ACC • Address bus • Data bus • Control bus	4
5(b)(ii)	Any two from: • To temporarily store • ... the address of the next data/instructions required • ... the address of the location in memory where data is to be stored • Stores the address that is to be collected by the address bus	2

Learning Objectives

3.2 Input and output devices

Candidates should be able to:

- 1 Understand what is meant by an input device and why it is required
- 2 Understand what is meant by an output device and why it is required

Notes and guidance

- Including:
 - barcode scanner
 - digital camera
 - keyboard
 - microphone
 - optical mouse
 - QR code scanner
 - touch screen (resistive, capacitive and infra-red)
 - two-dimensional (2D) and three-dimensional (3D) scanners
- Including:
 - actuator
 - digital light processing (DLP) projector
 - inkjet printer
 - laser printer
 - light emitting diode (LED) screen
 - liquid crystal display (LCD) projector
 - liquid crystal display (LCD) screen
 - speaker
 - 3D printer

Learning Objectives

3.2 Input and output devices

3 (a) Understand what is meant by a sensor and the purposes of sensors

- Limited to:
 - acoustic
 - accelerometer
 - flow
 - gas
 - humidity
 - infra-red
 - level
 - light
 - magnetic field
 - moisture
 - pH
 - pressure
 - proximity
 - temperature

(b) Identify the type of data captured by each sensor and understand when each sensor would be used, including selecting the most suitable sensor for a given context

3.2 Input and Output Devices



Keyboard - Input Device

It allows you to type and input this information for the computer to understand.



Screen - Output Device

It shows you the information on the screen. The output displays the information.

Which are Inputs & which are Outputs?





3.2: Input Devices



3.2 Input and Output Devices

Watch the video for Input Devices

<https://www.youtube.com/watch?v=rkhT6s5g0pE>



3.2 Input and Output Devices

Watch the video for Input Devices

Specific input devices you need to know about for the exam

Barcode scanner		Optical mouse	
Digital camera		QR code scanner	
Keyboard		Touch screen (resistive, capacitive, infrared)	
Microphone		Two-dimensional (2D) and three-dimensional (3D) scanners	

3.2 Input and Output Devices

Watch the video for Input Devices

A note from the exam board

This section of the syllabus has been greatly simplified for exams taking place from 2023 onwards.

In the old syllabus, candidates needed to know details of how each input device operates – for example, how an input device scans a barcode and converts it into digital data.

Now, candidates only need to know:

- what each device does;
- why it does it; and
- when it might be used.

There are still many textbooks, revision books and online videos that go into depth about how each device physically functions – you will **not** be examined on this.



3.2 Input and Output Devices

3.2.1.1: Barcode scanner

Barcode scanner

What does it do?

It shines a red laser at a barcode to illuminate it. The white lines reflect the light back, whereas the black lines absorb some of the light and reflect less. The strength of the reflection from each line is interpreted by a **microprocessor** and converted into a binary value.

What is it typically used for?

- Scanning goods at a supermarket checkout.
- Checking library books in and out.
- Tracking packages out for delivery.

Why is it required?

- Faster checkout queues.
- Reduces errors caused by manual input.
- More detailed tracking information.
- Automatic stock control.



3.2 Input and Output Devices

3.2.1.2: Digital cameras

Digital camera

What does it do?

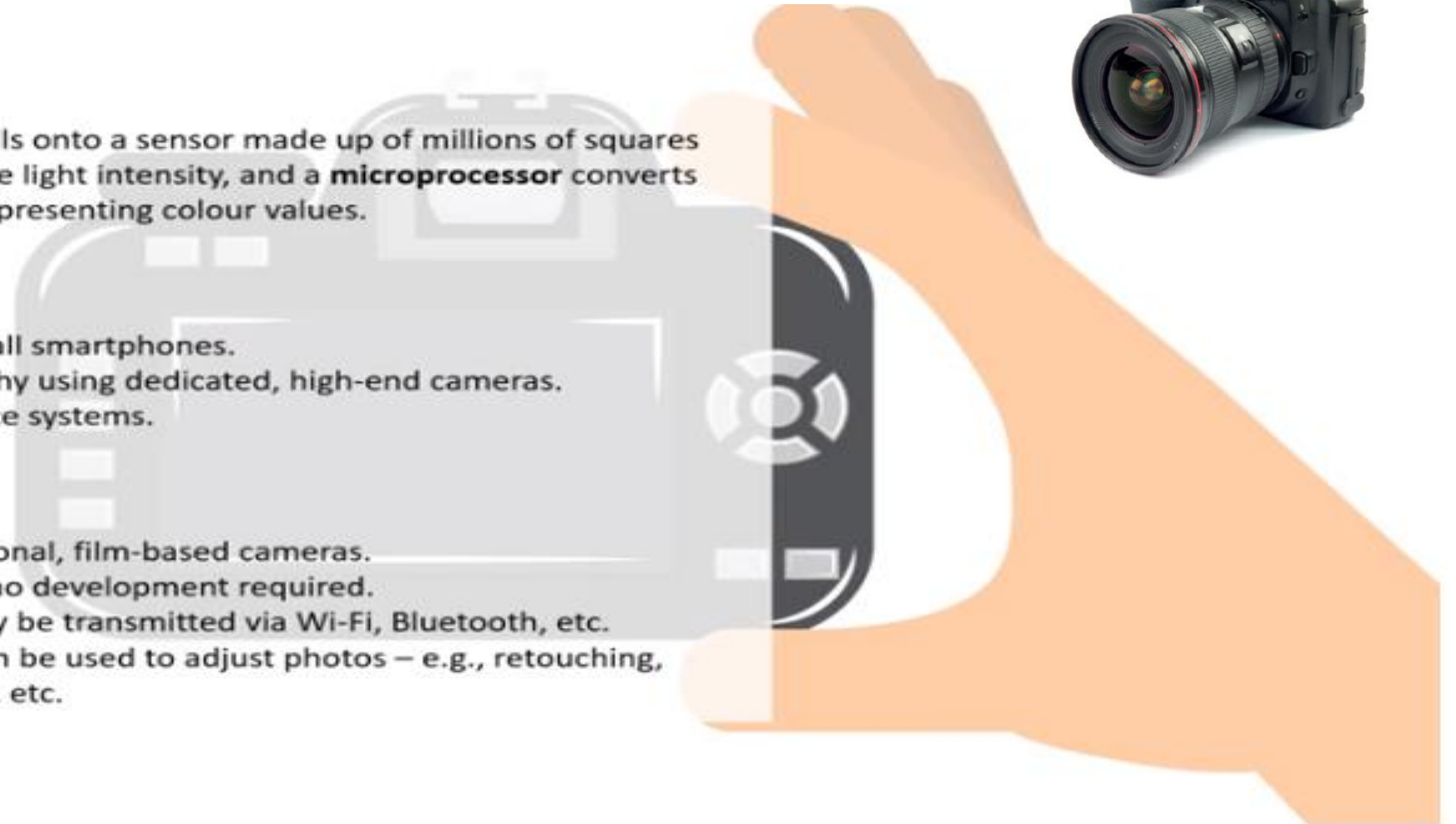
As light enters the camera, it falls onto a sensor made up of millions of squares (**pixels**). Each pixel measures the light intensity, and a **microprocessor** converts these values into digital data representing colour values.

What is it typically used for?

- Integrated into almost all smartphones.
- Professional photography using dedicated, high-end cameras.
- Security and surveillance systems.

Why is it required?

- Largely replaced traditional, film-based cameras.
- Instant photographs – no development required.
- Digital photos can easily be transmitted via Wi-Fi, Bluetooth, etc.
- Embedded software can be used to adjust photos – e.g., retouching, filters, red eye removal, etc.



3.2 Input and Output Devices

3.2.1.3: Keyboard

Keyboard

What does it do?

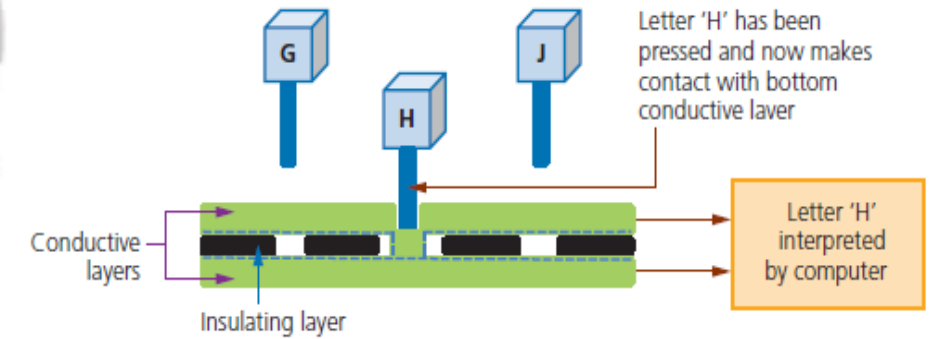
Traditional and mechanical **keyboards** operate slightly differently but, in essence, achieve the same results. Each character on a **keyboard** has a corresponding **character set** value. Key presses are converted into a digital signal, which the computer then interprets.

What is it typically used for?

- Connected to most desktop computers via USB connection or wireless/Bluetooth
- Built into laptop computers.
- Tablets and smartphones use virtual keyboards.

Why is it required?

- One of the most common forms of text-based data input.



3.2 Input and Output Devices

3.2.1.4: Optical Mouse

Optical mouse

What does it do?

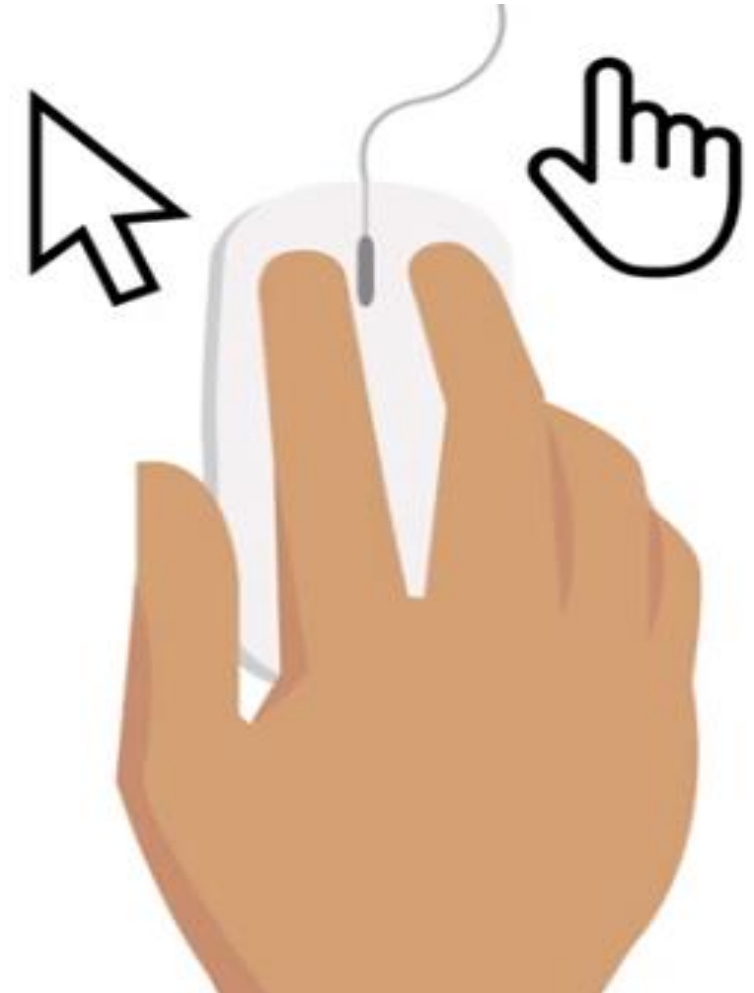
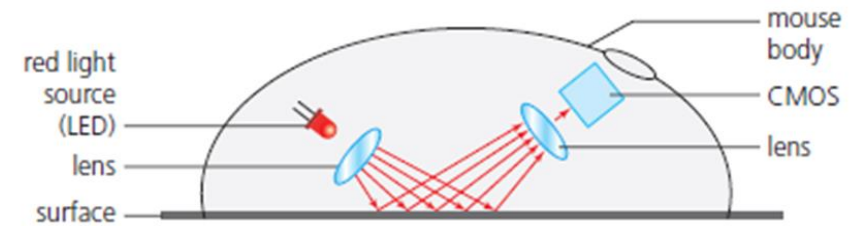
It uses a red LED and a sensor to determine the movement of the mouse relative to the surface beneath. A **microprocessor** analyses both the speed and distance of the movement and replicates it on-screen via a virtual cursor.

What is it typically used for?

- Navigation and interaction with almost all elements of a **graphical user interface (GUI)**.

Why is it required?

- Easy interaction with applications and other GUI elements.
- No moving parts, unlike mechanical mice – more reliable.
- No need for a special surface – i.e., a mouse mat.



3.2 Input and Output Devices

3.2.1.5: Microphone

Microphone

What does it do?

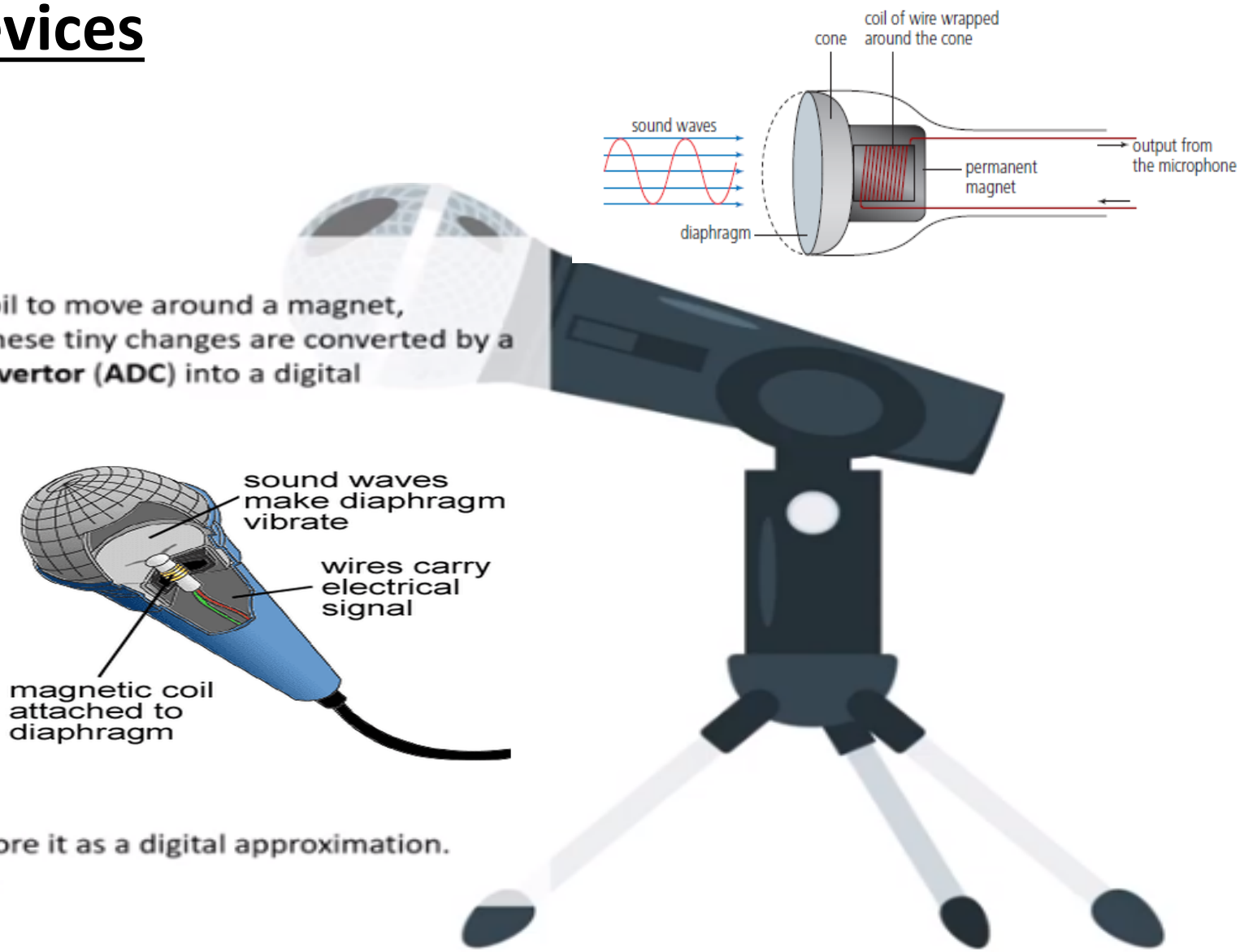
Vibrations caused by sound waves cause a coil to move around a magnet, resulting in electromagnetic field changes. These tiny changes are converted by a **microprocessor** and an **analog-to-digital convertor (ADC)** into a digital representation of the sound.

What is it typically used for?

- Music recordings.
- Dictation.
- Radio, TV and films.
- Telephone calls.
- Security.
- Online communication.

Why is it required?

- Capture any real-world sound and store it as a digital approximation.
- Digital sounds can then be modified.



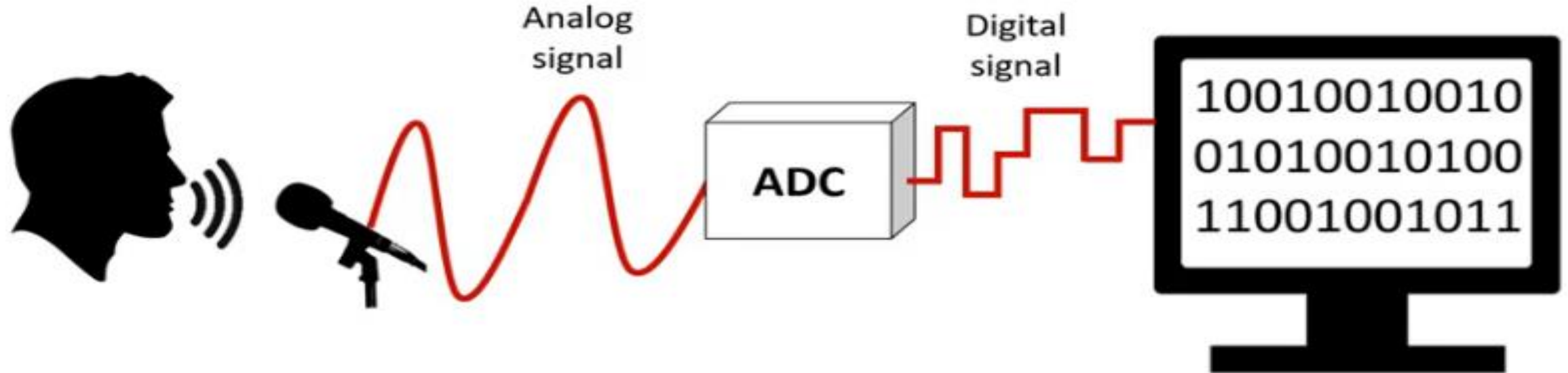
3.2 Input and Output Devices

3.2.1.5: Microphone

From analog to digital

To make sense of physical, real-world quantities, all sensors convert the information they collect into a digital format – this is typically achieved using some form of **analog-to-digital converter (ADC)**.

These devices convert physical, real-world information into discrete digital values.



3.2 Input and Output Devices

3.2.1.6: Quick response (QR) codes

QR code scanner

What does it do?

QR codes are read using a camera, typically on a mobile device. An app then processes the image, converting the squares into readable data.

What is it typically used for?

- Advertising products.
- Quick links to websites – no need to type in the URL.
- Sharing contact details instantly.
- Electronic travel passes, event tickets, etc.

Why is it required?

- Holds far more information than a traditional barcode:
 - Barcode: 30 digits
 - QR code: 4296 characters (7089 digits)



3.2 Input and Output Devices

3.2.1.7: Touch Screen (Resistive, Capacitive and Infra-red)

Touch screen: Resistive, capacitive and infrared

What does it do?

A **touch screen** is both an **input** and **output device**, although the exam board refers to it only as an **input device**.

Touch screens have become an incredibly popular way of receiving user input, replicating the same functions as a **mouse**.

There are three main types of **touchscreen** technology that you need to know about:

- Capacitive
- Infrared
- Resistive

As you no longer need to know how each of these technologies work in detail, we will just briefly go over the main differences.



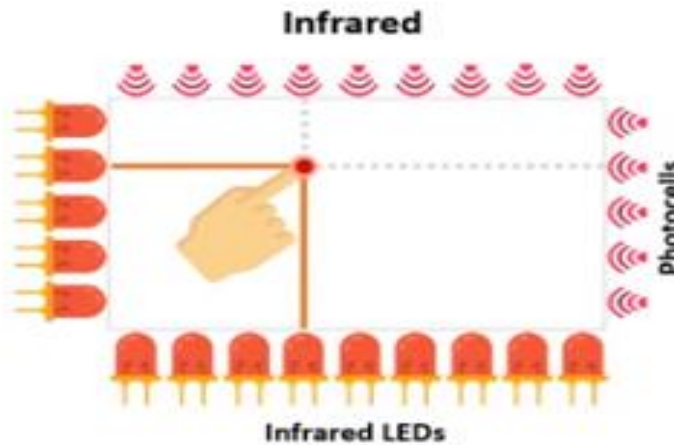
3.2 Input and Output Devices

3.2.1.7: Touch Screen (Resistive, Capacitive and Infra-red)

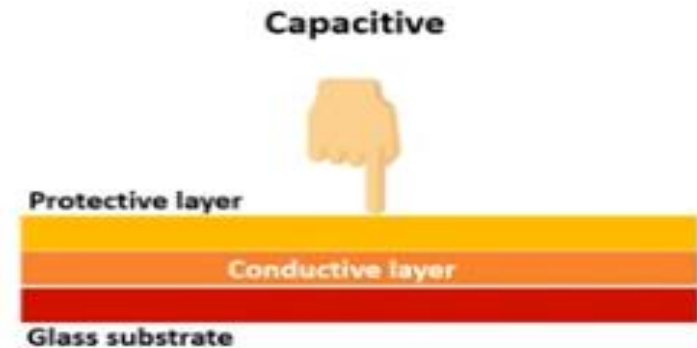
Touch screen: Resistive, capacitive and infrared



- Consists of two conductive layers.
- The top layer is made of a flexible polyester material.
- The two layers are separated by an insulating layer (air gap/invert gas).
- When the screen is pressed, the two layers contact, completing a circuit.



- LEDs shine infrared light across a screen, forming a matrix.
- When the screen is touched, the beams are interrupted.



- Composed of a protective layer, transparent conductive layer and a glass substrate.
- Touching the screen with a bare finger or special stylus peripheral changes the electrostatic field of the conductive layer.

3.2 Input and Output Devices

3.2.1.7: Touch Screen (Resistive, Capacitive and Infra-red)

Touch screen: Resistive, capacitive and infrared

What resistive touch screens are used for and why they are required

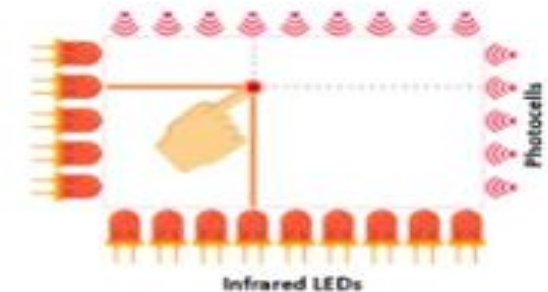
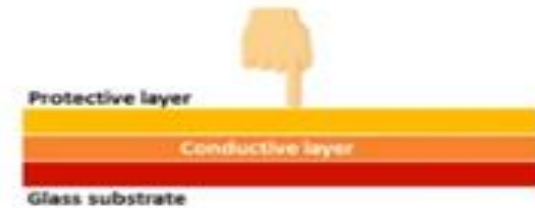
- + Cheap to produce. Can be activated with virtually any object – e.g., finger, stylus, gloved hand, etc. Very resistant to surface contaminants.
- Lower image clarity and precision compared to other types.
- i Cash machines, information kiosks and medical equipment.

What capacitive touch screens are used for and why they are required

- + Excellent image clarity. Very high precision. Durable screen that allows for multiple touches at the same time.
- Requires a bare finger or capacitive stylus for activation.
- i Tablets, laptops and smartphones.

What infrared touch screens are used for and why they are required

- + Excellent image quality. Unlimited *touch-life*. Scales very well.
- Sensitive to interference from ambient light, water, snow, etc.
- i Very large-scale commercial displays, information kiosks and medical equipment.



3.2 Input and Output Devices

3.2.1.8: two-dimensional (2D) and three-dimensional (3D) scanners



Two-dimensional (2D) and three-dimensional (3D) scanners

What does it do?

2D or flatbed scanners shine a bright line across a document to measure the levels of reflected light and create a digital version of it.

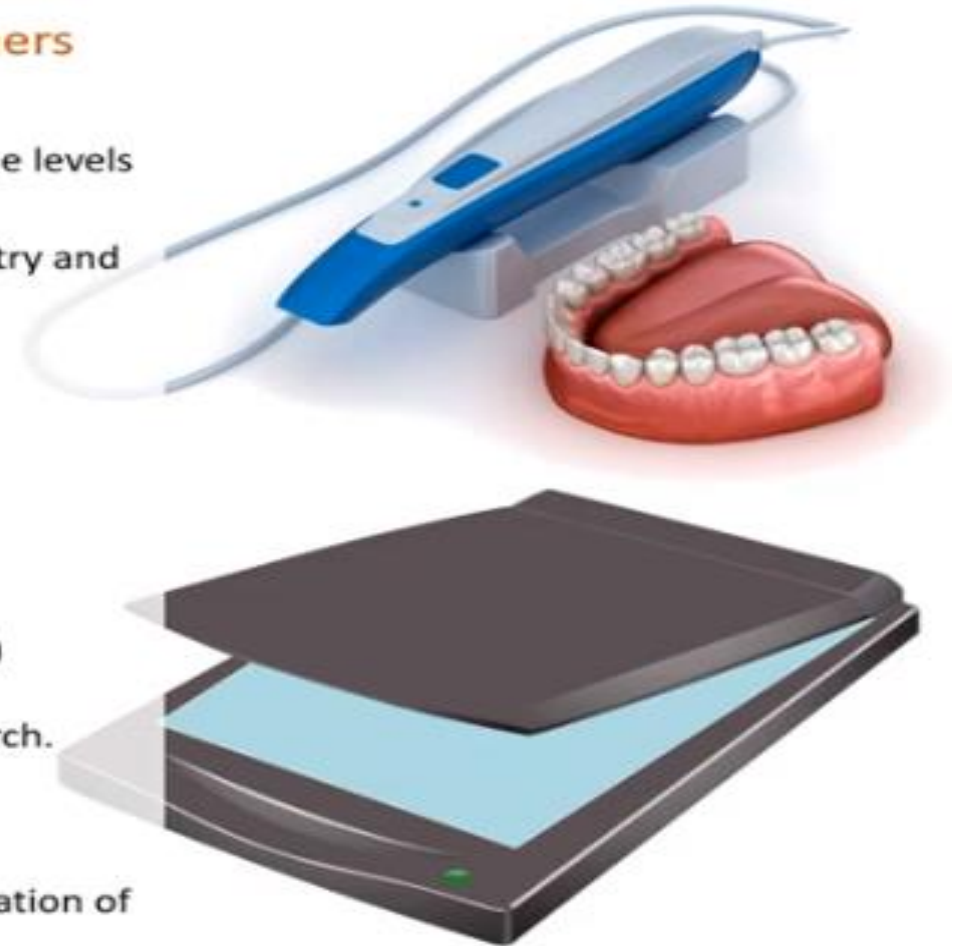
3D scanners shine a laser over the surface of an object to record its geometry and dimensions and create a digital model of it.

What is it typically used for?

- 2D scanners:
 - Creating digital versions of documents and photographs.
 - Reading passports at airports.
- 3D scanners:
 - Creating 3D models for use with computer-aided design (CAD) software or for creating working replicas with a 3D printer.
 - Dentistry, product development, quality inspection and research.

Why is it required?

- A simple, cheap and effective way of generating a digital representation of 2D or 3D objects for further manipulation.



3.2.1-Input Devices

3.2.1.8: two-dimensional (2D) and three-dimensional (3D) scanners

2D scanners:

Application of 2D scanners at an airport

- 2D scanners are used at airports to read passports. They make use of OCR technology
- to produce digital images which represent the passport pages. Because of the
- OCR technology, these digital images can be manipulated in a number of ways.

For example: the OCR software is able to review these images, select the text part, and then automatically put the text into the correct fields of an existing database. It is possible for the text to be stored in an ASCII format – it all depends on how the data is to be used.

- At many airports the two-dimensional photograph in the passport is scanned and stored as a JPEG image. The passenger's face is also photographed using a digital camera (a 2D image is taken so it can be matched to the image taken from the passport). The two digital images are compared using face recognition/detection software. Key parts of the face are compared. Lets see the video below How does Facial Recognition works?

3.2.1-Input Devices

3.2.1.8: two-dimensional (2D) and three-dimensional (3D) scanners

3D Scanners:

Existing 3D scanning is an exciting new technology that has had a positive impact in both medical and industrial fields.

Medical - Scanning the human body/ parts allows doctors to take a closer look at possible problems virtually which can aid in giving a correct diagnosis

e.g. taking an MRI scan of your knee to look for a small tendon tear.

Industrial - Existing tools or parts can be scanned into digital format. Once scanned the object can be altered and tested using specialist 3D software.



3.2 Input and Output Devices

3.2.1.8: two-dimensional (2D) and three-dimensional (3D) scanners



Input devices

Input device: Any device that allows you to pass information from the outside world into a computer system.

- **Barcode scanner:** Uses a red laser to read a series of black and white stripes.
 - Scanning goods, checking out books, tracking deliveries.
 - Reduces errors caused by manual input.
- **Digital camera:** Captures light via sensors and converts it into a digital image.
 - Integrated into smartphones and security systems.
 - Replaces slower traditional cameras and allows the image to be manipulated.
- **Keyboard:** Each key is mapped against a character. Key presses are detected, and a signal is sent to the computer.
 - Connected via wired USB, wireless or Bluetooth. Built into laptops. Tablets and smartphones use virtual keyboards.
 - One of the most common forms of text-based data input.
- **Optical mouse:** Uses a red LED and sensors to determine movement relative to the surface beneath.
 - Easy navigation around a graphical user interface (GUI).
 - No moving parts – more robust than a mechanical mouse.
- **Microphone:** Vibrations caused by sound waves are captured and converted into a digital representation of sound.
 - Music recordings, dictation, radio, TV, film, online communication, etc.

3.2 Input and Output Devices

3.2.1.8: two-dimensional (2D) and three-dimensional (3D) scanners



Input devices

- **QR code scanner:** Read using a camera, processed by an app and converted into readable data.
 - Advertising products, quick access to websites, storing e-tickets, etc.
 - Can hold far more information than traditional barcodes.
- **Touch screens:** A popular way of receiving user input, replicates the functions of a mouse.
 - **Resistive:** Consists of two conductive layers separated by an insulating layer.
 - Cheap to produce, can be activated using virtually any object.
 - Cash machines, kiosks, medical equipment.
 - **Capacitive:** Consists of a protective layer, a transparent conductive layer and a glass substrate.
 - Excellent image quality, high precision, supports multiple touches.
 - Tablets, laptops, smartphones.
 - **Infra-red:** LEDs shine infrared light to form a matrix over the screen.
 - Excellent image quality, unlimited *touch-life*, scales very well.
 - Very large commercial displays, kiosks, medical equipment.
- **Scanners**
 - **2D:** Shines a bright line across a document and measures light reflections to create a digital version.
 - Scan documents and photos.
 - **3D:** Shines a laser over the surface of an object and captures its geometry and dimensions to create a 3D model.
 - Dentistry, product development, quality inspection, research.

3.2 Input and Output Devices

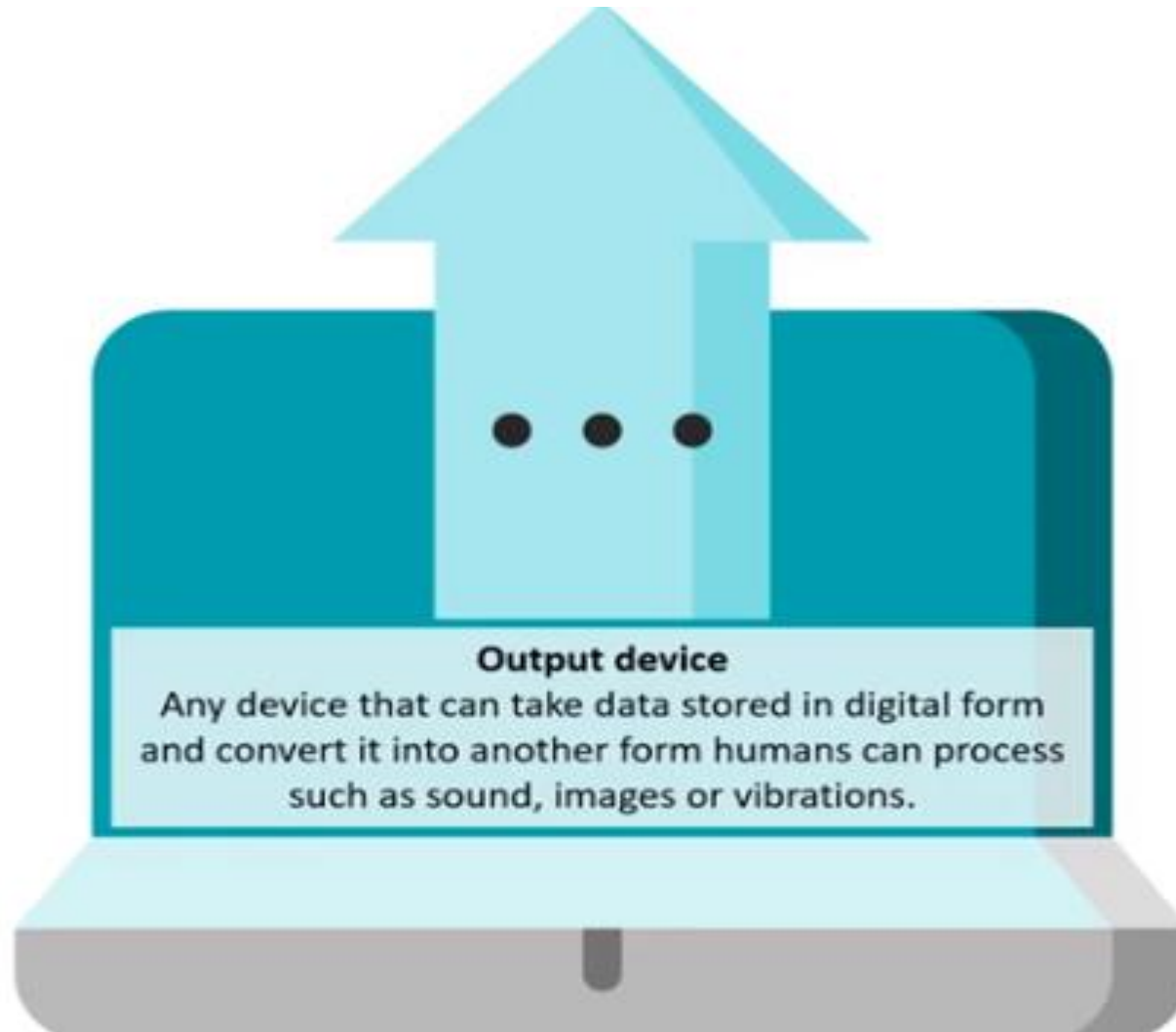
Watch the video for Onput Devices

<https://www.youtube.com/watch?v=l10vxFNbp1c>



3.2 Input and Output Devices

Output devices



3.2 Input and Output Devices

Output Devices

Examples of output devices



3.2 Input and Output Devices

Output Devices

Specific output devices you need to know about for the exam

Actuator		Light-emitting diode (LED) screen	
Digital light processing (DLP) projector		Liquid crystal display (LCD) projector/screen	
Inkjet printer		Speaker	
Laser printer		3D printer	

3.2 Input and Output Devices

Output Devices

A note from the exam board

This section of the syllabus has been greatly simplified for exams taking place from 2023 onwards.

In the old syllabus, candidates needed to know details of how each output device operates – for example, how a laser printer actually replicates a digital image on paper.

Now, candidates only need to know:

- what each device does;
- why it does it; and
- when it might be used.

There are still many textbooks, revision books and online videos that go into depth about how each device physically functions – you will **not** be examined on this.



3.2 Input and Output Devices

3.2.2.1: Actuator

Actuator

What does it do?

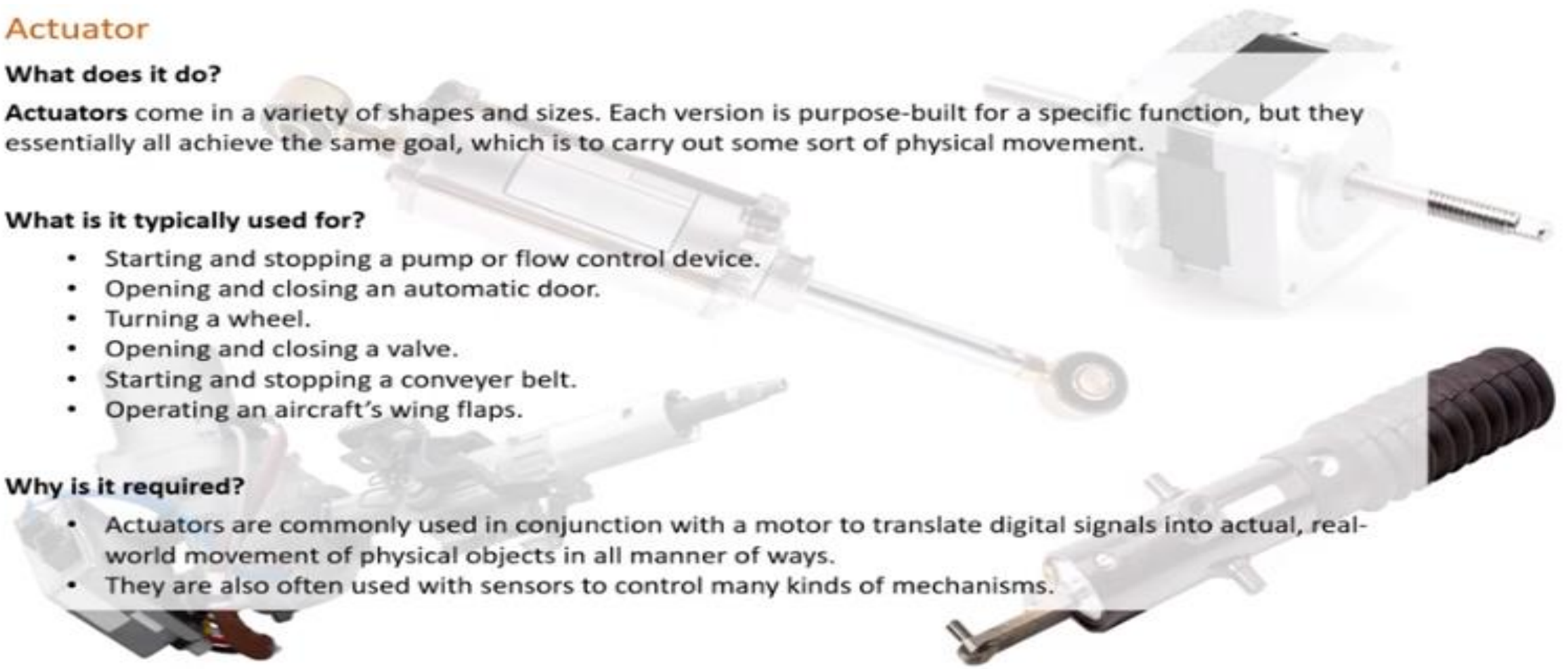
Actuators come in a variety of shapes and sizes. Each version is purpose-built for a specific function, but they essentially all achieve the same goal, which is to carry out some sort of physical movement.

What is it typically used for?

- Starting and stopping a pump or flow control device.
- Opening and closing an automatic door.
- Turning a wheel.
- Opening and closing a valve.
- Starting and stopping a conveyor belt.
- Operating an aircraft's wing flaps.

Why is it required?

- Actuators are commonly used in conjunction with a motor to translate digital signals into actual, real-world movement of physical objects in all manner of ways.
- They are also often used with sensors to control many kinds of mechanisms.



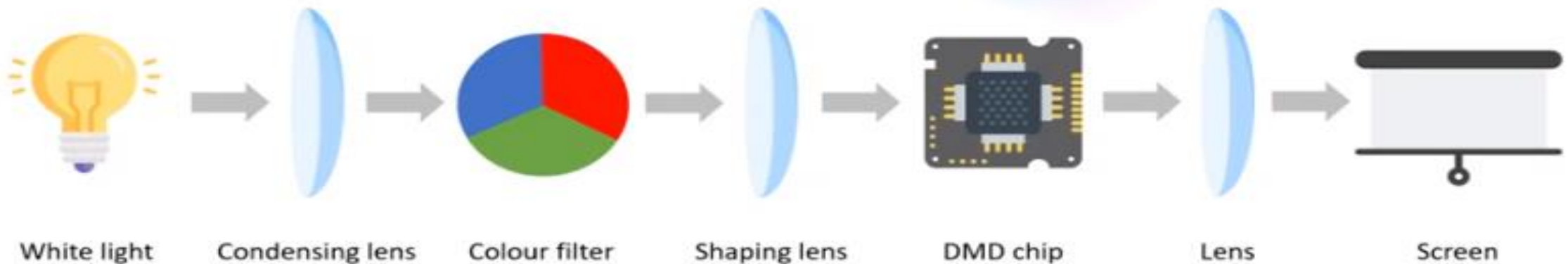
3.2 Input and Output Devices

3.2.2.2: Digital Light Processing (DLP) projector

Digital light processing (DLP) projector

What does it do?

DLP systems use millions of micro-mirrors arranged in a matrix grid and placed on a small microchip within the projector itself. White light is then shone through a rotating colour filter to produce the image, while an angled mirror changes the colour intensity.



3.2 Input and Output Devices

3.2.2.2: Digital Light Processing (DLP) projector

Digital light processing (DLP) projector

What does it do?

DLP systems use millions of micro-mirrors arranged in a matrix grid and placed on a small microchip within the projector itself. White light is then shone through a rotating colour filter to produce the image, while an angled mirror changes the colour intensity.

What is it typically used for and why is it required?

- Projecting a computer output onto a wall, whiteboard, projection screen, etc.
- Personal/home and professional settings.

Advantages over LCD projectors:

- Higher contrast ratios.
- Smoother video.
- Smaller, lighter and more portable.
- Better suited to dusty or smoky atmospheres.



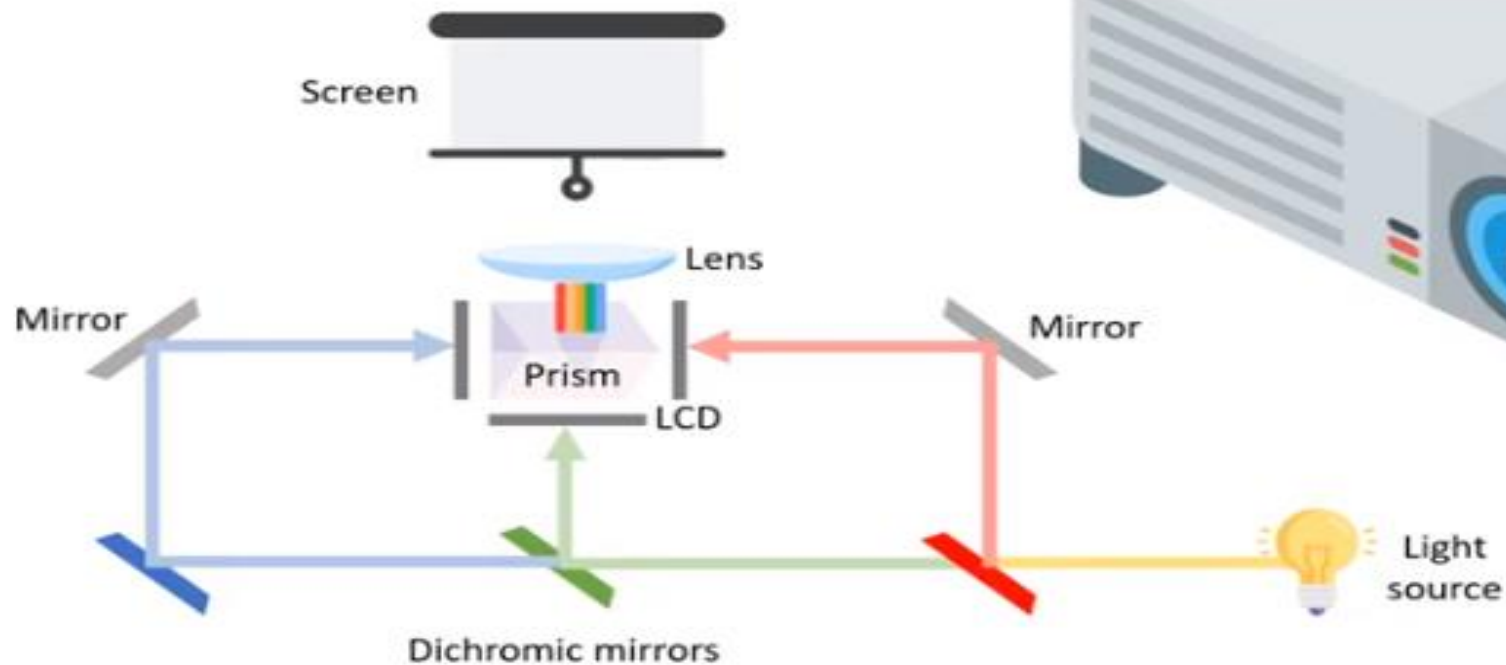
3.2 Input and Output Devices

3.2.2.3: liquid crystal display (LCD) projector

Liquid crystal display (LCD) projector

What does it do?

LCD projectors use three mirror filters to separate an image into its red, green and blue wavelengths. These images are then passed through a prism and recombined. Finally, the full colour image is passed through a lens and projected.



3.2 Input and Output Devices

3.2.2.3: liquid crystal display (LCD) projector

Liquid crystal display (LCD) projector

What does it do?

LCD projectors use three mirror filters to separate an image into its red, green and blue wavelengths. These images are then passed through a prism and recombined. Finally, the full colour image is passed through a lens and projected.

What is it typically used for and why is it required?

- Projecting a computer output onto a wall, whiteboard, projection screen, etc.
- Personal/home and professional settings.

Advantages over DLP projectors:

- Sharper image.
- Better colour reproduction.
- Typically less expensive.
- Quieter operation.
- Lower power usage and heat production.



3.2 Input and Output Devices

3.2.2.4: inkjet printer

Inkjet printer

What does it do?

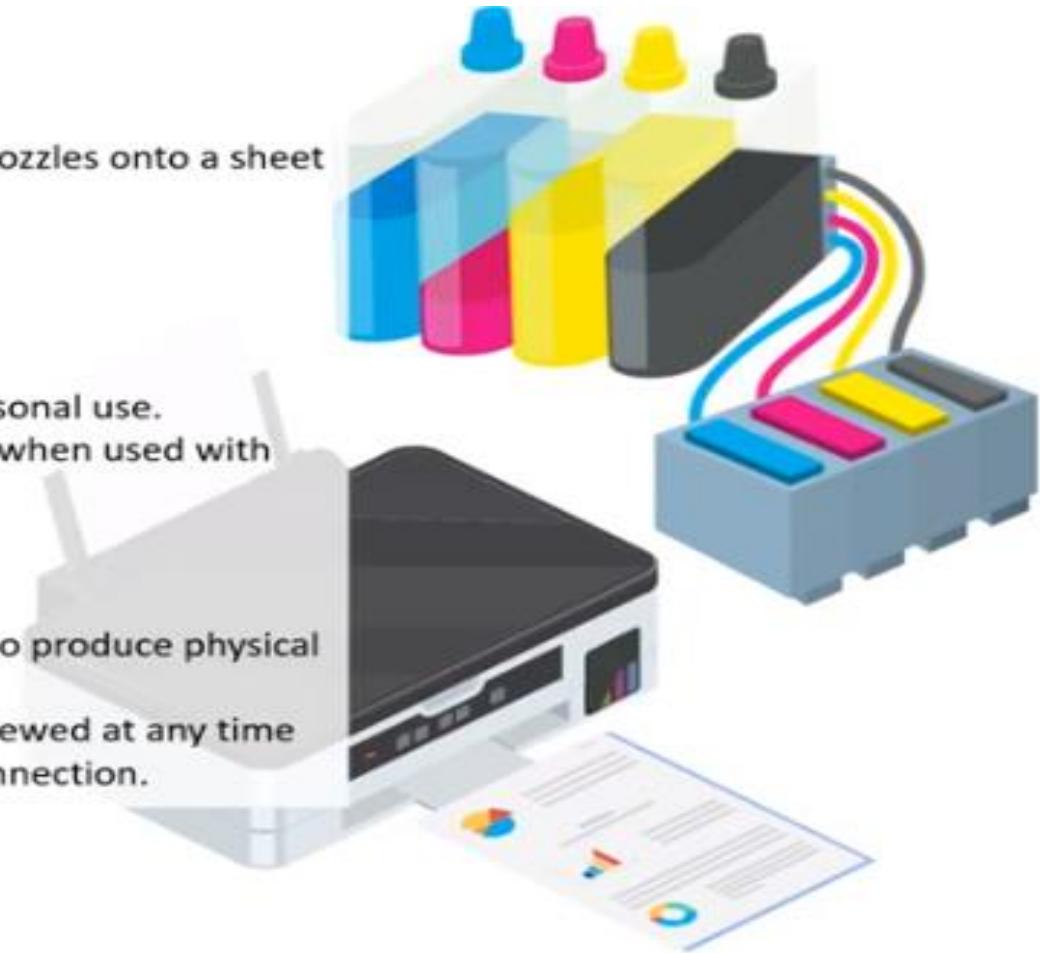
Inject printers force tiny droplets of liquid ink through very fine nozzles onto a sheet of paper to replicate a digital document or image.

What is it typically used for?

- Mostly used for small-scale print jobs.
- Usually cheaper and smaller, so they are popular for personal use.
- Excellent for printing out high-quality photos, especially when used with specialist photo paper.

Why is it required?

- Even now, in this age of technology, there is still a need to produce physical copies of information stored in an electronic format.
- Printouts or hard copies can be easily transported and viewed at any time without the need for an electronic device or internet connection.



3.2 Input and Output Devices

3.2.2.5: laser printer

Laser printer

What does it do?

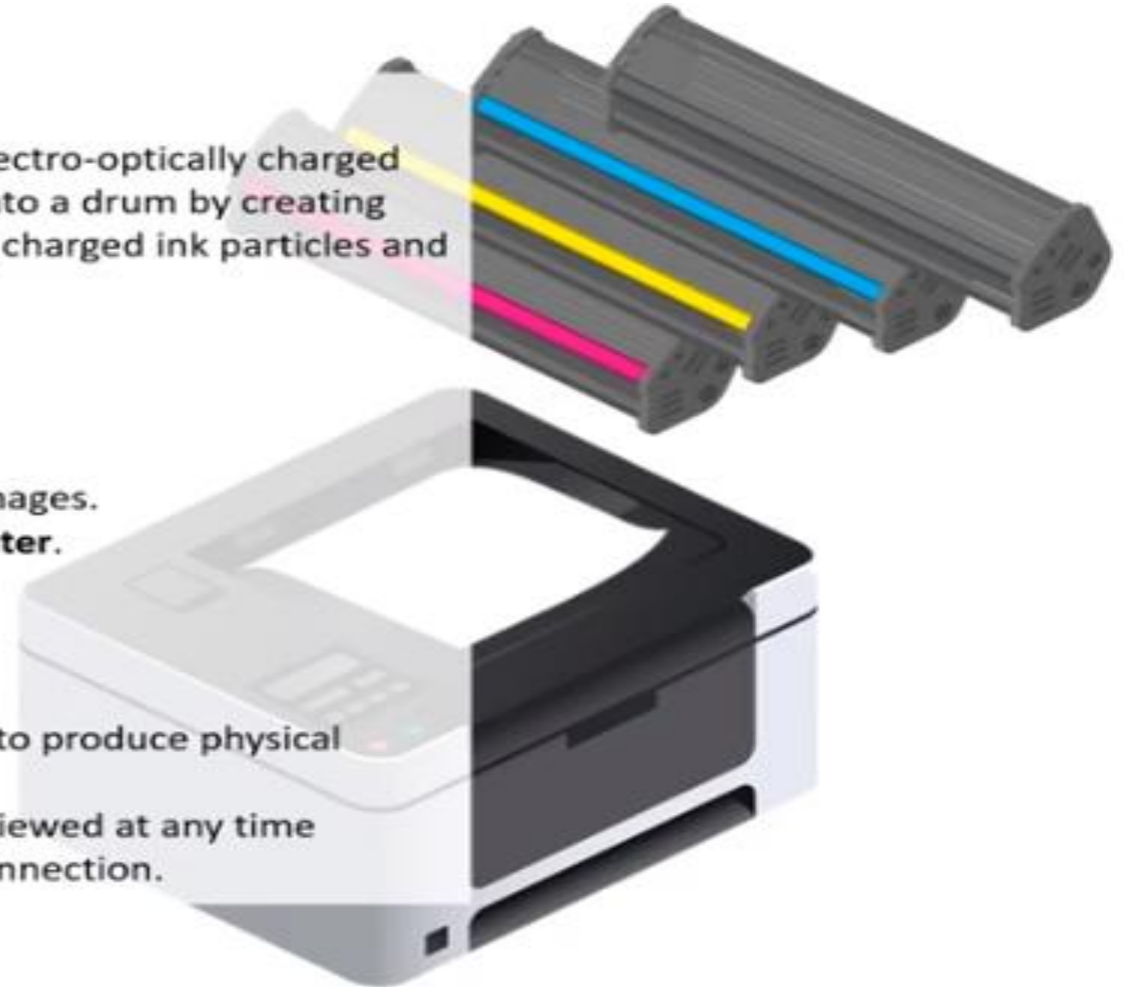
Laser printers use toner cartridges, which contain powdered, electro-optically charged ink. Using a laser, a mirror image of the printer page is drawn onto a drum by creating negatively charged areas. The drum then picks up the positively charged ink particles and bonds them to the paper using heat.

What is it typically used for?

- High-quality, inexpensive printing.
- Very fast and accurate for non-colour documents and images.
- Lower running cost per page compared to an **inkjet printer**.
- Often found in schools and offices.

Why is it required?

- Even now, in this age of technology, there is still a need to produce physical copies of information stored in an electronic format.
- Printouts or hard copies can be easily transported and viewed at any time without the need for an electronic device or internet connection.



3.2 Input and Output Devices

3.2.2.6: liquid crystal display (LCD) screen

Liquid crystal display (LCD) screen

What does it do?

Liquid crystal displays (LCDs) are made up of millions of tiny liquid crystals. These crystals form a matrix of pixels, which can be affected by changes in applied electric fields. **LCD** screens require a form of backlighting, often provided by a fluorescent lamp or **LEDs**.

What is it typically used for?

- TVs, monitors, laptops, mobile devices, etc.
- Gradually being phased out in favour of newer technologies like LED and OLED.

Why is it required?

- We need to be able to output digital information in a visual way.
- Although display technology is constantly evolving, there will probably always be a need for screens in some form.



3.2 Input and Output Devices

3.2.2.7: light emitting diode (LED) screen

Light-emitting diode (LED) screen

What does it do?

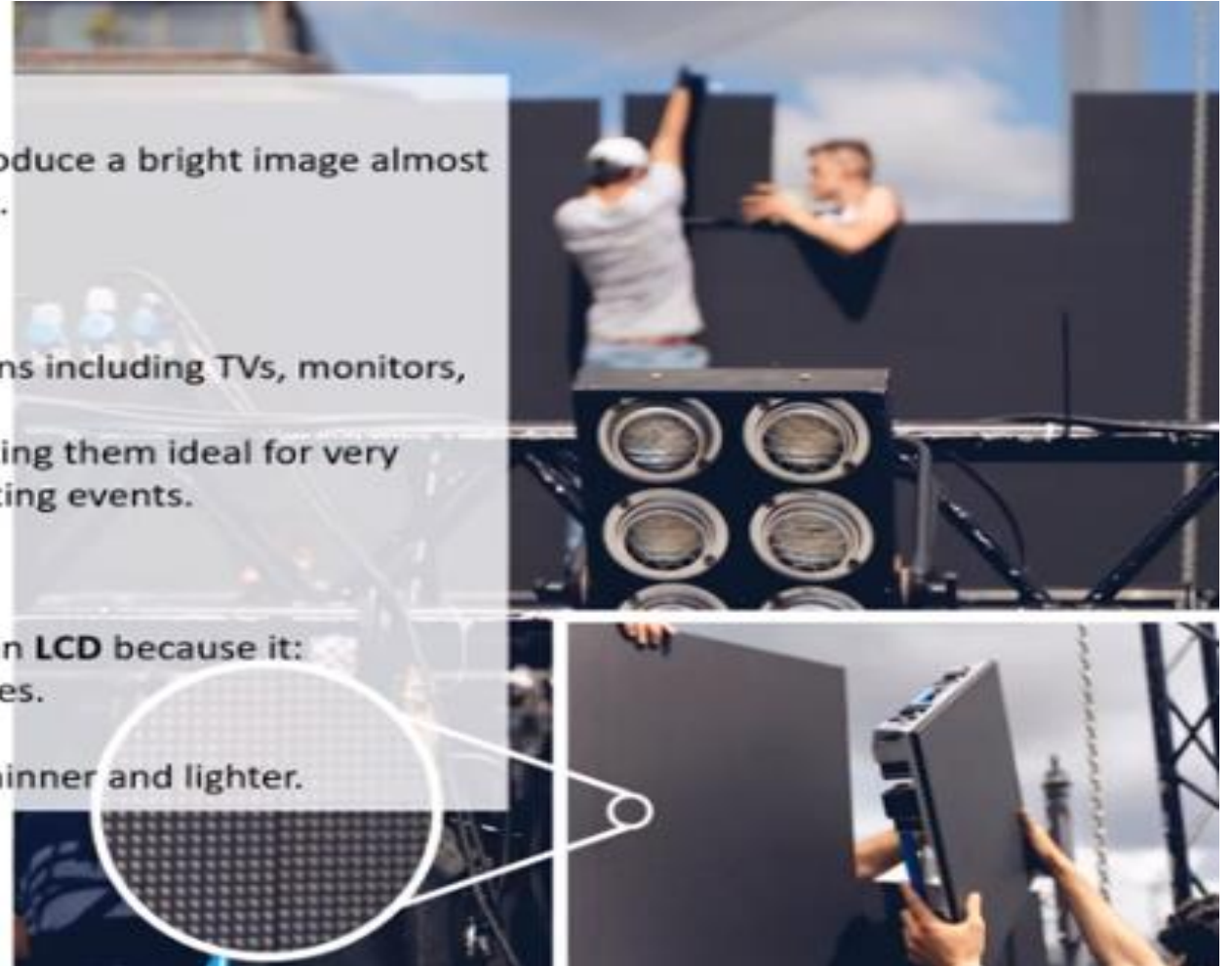
These displays use tiny **light-emitting diodes (LEDs)** to produce a bright image almost instantly without the need for any additional backlighting.

What is it typically used for?

- Replacing **LCD** displays in more and more situations including TVs, monitors, laptops and mobile devices.
- Multiple **LED** screens can be joined together, making them ideal for very large displays used in concerts, festivals and sporting events.

Why is it required?

- **LED** technology is generally considered better than **LCD** because it:
 - Offers better image quality and viewing angles.
 - Has a longer lifespan.
 - Requires no additional backlighting, so it's thinner and lighter.



3.2 Input and Output Devices

3.2.2.8: speaker

Speaker

What does it do?

Essentially, a **speaker** is the counterpart to a **microphone** (an **input device**) and works in the opposite way.

A digital representation of sound is passed through a **digital-to-analog convertor (DAC)**, which can then be further boosted by an amplifier before being passed to a **speaker** to create analog sound waves.

What is it typically used for?

- TV, film and video games
- Live music
- Music recording playback
- Alarms

Why is it required?

- Speakers provide us with a way to take digital sounds or recordings and output them in a form that can be heard by the human ear.



3.2 Input and Output Devices

3.2.2.9: 3D printer

3D printer

What does it do?

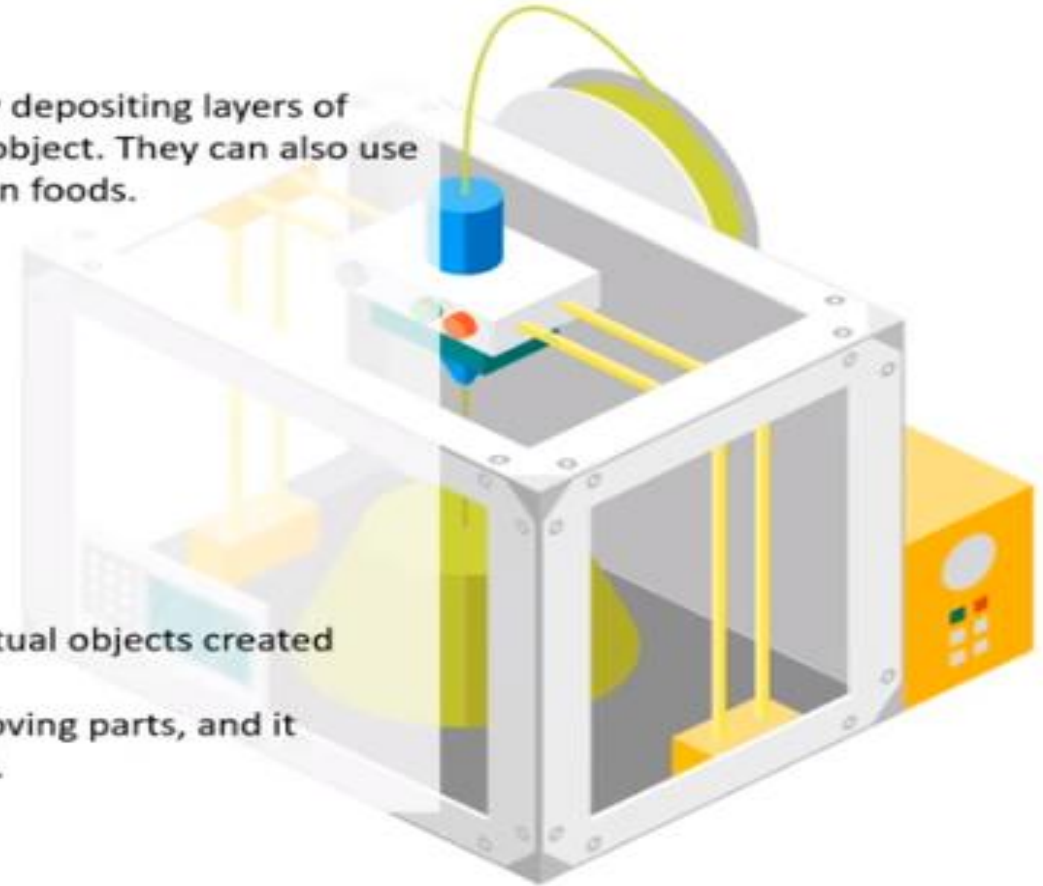
Based on **inkjet** and **laser printer** technology, **3D printers** work by depositing layers of material (often plastic) on top of each other to slowly build a 3D object. They can also use powered resin or metal, paper, ceramic, concrete – or even certain foods.

What is it typically used for?

- Inexpensive prosthetics
- Spare parts
- Rapid prototyping
- Personalised items

Why is it required?

- 3D printing allows us to produce real-world models of virtual objects created using **computer-aided design (CAD)** software.
- It can create objects with intricate internal spaces and moving parts, and it provides a means of manufacturing with minimum waste.



3.2 Input and Output Devices

Output Devices



Output devices

Output device: Any device that can take data stored in digital form and convert it into another form humans can process such as sound, images or vibrations.

- **Actuator:** Allow physical movements to be carried out, often used in conjunction with a motor and sensor.
 - Opening and closing valves and doors.
 - Starting and stopping motorised objects like conveyor belts.
- **Digital light processing (DLP) projector:** Millions of micro-mirrors arranged in a matrix grid, placed on a small microchip within the projector itself.
 - Higher contrast and smoother video than LCD projectors.
 - Small, light and portable.
- **Liquid crystal display (LCD) projector:** Uses three separate mirror filters and a prism to create an image.
 - Provide a sharper image with better colour reproduction than LED projectors.
 - Quieter, use less power and produce less heat.
- **Inkjet printer:** Uses liquid ink forced through fine nozzles.
 - Produces high-quality photo prints.
 - Tends to be cheaper, popular for home setups.

3.2 Input and Output Devices

Output Devices



Output devices

- **Laser printer:** Uses dry powder fused onto paper using heat.
 - Very fast and accurate for bulk printing non-colour documents.
 - Lower running cost per page than inkjet printers.
 - Often used in schools and offices.
- **Liquid crystal display (LCD) screen:** Made up of millions of liquid crystals, requires backlighting.
 - TVs, monitors, laptops.
 - Generally considered an older form of display technology.
- **Light-emitting diode (LED) screen:** Uses tiny LEDs to produce bright images without the need for additional backlighting.
 - Superior picture quality compared to LCDs.
 - Slimmer and lighter.
- **Speaker:** Converts digital representations of sound into analog waves that can be heard by the human ear.
 - TV, film, video games.
 - Live music, recording playback.
 - Alarms and sirens.
- **3D printer:** Deposits layers of material to build a 3D object.
 - Usable materials include plastic, powdered metal and resin, paper, ceramic, concrete and some foods.
 - Used to produce prosthetics, spare parts, models and personalised items.

3.2 Input and Output Devices

Example 1 output and input devices together

Blurring the lines



Not exactly. Modern controllers also have built-in speakers and haptic feedback technology – these make it an output device as well.

As far as the exams are concerned, you can consider a game controller an input device primarily.

However, be aware that the lines can sometimes be blurry, so always be sure to justify your answer.

3.2 Input and Output Devices

Example 2 output and input devices together

Appropriate devices for specific situations



What input and output devices might be used as part of this cashpoint system?

Input devices

- Card reader
- Keypad
- Buttons
- Camera

Output devices

- Monitor
- Printer
- Speakers
- Actuator/motor

3.2 Input and Output Devices

3.2.3: Sensors

https://www.youtube.com/watch?v=ZmdXZMO_lx0



3.2.3: Sensors

Sensors are **input devices**. They come in different shapes and sizes, but they all measure physical properties of the environment and translate these measurements into digital information that a computer can use.



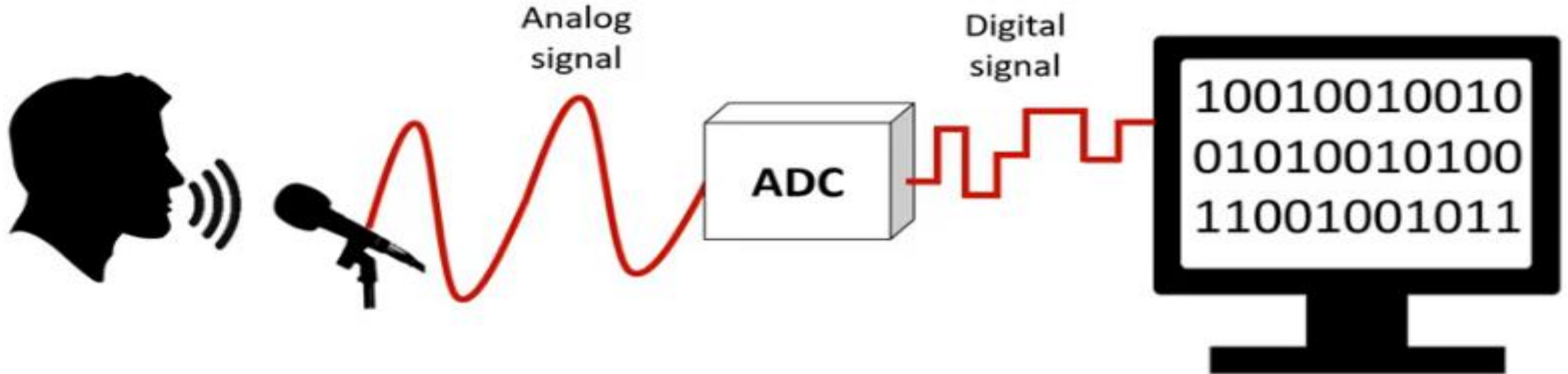
3.2 Input and Output Devices

3.2.3: Sensors

From analog to digital

To make sense of physical, real-world quantities, all sensors convert the information they collect into a digital format – this is typically achieved using some form of **analog-to-digital converter (ADC)**.

These devices convert physical, real-world information into discrete digital values.

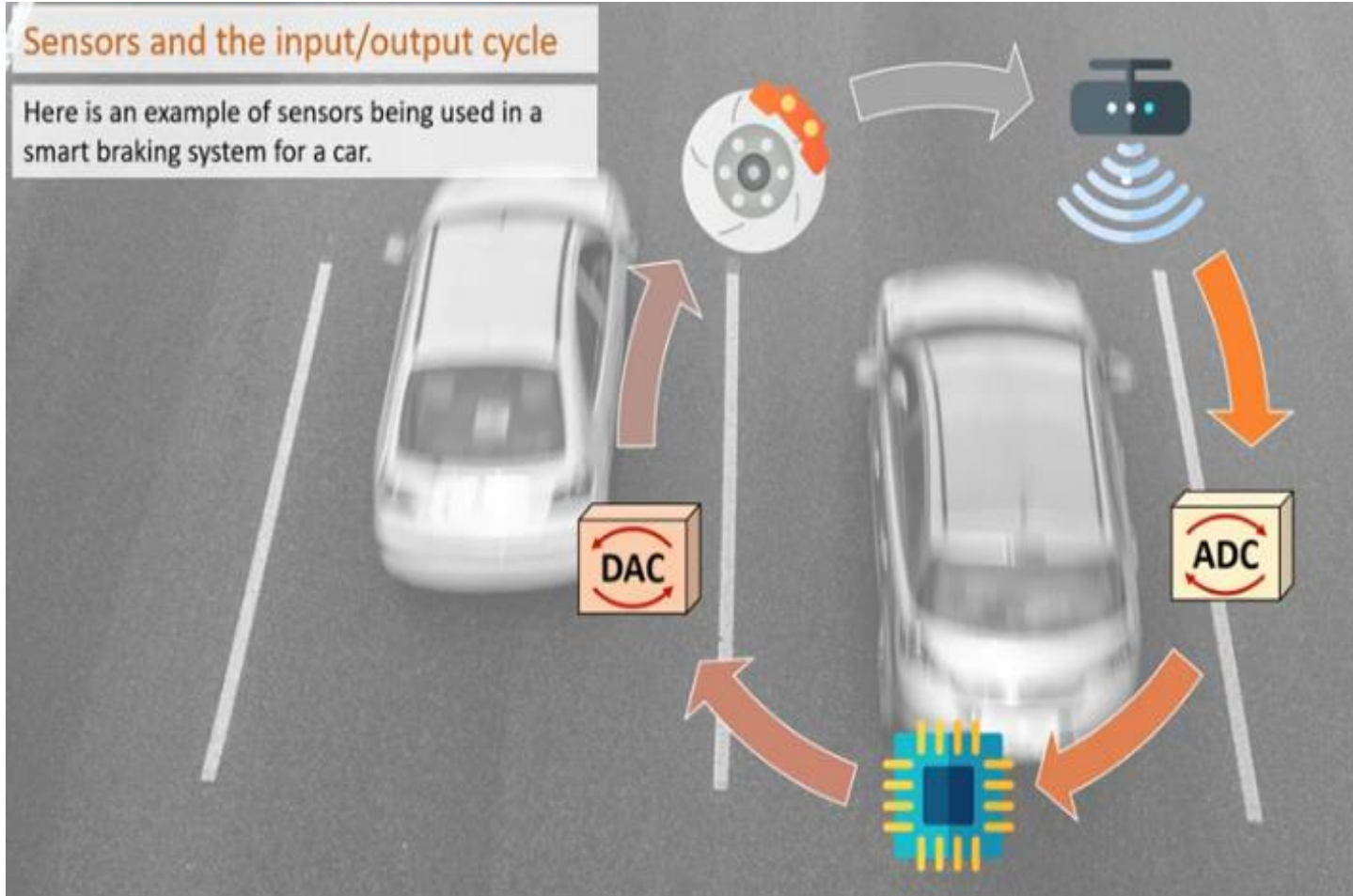


3.2 Input and Output Devices

3.2.3: Sensors

Sensors and the input/output cycle

Here is an example of sensors being used in a smart braking system for a car.

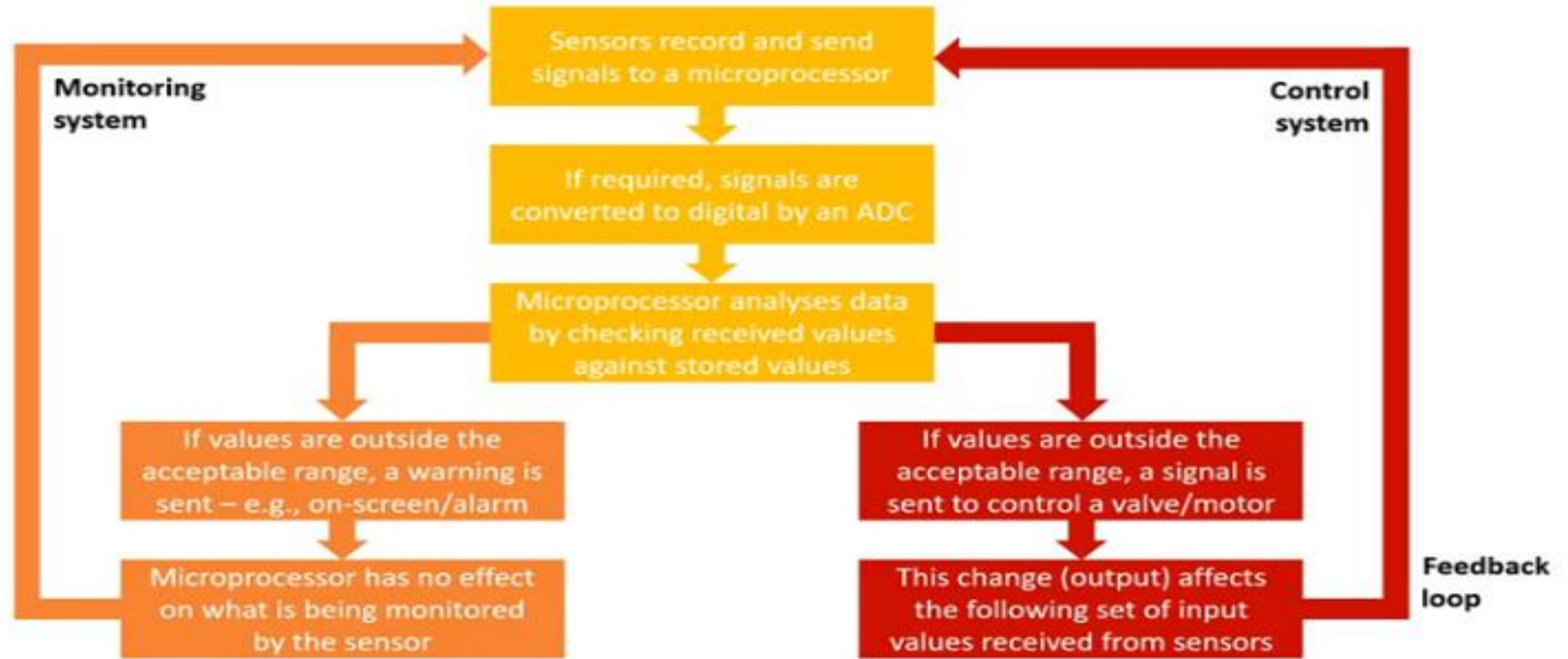


1. A sensor takes constant readings of the car's distance to objects ahead of it.
2. These analog, real-world values are converted to a format a microprocessor can understand using an analog-to-digital convertor (ADC).
3. A microprocessor takes this input and calculates the speed the car is travelling and safe braking distances. If it detects the car is travelling too fast and a collision is likely, it issues commands to brake.
4. Digital brake signals are converted into an analog form – these can then be sent to a motor (actuator), which will apply the brake via a digital-to-analog convertor (DAC).
5. The brake slows the car down.
6. The sensor continues to send out values. These will have changed due to the slowing of the car – this is known as feedback.

3.2 Input and Output Devices

3.2.3: Sensors

Using sensors for monitoring and control

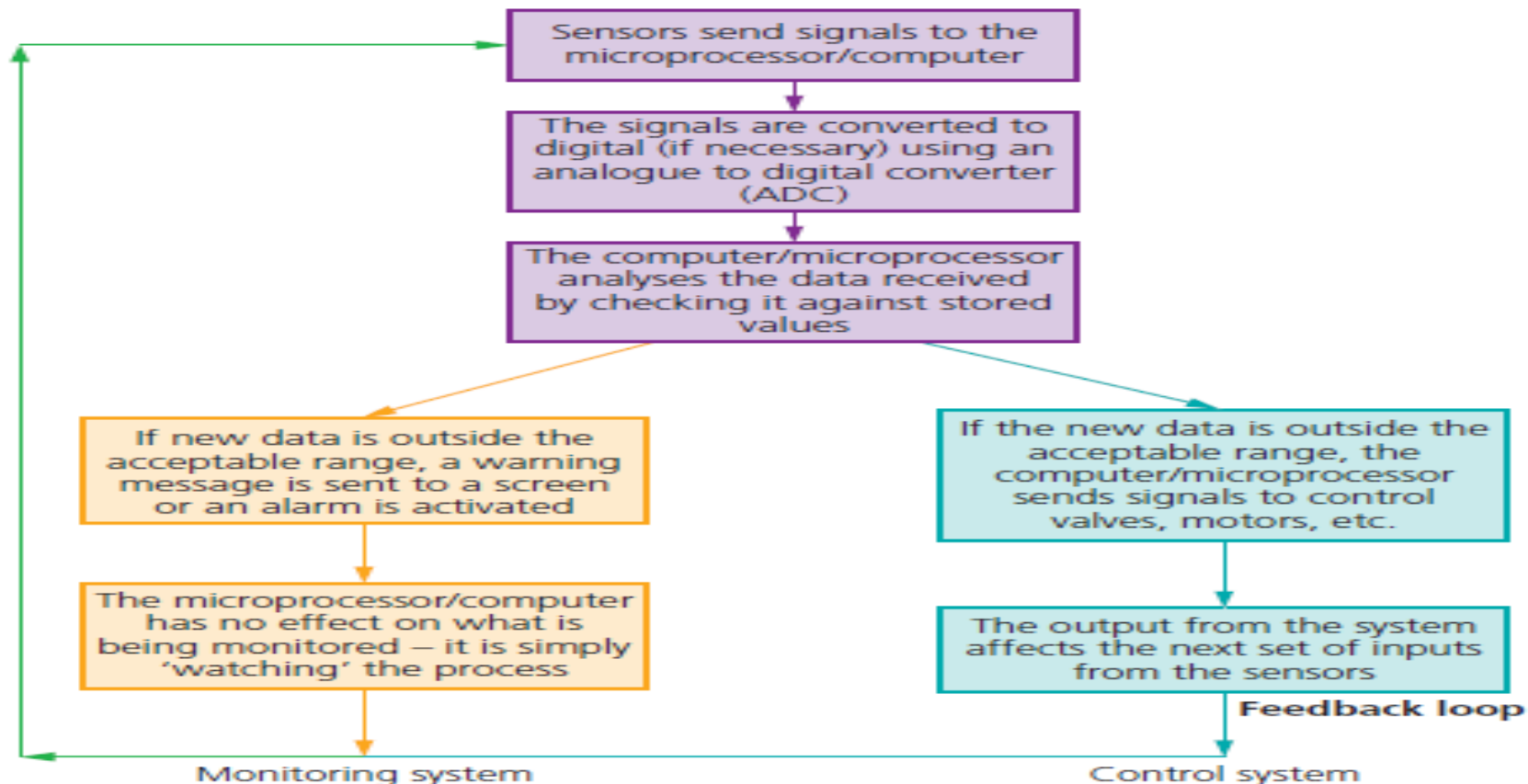


3.2 Input and Output Devices

3.2.3: Sensors

Sensors are used in both monitoring and control applications. There is a subtle difference between how these two methods work (the flowchart is a simplification of the process):

Monitoring and control systems using sensors:



3.2 Input and Output Devices

3.2.3: Sensors

Sensors

Sensors can be designed to detect the levels of a wide variety of conditions.

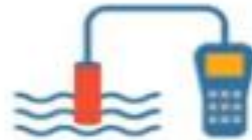
You are required to know about the following sensors for the exam.



Acoustic



Accelerometer



Flow



Gas



Humidity



Infrared



Level



Light



Magnetic field



Moisture



pH



Pressure



Proximity



Temperature

3.2 Input and Output Devices

3.2.3: Sensors

Sensors

All sensors are input devices. They come in a variety of different shapes and sizes.

Sensors measure physical properties of the environment. They then need to translate the information they collect into a digital form that a computer can use, achieved via an analog-to-digital convertor (ADC).

Sensors that you may be examined on are limited to:

- Acoustic
- Accelerometer
- Flow
- Gas
- Humidity
- Infrared
- Level
- Light
- Magnetic field
- Moisture
- pH
- Pressure
- Proximity
- Temperature



3.2 Input and Output Devices

3.2.3: Sensors

Sensor	Description of sensor	Example applications
Temperature	measures temperature of the surroundings by sending signals; these signals will change as the temperature changes	• control of a central heating system • control/monitor a chemical process • control/monitor temperature in a greenhouse
Moisture	measures water levels in, for example, soil (it is based on the electrical resistance of the sample being monitored)	• control/monitor moisture levels in soil in a greenhouse • monitor the moisture levels in a food processing factory
Humidity	this is slightly different to moisture; this measures the amount of water vapour in, for example, a sample of air (based on the fact that the conductivity of air will change depending on the amount of water present)	• monitor humidity levels in a building • monitor humidity levels in a factory manufacturing microchips • monitor/control humidity levels in the air in a greenhouse
Light	these use photoelectric cells that produce an output (in the form of an electric current) depending on the brightness of the light	• switching street lights on or off depending on light levels • switch on car headlights automatically when it gets dark
Infrared (active)	these use an invisible beam of infrared radiation picked up by a detector; if the beam is broken, then there will be a change in the amount of infrared radiation reaching the detector (sensor)	• turn on car windscreen wipers automatically when it detects rain on the windscreen • security alarm system (intruder breaks the infrared beam)
Infrared (passive)	these sensors measure the heat radiation given off by an object, for example, the temperature of an intruder or the temperature in a fridge	• security alarm system (detects body heat) • monitor the temperature inside an industrial freezer or chiller unit
Pressure	a pressure sensor is a transducer and generates different electric currents depending on the pressure applied	• weighing of lorries at a weighing station • measure the gas pressure in a nuclear reactor
Acoustic/sound	these are basically microphones that convert detected sound into electric signals/pulses	• pick up the noise of footsteps in a security system • detect the sound of liquids dripping at a faulty pipe joint

3.2 Input and Output Devices

3.2.3: Sensors

Gas	most common ones are oxygen or carbon dioxide sensors; they use various methods to detect the gas being monitored and produce outputs that vary with the oxygen or carbon dioxide levels present	• monitor pollution levels in the air at an airport • monitor oxygen and carbon dioxide levels in a greenhouse • monitor oxygen levels in a car exhaust
pH	these measure acidity through changes in voltages in, for example, soil	• monitor/control acidity levels in the soil in a greenhouse • control acidity levels in a chemical process
Magnetic field	these sensors measure changes in magnetic fields – the signal output will depend on how the magnetic field changes	• detect magnetic field changes (for example, in mobile phones and CD players) • used in anti-lock braking systems in cars
Accelerometer	these are sensors that measure acceleration and motion of an application, i.e. the change in velocity (a piezoelectric cell is used whose output varies according to the change in velocity)	• used in cars to measure rapid deceleration and apply air bags in a crash • used by mobile phones to change between portrait and landscape mode
Proximity	these sensors detect the presence of a nearby object	• detect when a face is close to a mobile phone screen and switches off screen when held to the ear
Flow (rate)	these sensors measure the flow rate of a moving liquid or gas and produce an output based on the amount of liquid or gas passing over the sensor	• used in respiratory devices and inhalers in hospitals • measure gas flows in pipes (for example, natural gas)
Level	these sensors use ultrasonics (to detect changing liquid levels in, for example, a tank) or capacitance/ conductivity (to measure static levels (for example, height of water in a river) – note, level sensors can also be optical or mechanical in nature	• monitor levels in a petrol tank in a car • in a pharmaceutical process where powder levels in tablet production need to be monitored • leak detection in refrigerant (air conditioning)

3.2 Input and Output Devices

3.2.3: Sensors

<https://www.youtube.com/watch?v=gtDqaoCaUVk>



3.2 Input and Output Devices

3.2.3: Sensors

Sensors

Sensors can be designed to detect the levels of a wide variety of conditions.

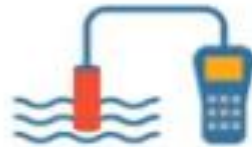
You are required to know about the following sensors for the exam.



Acoustic



Accelerometer



Flow



Gas



Humidity



Infrared



Level



Light



Magnetic field



Moisture



pH



Pressure



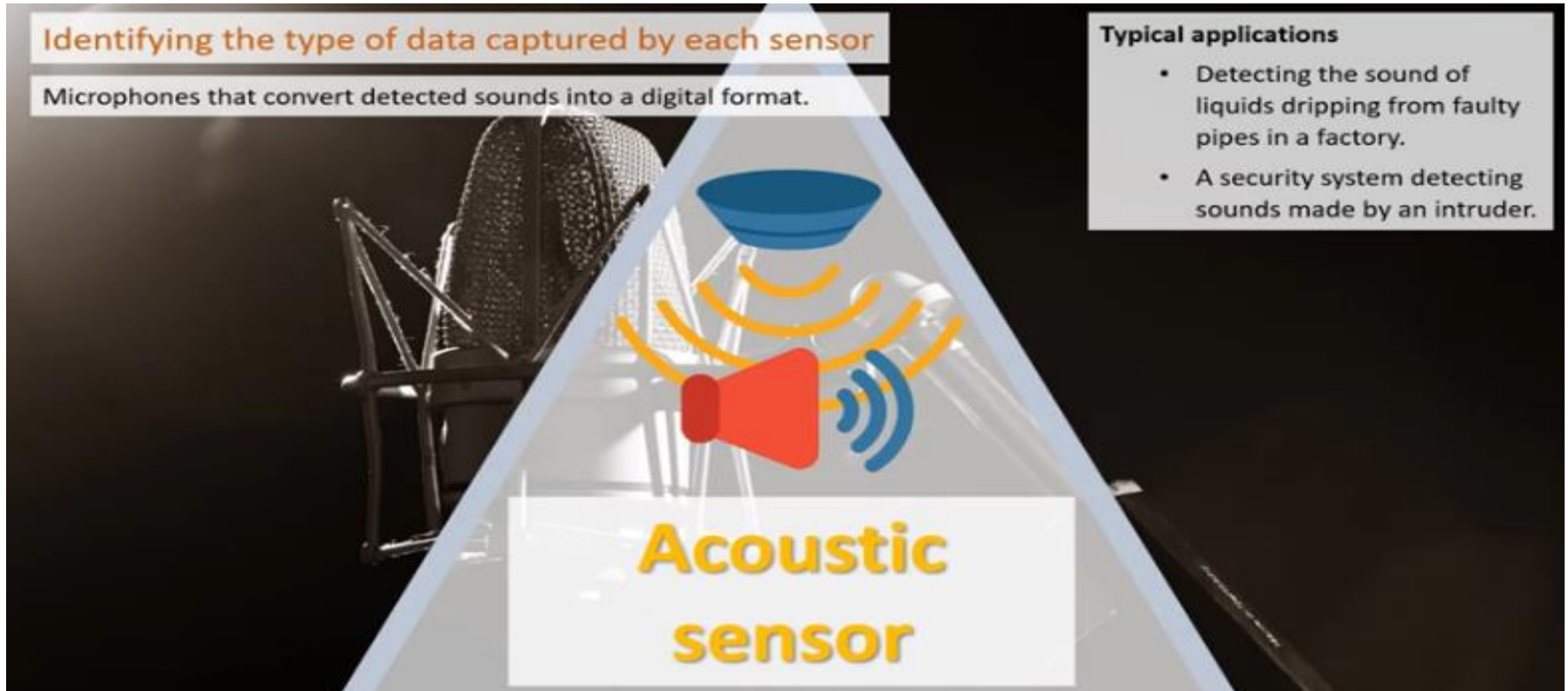
Proximity



Temperature

3.2 Input and Output Devices

3.2.3.1: acoustic



Identifying the type of data captured by each sensor

Microphones that convert detected sounds into a digital format.

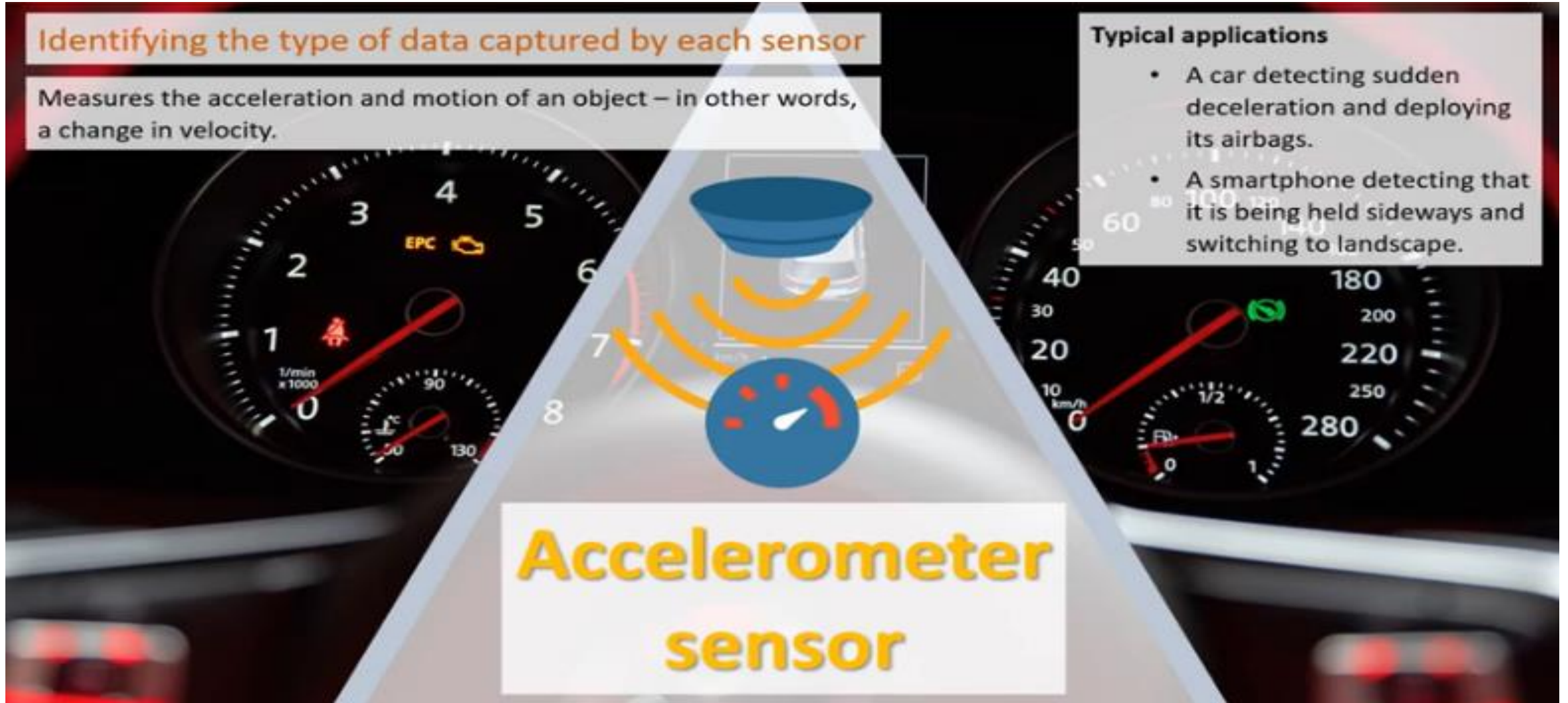
Acoustic sensor

Typical applications

- Detecting the sound of liquids dripping from faulty pipes in a factory.
- A security system detecting sounds made by an intruder.

3.2 Input and Output Devices

3.2.3.2: accelerometer



Identifying the type of data captured by each sensor

Measures the acceleration and motion of an object – in other words, a change in velocity.

Typical applications

- A car detecting sudden deceleration and deploying its airbags.
- A smartphone detecting that it is being held sideways and switching to landscape.

Accelerometer sensor

3.2 Input and Output Devices

3.2.3.3: flow Sensor

Identifying the type of data captured by each sensor

Measures the flow rate of a liquid or gas in motion and produces an output based on the amount passing over the sensor.

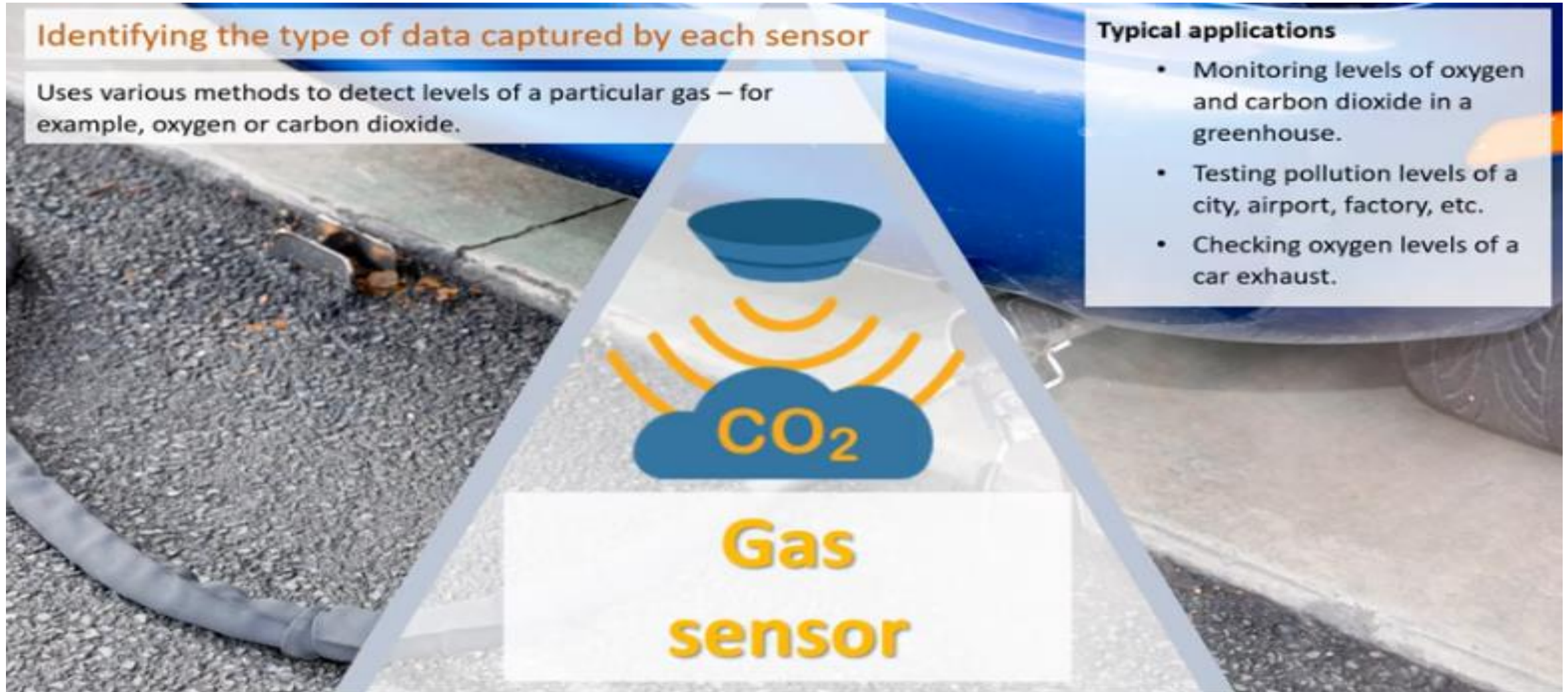
Typical applications

- Measures flow in pipes – for example, natural gas.
- A respiratory device checking oxygen levels.

**Flow
sensor**

3.2 Input and Output Devices

3.2.3.4: Gas Sensor



Identifying the type of data captured by each sensor

Uses various methods to detect levels of a particular gas – for example, oxygen or carbon dioxide.

Typical applications

- Monitoring levels of oxygen and carbon dioxide in a greenhouse.
- Testing pollution levels of a city, airport, factory, etc.
- Checking oxygen levels of a car exhaust.

Gas sensor

3.2 Input and Output Devices

3.2.3.5: Humidity Sensor



Identifying the type of data captured by each sensor

Not to be confused with a sensor that detects moisture, these sensors measure water vapour – e.g., in an air sample.

Typical applications

- Monitoring humidity levels in a factory producing microchips.
- Checking the level of humidity in a greenhouse.

Humidity sensor

3.2 Input and Output Devices

3.2.3.6: infra-red Sensor (active)

Identifying the type of data captured by each sensor

Uses an invisible beam of infrared radiation picked up by a detector. If the beam is broken by an object, the detector senses the change.

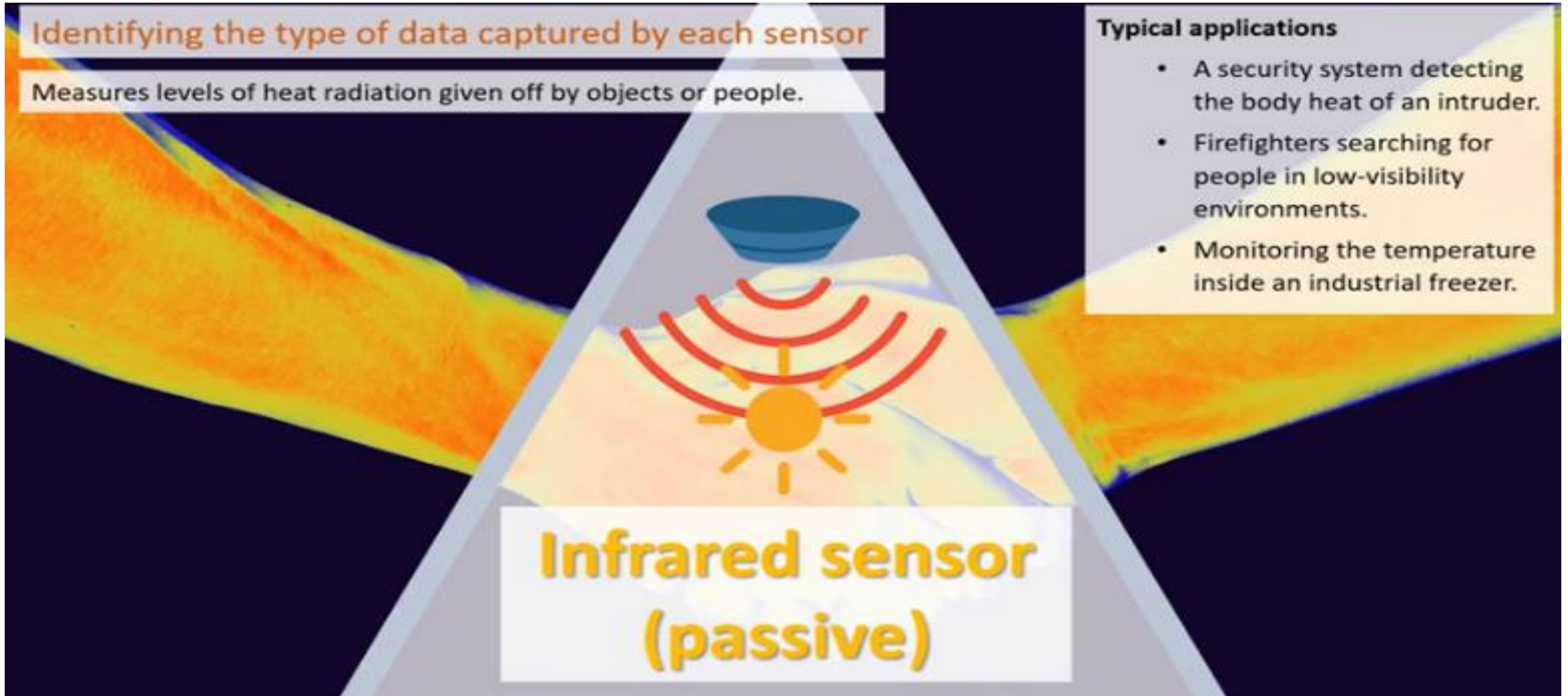


Typical applications

- A security system detecting when an intruder breaks the infrared beam.
- A car automatically turning on its windscreen wipers when rain is detected.

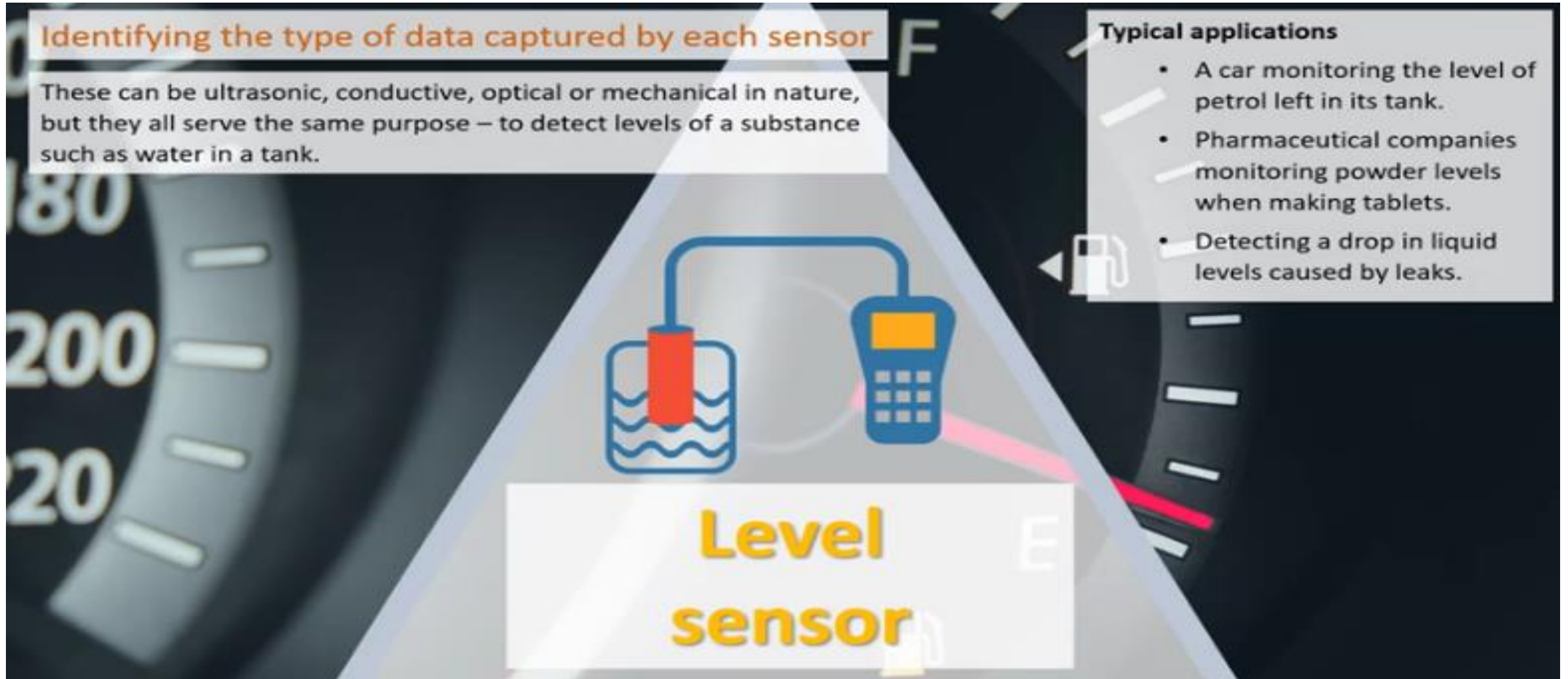
3.2 Input and Output Devices

3.2.3.7: infra-red Sensor (passive)



3.2 Input and Output Devices

3.2.3.8: level sensor



The diagram illustrates a level sensor system. A blue rectangular tank contains a red cylindrical sensor. A blue cable connects the sensor to a blue handheld electronic device with a yellow screen and a keypad. The entire setup is shown within a light blue triangular frame. In the background, there are blurred images of a speedometer and a fuel gauge.

Identifying the type of data captured by each sensor

These can be ultrasonic, conductive, optical or mechanical in nature, but they all serve the same purpose – to detect levels of a substance such as water in a tank.

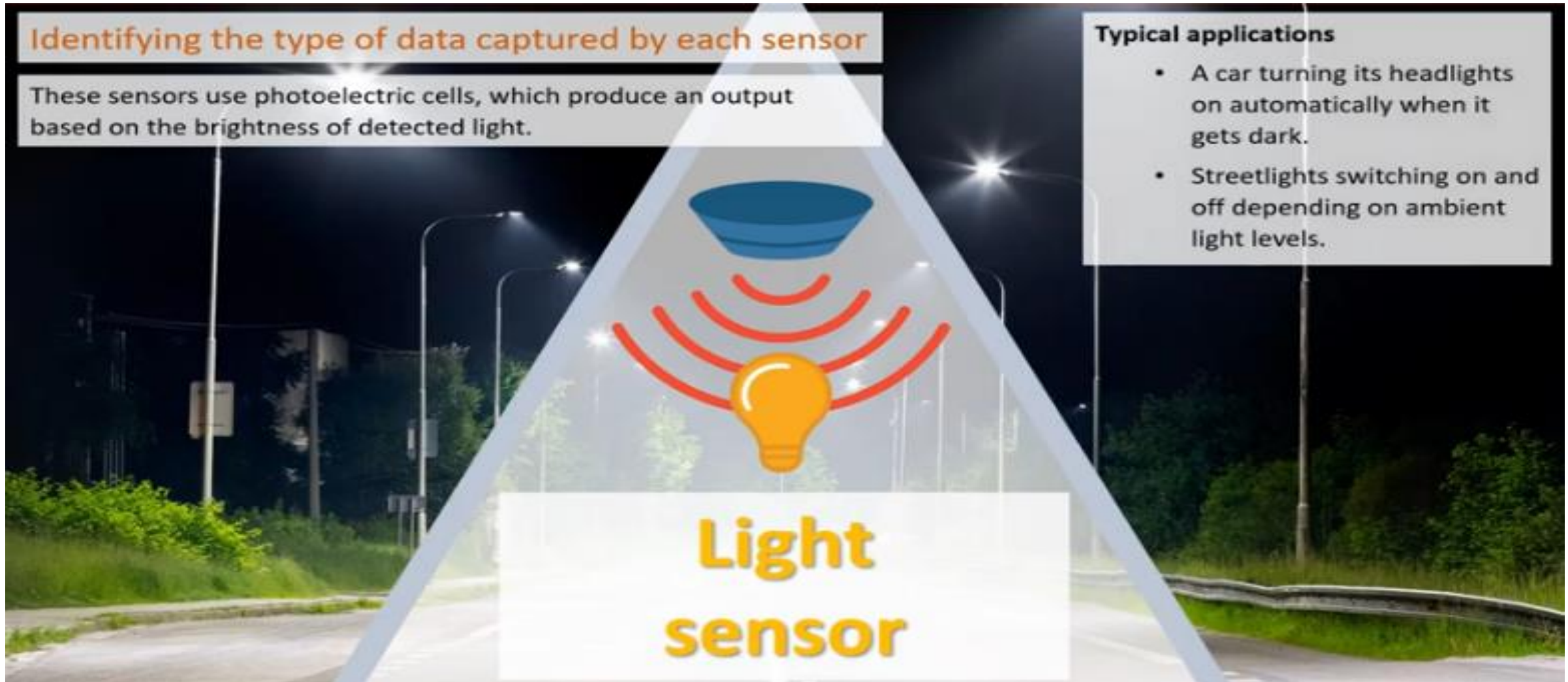
Typical applications

- A car monitoring the level of petrol left in its tank.
- Pharmaceutical companies monitoring powder levels when making tablets.
- Detecting a drop in liquid levels caused by leaks.

Level sensor

3.2 Input and Output Devices

3.2.3.9: light sensor



Identifying the type of data captured by each sensor

These sensors use photoelectric cells, which produce an output based on the brightness of detected light.

Typical applications

- A car turning its headlights on automatically when it gets dark.
- Streetlights switching on and off depending on ambient light levels.

Light sensor

3.2 Input and Output Devices

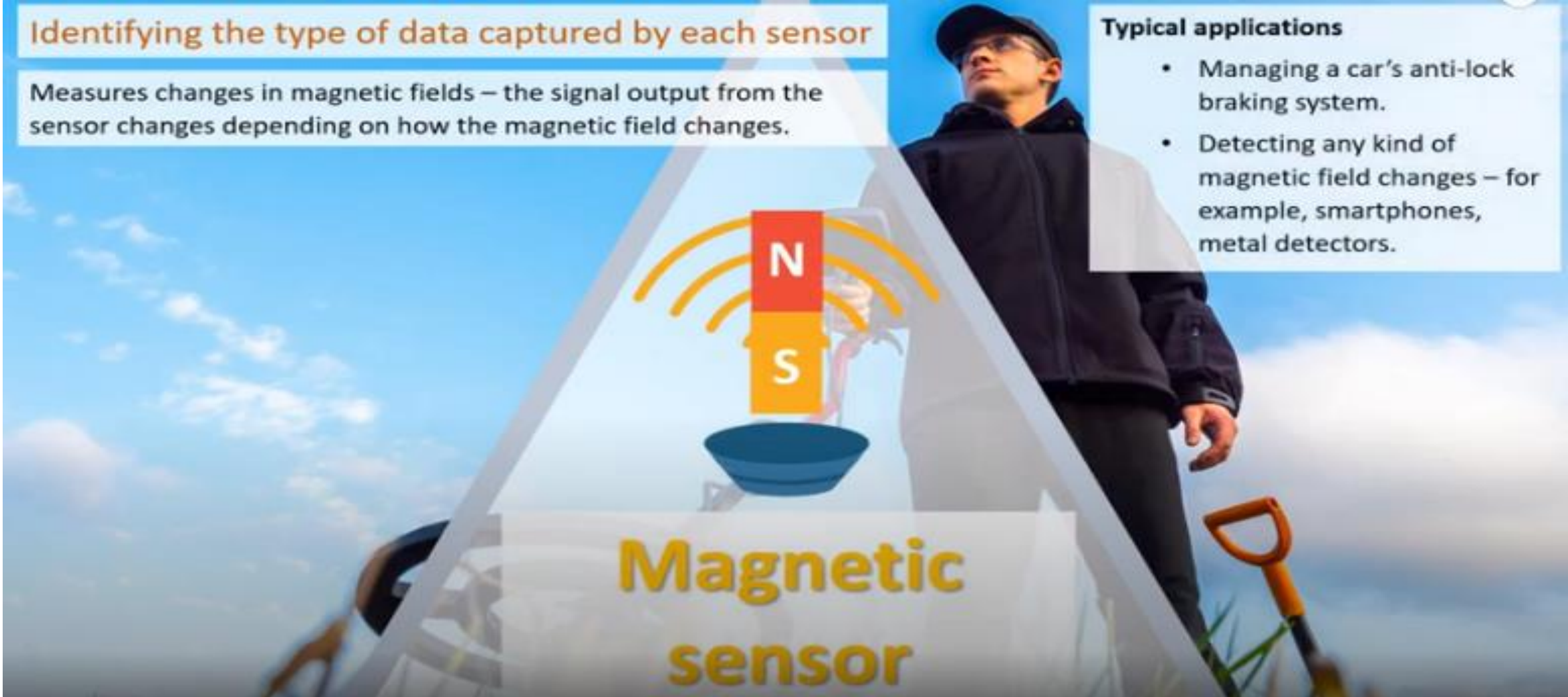
3.2.3.10: magnetic field sensor

Identifying the type of data captured by each sensor

Measures changes in magnetic fields – the signal output from the sensor changes depending on how the magnetic field changes.

Typical applications

- Managing a car's anti-lock braking system.
- Detecting any kind of magnetic field changes – for example, smartphones, metal detectors.



**Magnetic
sensor**

3.2 Input and Output Devices

3.2.3.11: moisture sensor



Identifying the type of data captured by each sensor

Measures levels of moisture – for example, in soil – based on the electrical resistance of the sample.

Typical applications

- Detecting wood rot.
- Monitoring moisture levels in food processing factories.
- Checking the levels of moisture in soil.

Moisture sensor

3.2 Input and Output Devices

3.2.3.12: pH sensor



Identifying the type of data captured by each sensor

Measures levels of acidity – for example, in soil – by detecting changes in voltages.

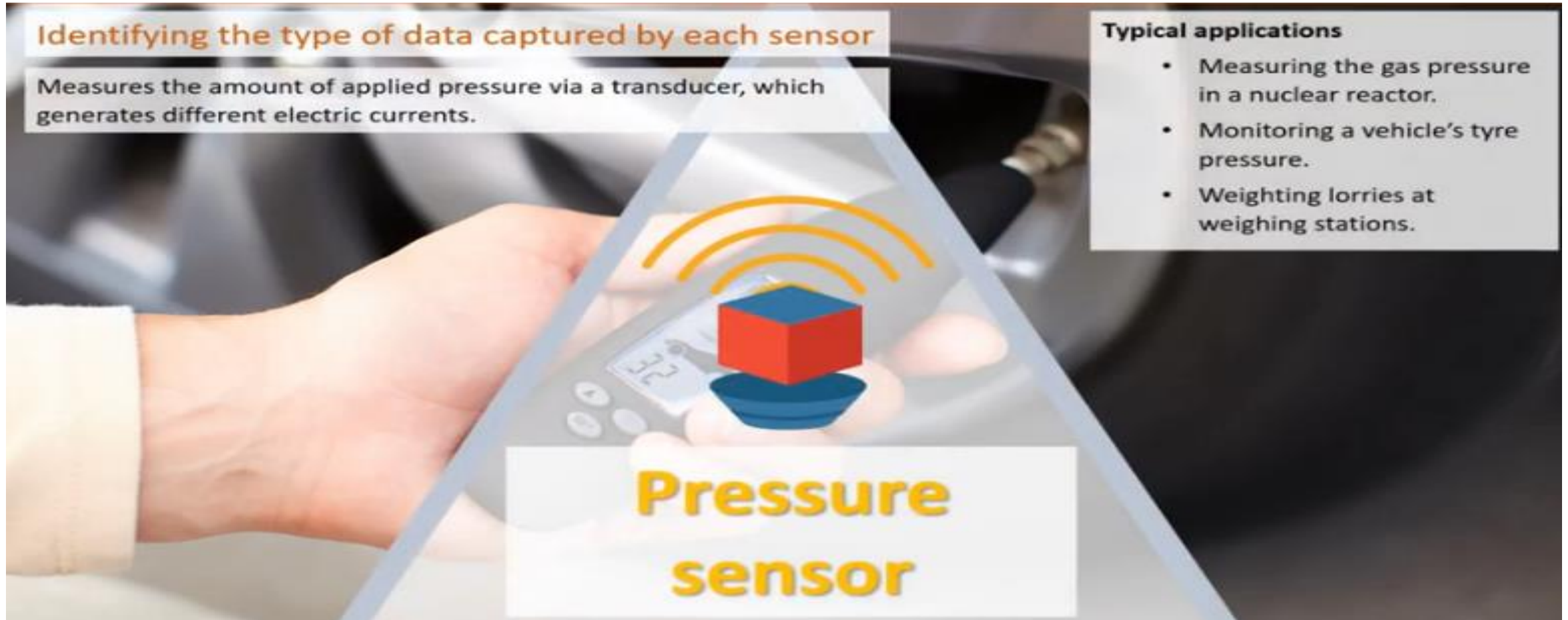
Typical applications

- Monitoring acidity levels of soil for agriculture.
- Checking the level of acidity in a chemical process.

pH sensor

3.2 Input and Output Devices

3.2.3.13: pressure sensor



Identifying the type of data captured by each sensor

Measures the amount of applied pressure via a transducer, which generates different electric currents.

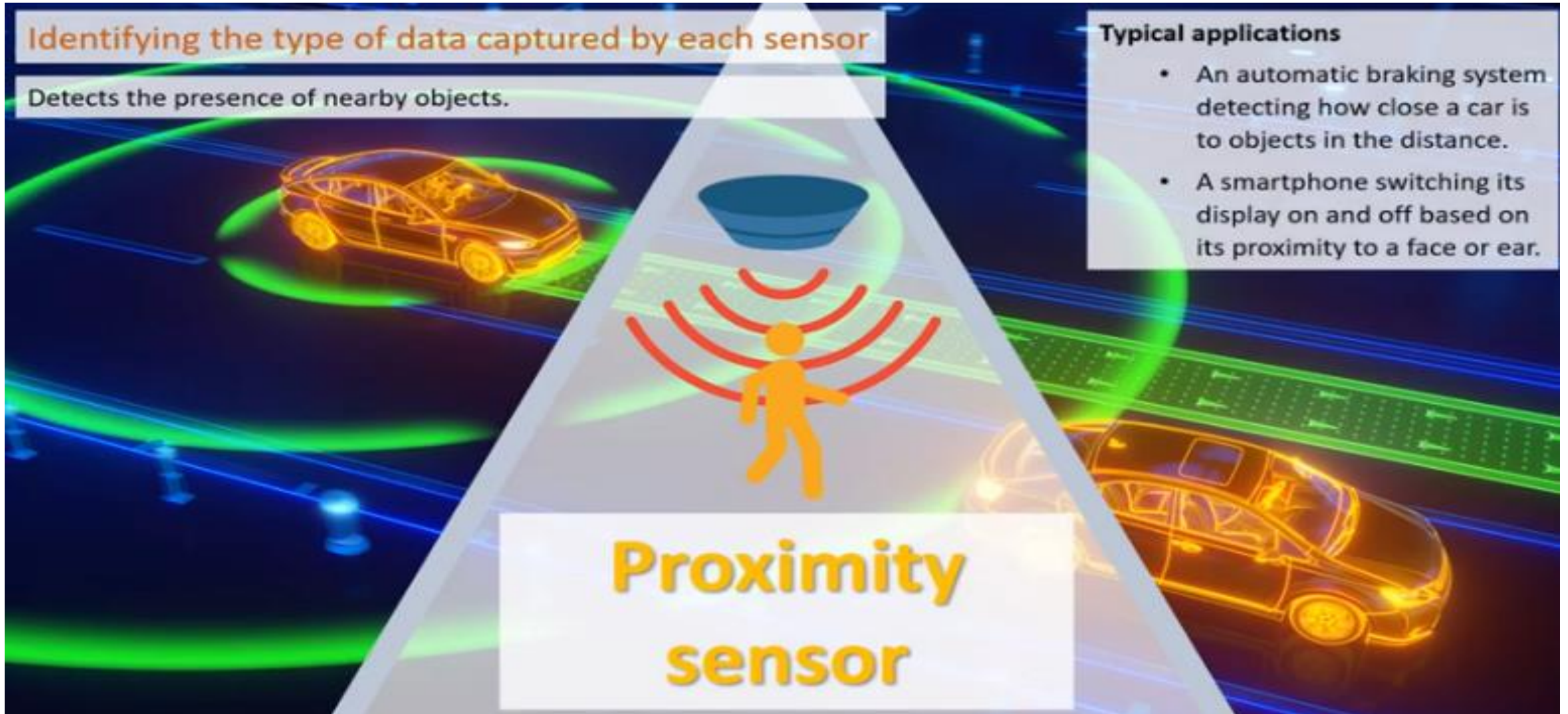
Typical applications

- Measuring the gas pressure in a nuclear reactor.
- Monitoring a vehicle's tyre pressure.
- Weighting lorries at weighing stations.

Pressure sensor

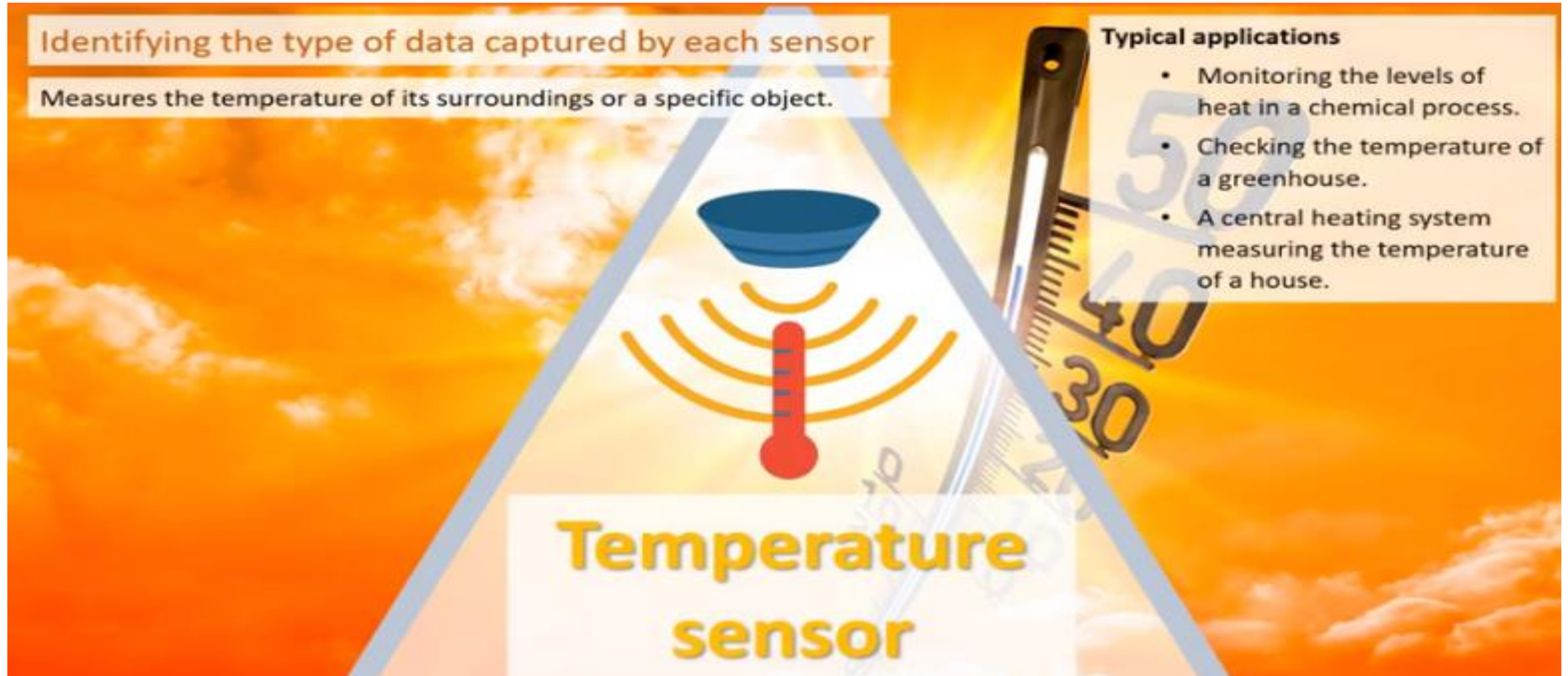
3.2 Input and Output Devices

3.2.3.14: proximity sensor



3.2 Input and Output Devices

3.2.3.15: temperature sensor

An infographic with an orange background featuring a stylized sun and clouds. In the center is a blue temperature sensor with a red thermometer-like tube and yellow concentric arcs representing heat waves. A large thermometer is also visible on the right side of the infographic.

Identifying the type of data captured by each sensor

Measures the temperature of its surroundings or a specific object.

Typical applications

- Monitoring the levels of heat in a chemical process.
- Checking the temperature of a greenhouse.
- A central heating system measuring the temperature of a house.

Temperature sensor

3.2 Input and Output Devices

3.2.3: Sensors: Question 1

Selecting the most suitable sensor for a given context

Security monitoring system

What sensors do you think might be involved in this situation?

Infrared

Acoustic

Proximity

Pressure



3.2 Input and Output Devices

3.2.3: Sensors: Answer 1

Selecting the most suitable sensor for a given context

Security monitoring system

What sensors do you think might be involved in this situation?

Infrared

Proximity

Acoustic

Pressure



3.2 Input and Output Devices

3.2.3: Sensors: Question 2



3.2 Input and Output Devices

3.2.3: Sensors: Answer 2



3.2 Input and Output Devices

3.2.3: Sensors

Sensors

Sensors measure the real-world, physical properties of their environment and convert these analog readings into discrete digital values:



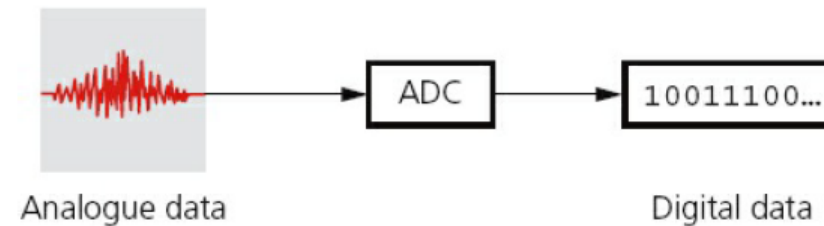
- **Acoustic:** Sound levels.
- **Accelerometer:** Acceleration, vibration, tilt.
- **Flow:** Rate of gas, liquid or powder flow.
- **Gas:** Presence of gas – e.g., carbon monoxide.
- **Humidity:** Levels of water vapour.
- **Infra-red:** Detecting motion or heat.
- **Level:** Liquid levels.
- **Light:** Light levels.
- **Magnetic field:** Presence and strength.
- **Moisture:** Presence and levels of moisture.
- **pH:** Acidity or alkalinity.
- **Pressure:** Gas, liquid or physical pressure.
- **Proximity:** Distance – e.g., parking sensor.
- **Temperature:** Heat levels.

3.2 Input and Output Devices

3.2.3: Sensors

A sensor is a hardware device that will take measurement of physical properties like temperature, pressure or acidity.

The data that is taken in / measured by a sensor is usually in **analogue form**. computers cannot make any sense of these physical quantities and the data needs to be converted into a digital format. This is usually achieved by an **ANALOGUE TO DIGITAL CONVERTER (ADC)**. This device converts physical values into discrete digital values.



When the computer is used to control devices, such as a motor or a valve, it is necessary to use a **DIGITAL TO ANALOGUE CONVERTER (DAC)** since these devices need analogue data to operate. There are many different sensors available each designed to measure different physical properties. Some of these sensors are included in the table below:

IGCSE Exam: Question 1

(b) A printer is one example of an output device.

Give **three** other examples of output devices.

Example 1

Example 2

Example 3

[3]

(c) Give **three** examples of input devices.

Example 1

Example 2

Example 3

[3]

IGCSE Exam: Answer 1

11(b)	Any three from e.g.: • Monitor // screen • Speaker • Headphones • Light // LED • (2D/3D) cutter • DLP // LCD projector • Actuator	3
11(c)	Any two from e.g.: • Keyboard • Mouse // trackerball • Microphone • (2D/3D) scanner • Barcode reader • QR code reader • Digital camera // webcam • Interactive whiteboard • Touchscreen • Sensor	3

IGCSE Exam: Question 2

- 5 A security light system is used by a factory. The light only comes on when it is dark and when movement is detected. The light will stay on for 1 minute before switching off.

Sensors and a microprocessor are used to control the security light system.

- (a) Identify **two** sensors that would be used in the security light system.

Sensor 1

Sensor 2

[2]

- (b) Describe how the sensors and the microprocessor control the security light system.

.....

.....

.....

..... [8]

IGCSE Exam: Answer 2

5(a)	– Light sensor – Motion sensor // infra-red sensor	2
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Question	Answer	Marks
5(b)	Eight from: – Sensors send data to microprocessor – Data is converted to digital (using ADC) – Microprocessor compares data to stored value(s) ... – ... if one value or neither values are within range/out of range/match no action is taken – ... If both values are out of range/in range/match microprocessor sends signal to switch light on ... – ... 1-minute timer is started – Actuator used to switch on/off light – When timer reaches 1 minute, microprocessor sends signal to switch light off – Whole process is continuous	8

IGCSE Exam: Question 3

- (c) An infrared touch screen is used to view and navigate the supermarket stock system.

Explain how the infrared touch screen detects a user's touch.

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[4]

IGCSE Exam: Answer 3

(c)

Question (c) requires you to give an explanation of how an infrared touch screen works. The question is worth 4 marks so at least 4 points should be given.

Each of the following points would be worth 1 mark each (Max 4)

Answers

- (Infrared) rays are sent across screen (from edges)
- Has sensors around edge // sensors capture beams
- (Infrared) rays form a grid across the screen
- (Infrared) ray is broken (by a finger blocking a beam)
- Calculation is made (on where beam is broken) to locate the touch

IGCSE Exam: Question 4

10 Alexandra has a new mobile device.

It has a touch screen that uses capacitive technology.

(a) Describe how a capacitive touch screen registers Alexandra's touch.

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..... [4]

IGCSE Exam: Answer 4

10

(a) Question 10 (a) asks for an explanation of how capacitive touch screens work. Any four of the following points would gain 1 mark:

Answers

- Conductive layer
- An electrostatic / electric field is created
- Sensors around the screen monitor the electrostatic field
- Then touched (electrostatic) charge is transferred to finger
- Location of touch is calculated

IGCSE Exam: Question 5

- 2 The majority of mobile phones use touch screens. Three common technologies are used by different mobile phone manufacturers.

Choose one of the following mobile phone technologies:

- resistive
- capacitive
- infrared

Chosen technology

- (i) Describe how your chosen technology works to allow a user to make selections by touching the screen.

.....

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.....[2]

IGCSE Exam: Answer 5

- (i) Either of the three options, **resistive**, **capacitive** or **infra-red** must be chosen maximum of **two** marks from chosen technology:

resistive

- uses multiple layers of material ...
- ... that transmit electric currents
- when the top layer/screen is pushed/touched into the lower/bottom layer ...
- ... the electric current changes and location of “touch” is found

capacitive

- current sent/flows out from all 4 corners of the screen
- when finger/stylus touches screen, the current changes
- the location of “touch” is calculated

infra-red

- an “invisible” grid on the screen (pattern of infra-red LED beams)
- sensors detect where the screen has been touched through a break in an infrared beam(s)
- the position where the screen touched is calculated

IGCSE Exam: Question 6

3 Four input devices, four descriptions and four applications are shown below.

Draw a line to connect each input device to its correct description. Then connect each description to its correct application.

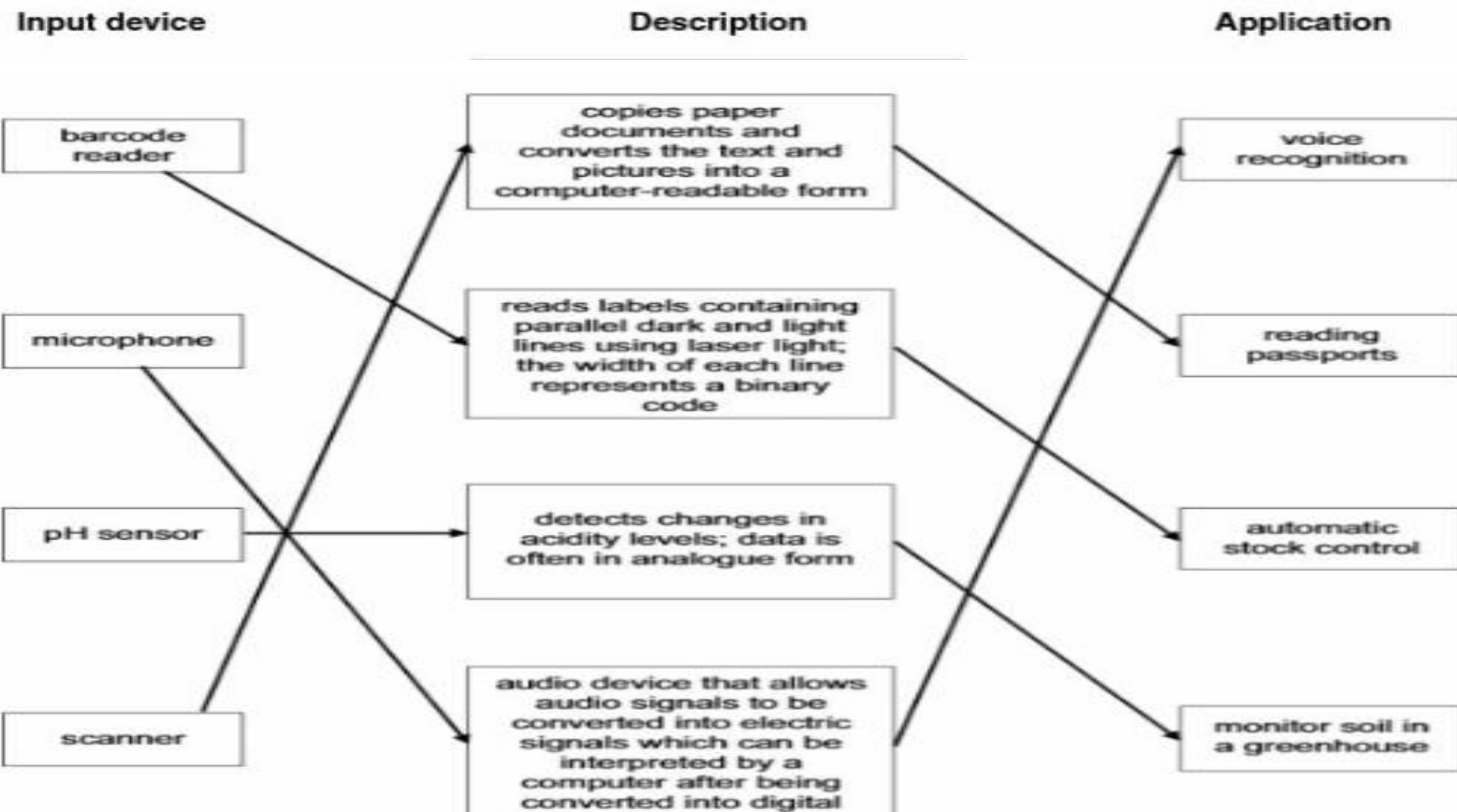
Input device	Description	Application
barcode reader	copies paper documents and converts the text and pictures into a computer-readable form	voice recognition
microphone	reads labels containing parallel dark and light lines using laser light or LEDs; the width of each line represents a binary code	reading passports
pH sensor	detects changes in acidity levels; data is often in analogue form	automatic stock control
scanner	device that allows audio signals to be converted into electric signals; these can be interpreted by a computer after being converted into digital form	monitor soil in a greenhouse

[6]

IGCSE Exam: Answer 6

3 Four input devices, four descriptions and four applications are shown below.

Draw a line to connect each input device to its correct description. Then connect each description to its correct application.



[6]

IGCSE Exam: Question 7

A supermarket uses a barcode scanner to read the barcodes on its products.

(a) Describe how the barcode scanner reads the barcode.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

IGCSE Exam: Answer 7

(a) Question 2 (a) Asks you to describe how a bar code scanner reads a barcode.

Each of the following points would be worth one mark (Max 4)

Answers

1. Shines light / (Red) laser at barcode
2. Light is called an illuminator
3. Light is reflected back // White lines reflect light // Black lines reflect less light
4. Sensors detect the light
5. Different reflections will be given different binary numbers
6. Microprocessor interprets the data

IGCSE Exam: Question 8

- (a) Nancy has captured images of her holiday with her camera. The captured images are stored as digital photo files on her camera.

Explain how the captured images are converted to digital photo files.

.....

.....

.....

.....

.....

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.....

.....

.....

.....

[4]

IGCSE Exam: Answer 8

Question 3 asks you to explain how captured images are converted into digital files. For this question you will also need to call upon knowledge from the binary and files section that discusses how bitmap images work.

Each of the following points are worth 1 mark (Max 3)

Answers

1. Image is converted from analogue to digital using a ADC
2. Image is turned into pixels
3. Each pixel is given a binary value
4. Pixels form a grid (To create the image)
5. Each pixel has a colour
6. Pixels are stored in a sequence
7. Meta data is stored
8. An example of a suitable file photo format e.g. JPEG

IGCSE Exam: Question 9

(a) Five sensors and five applications are shown below.

Draw a line to link each sensor to its most appropriate application.

Sensor

Light
sensor

Moisture
sensor

Gas
sensor

pH
sensor

Pressure
sensor

Application

Monitor the pollution levels in a river

Control the switching off and on of
street lights

Detect intruders breaking into a
building

Monitor the amount of water left in
clothes in a dryer

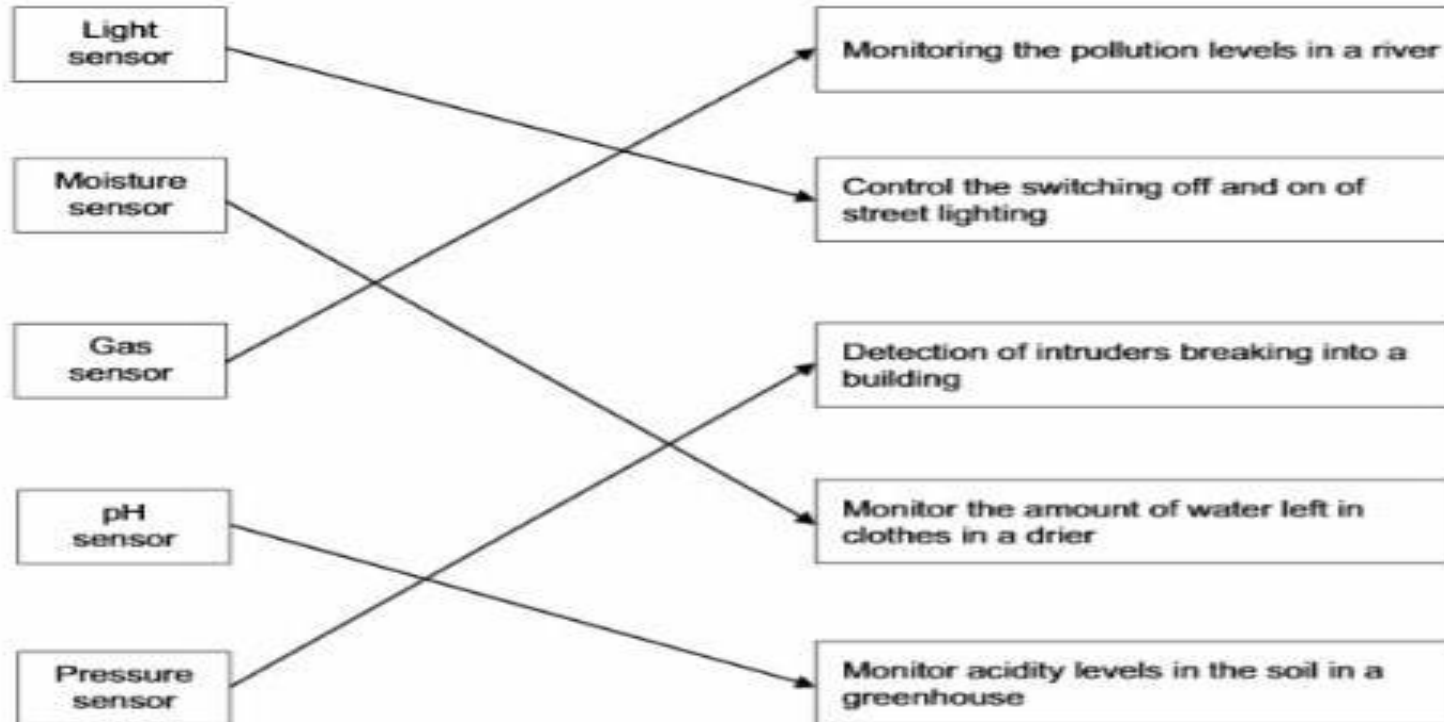
Monitor acidity levels in the soil in a
greenhouse

[4]

IGCSE Exam: Answer 9

5.

(a)



4/5 matches – 4 marks
3 matches – 3 marks
2 matches – 2 marks
1 match – 1 mark

[4]

IGCSE Exam: Question 10

8.

The processes in a chemical factory are monitored by sensors connected to a microprocessor.

- (a) Identify **two** different sensors used in this application. Give an example of how each sensor could be used in the chemical factory.

Sensor 1

Use

.....

Sensor 2

Use

.....

[4]

IGCSE Exam: Answer 10

Answer	Marks
1 mark for appropriate sensor and 1 further mark for appropriate use. Two from: ☐ Gas (sensor) e.g. to measure the levels of oxygen/carbon dioxide / nitrogen in the factory to make sure they are not too high / low ☐ Temperature (sensor) e.g. to measure the temperature of the chemicals to make sure it is not too high/low ☐ Motion / Infra-red (sensor) e.g. to detect any persons in an unauthorised area of the factory ☐ Pressure (sensor) e.g. to measure the pressure of chemicals flowing through pipes to check that level are not too high / low ☐ pH (sensor) to measure the pH to make sure the acidity / alkalinity of the chemicals is correct ☐ Light (sensor) to measure the level of light to make sure it remains at a constant level for the chemical process	4

IGCSE Exam: Question 11

(b) Describe how the sensors and a microprocessor are used to monitor a process.

IGCSE Exam: Answer 11

Five from:

- ☐ Sensors send signals to microprocessor
- ☐ Analogue signals are converted to digital (using ADC)
- ☐ Microprocessor compares value to stored value ...
- ☐ ... If out of range / matches stored values ...
- ☐ ... signal sent to alert workers (e.g. sound alarm)
- ☐ ... microprocessor send signal to cause an action to occur e.g. cool a process down, heat a process up, add a chemical
- ☐ ... no action taken
- ☐ Output/record readings
- ☐ Monitoring is continuous

5

3.3: Question 12

Airline boarding passes can be read from a smartphone instead of a printout.



Identify what type of barcode **A** is an example of. Explain how the data stored in this type of barcode is read.

3.3: Answer 12

- 9 An advertisement in a magazine displays this barcode:



- (a) Identify this type of barcode.

..... QR/Quick response [1]

- (b) Explain how the data stored in this barcode is read.

Any four from:

- Read/scanned using app (on mobile device)
- It is the camera that is used to scan/capture the image
- The three large squares are used to define the alignment // uses alignment targets/modules
- Black squares reflect less light // white squares reflect more light
- The app/device processes the image
- Each small square/pixel is converted to a binary value

IGCSE Exam: Question 13

3 The diagram shows **five** output devices and **five** descriptions.

Draw a line between each output device and its description.

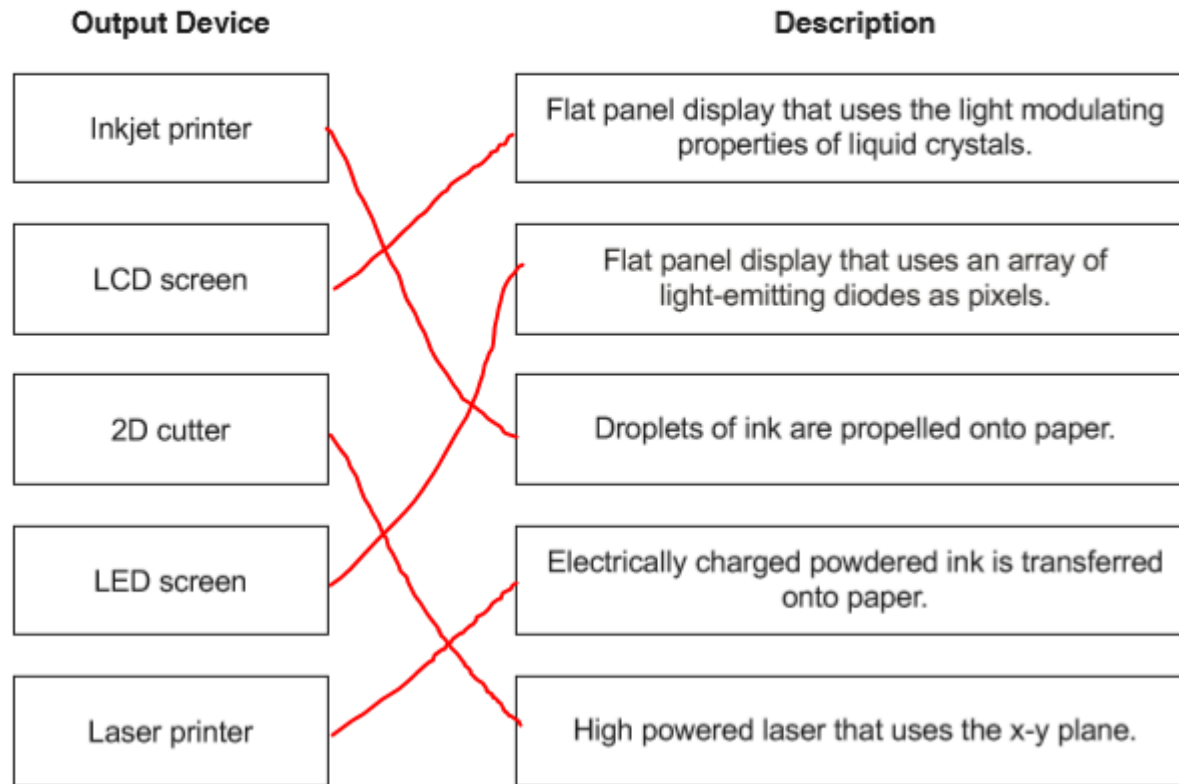
Output Device	Description
Inkjet printer	Flat panel display that uses the light modulating properties of liquid crystals.
LCD screen	Flat panel display that uses an array of light-emitting diodes as pixels.
2D cutter	Droplets of ink are propelled onto paper.
LED screen	Electrically charged powdered ink is transferred onto paper.
Laser printer	High powered laser that uses the x-y plane.

[4]

IGCSE Exam: Answer 13

The diagram shows **five** output devices and **five** descriptions.

Draw a line between each output device and its description.



[4]

IGCSE Exam: Question 14

- 1 A washing machine has a small display screen built into it.

One use of the display screen is to show an error code when a problem has occurred with a washing cycle.

- (a) State whether the display screen is an **input**, **output** or **storage device**.

.....[1]

IGCSE Exam: Answer 14

(a) Question 1 (a) is part of a larger question but (a) is the only section related to input / output devices. This is a simple question that asks you if a small display screen is an input, output or storage device.

Answer

Output device

IGCSE Exam: Question 15

- (b) State **one** suitable application for **each** printer below. A different application must be given for each printer.

Inkjet printer

.....

3D printer

.....

[2]

- (c) Name another type of printer and describe **one** way in which it is different from the printers named in **part (b)**.

Give an application for this printer.

Type of printer

Description

.....

.....

Application

.....

[3]

IGCSE Exam: Answer 15

Question (b) asks you to give example uses for inkjet and 3d printers.

Answers

Inkjet

- (Small quantities of) Documents
- Photos

3D Printer

- (Physical) prototype (From CAD)
- (Physical) model (from blueprint)

(c) Question (c) asks you to name another type of printer and then do give a description and example application.

Answers

Laser Printer

- Uses toner / powder ink
- uses (Positive and negative) charged drums // rotating drum
- uses static charge
- no moving head
- faster at printing
- high volume output / high speed
- producing flyers / leaflets / magazines

Other printer types also accepted e.g dot matrix.

IGCSE Exam: Question 16

6 Two examples of output devices are a 3D printer and a 3D cutter.

(a) The table contains **four** statements about 3D printers and 3D cutters.

Tick (✓) to show which statements apply to each output device, some statements may apply to both output devices.

Statement	3D printer (✓)	3D cutter (✓)
Outputs a physical 3D product		
Uses a high powered laser to create the output		
Creates 3D prototypes		
Uses layers of material to create the output		

[4]

(b) Identify the software used to create the computerised designs for 3D printing.

..... [1]

IGCSE Exam: Answer 16

(a)

Answers:

Question	Answer	Marks															
6(a)	1 mark for the correct ticks (✓) for each statement <table><tr><th>Statement</th><th>3D printer (✓)</th><th>3D cutter (✓)</th></tr><tr><td>Outputs a physical 3D product</td><td>✓</td><td>✓</td></tr><tr><td>Uses a high powered laser to create the output</td><td></td><td>✓</td></tr><tr><td>Creates 3D prototypes</td><td>✓</td><td>✓</td></tr><tr><td>Uses layers of material to create the output</td><td>✓</td><td></td></tr></table>	Statement	3D printer (✓)	3D cutter (✓)	Outputs a physical 3D product	✓	✓	Uses a high powered laser to create the output		✓	Creates 3D prototypes	✓	✓	Uses layers of material to create the output	✓		4
Statement	3D printer (✓)	3D cutter (✓)															
Outputs a physical 3D product	✓	✓															
Uses a high powered laser to create the output		✓															
Creates 3D prototypes	✓	✓															
Uses layers of material to create the output	✓																

(b)

Answer

Computer Aided Design / CAD

IGCSE Exam: Question 17

- (c) A Digital Light Projector (DLP) is another example of an output device.

Describe how a DLP displays an image.

.....

.....

.....

.....

.....

..... [3]

IGCSE Exam: Answer 17

(c) Question 6 (c) requires a description of how a Digital Light Projector produces an image. The question requires 3 key points.

1 mark will be awarded for each of the following points (Max 3)

Answers

- Uses a large number of tiny mirrors
- Mirrors are laid out in a grid / matrix
- Each mirrors creates a pixel in the image
- Mirrors can tilt toward or away from light source
- The mirrors reflect light toward a (projection) lens
- Colour is produced using a colour wheel // light passes through a colour wheel
- Can be used to display an image on a wall / screen

IGCSE Exam: Question 18

Five descriptions of different input or output devices are given in the table.

Complete the table by stating the **name** of each input or output device.

Description	Name of device
This is an input device that works by shining a light onto the surface of a document. The light source is automatically moved across the document and the reflected light is captured by mirrors and lenses.	
This is an input device where a laser or a light source is moved across an object. The width, height and depth of the object are measured to allow a model to be created.	
This is a large input device that is usually fixed to a wall. A user can calibrate the device to make sure the sensors align with a projected image. The user can use either their finger or a special pen to make selections.	
This is an output device that uses many small mirrors to reflect light towards a lens. This will display an image.	
This is an output device that creates an object by building layer upon layer of material.	

[5]

IGCSE Exam: Answer 18

Five descriptions of different input or output devices are given in the table.

Complete the table by stating the **name** of each input or output device.

Description of input or output device	Name of device
This is an input device that works by shining a light onto the surface of a document. The light source is automatically moved across the document and the reflected light is captured by mirrors and lenses.	2D Scanner
This is an input device where a laser or a light source is moved across an object. The width, height and depth of the object are measured to allow a model to be created.	3D scanner
This is a large input device that is usually fixed to a wall. A user can calibrate the device to make sure the sensors align with a projected image. The user can use either their finger or a special pen to make selections.	Interactive whiteboard
This is an output device that uses many small mirrors to reflect light towards a lens. This will display an image.	Projector
This is an output device that creates an object by building layer upon layer of material.	3D printer

IGCSE Exam: Input/output devices: Question with Answer 19

Hardware device	Input (✓)	Output (✓)	Storage (✓)
Solid state drive (SSD)			✓
Sensor	✓		
Headphones		✓	
Microphone	✓		
USB flash drive			✓
Actuator		✓	

IGCSE Exam: Input/output devices: Question with Answer 20

Statement	3D (✓)	Inkjet (✓)	Laser (✓)
Uses a moving print head	✓	✓	
Uses liquid ink		✓	
Produces output using materials such as plastic and resin	✓		
Uses piezoelectric or thermal technology	(✓)	✓	
Uses a rotating drum to transfer the image to the paper			✓
Uses layer upon layer of material to create the output	✓		

IGCSE Exam: Input/output devices: Question with Answer 21

Six statements are given about touch screen technology.

Tick (✓) to show if the statement applies to **Capacitive** or **Resistive** touch screen technology.

Statement	Capacitive (✓)	Resistive (✓)
Needs pressure to be applied to create a circuit		✓
May not register a touch if the user is wearing gloves	✓	
More commonly used in smartphones	✓	
More responsive to a touch	✓	
Needs an electrical field to be changed to register a touch	✓	
Cheaper to manufacture		✓

IGCSE Exam: Input/output devices: Question with Answer 22

Input and output devices are often connected to a personal computer.

(a) Identify **three** input devices that can be connected to a personal computer.

- Three from e.g.:
- | | | |
|---------|----------------------------|-------|
| 1 | - Keyboard | |
| | - Mouse | |
| | - Microphone | |
| 2 | - 2D scanner | |
| | - 3D scanner | |
| | - Touchscreen | |
| 3 | - Webcam // digital camera | |
| | - Joystick | |
| | - Trackpad | |
| | - Sensor | |
| | - Interactive whiteboard | |

[3]

(b) Identify **three** output devices that can be connected to a personal computer.

- Three from e.g.:
- | | | |
|---------|--------------------------|-------|
| 1 | - Monitor // touchscreen | |
| | - Inkjet printer | |
| | - Laser printer | |
| 2 | - 3D printer | |
| | - Speaker | |
| 3 | - Headphones | |
| | - LED Projector | |
| | - DLP | |
| | - 2D cutter | |
| | - 3D cutter | |

[3]

IGCSE Exam: Input/output devices: Question with Answer 23

8 A supermarket uses a barcode scanner to read the barcodes on its products.

(a) Describe how the barcode scanner reads the barcode.

Any four from:

- Shines light / (red) laser at barcode
 - Light is called an illuminator
 - Light is reflected back // White lines reflect light // Black lines reflect less light/absorbs light
 - Sensors / photoelectric cells detect the light
 - Different reflections / bars will give different binary values / digital values // pattern converted to digital values
 - A microprocessor interprets the data
-

IGCSE Exam: Input/output devices: Question with Answer 24

(b) Explain how the barcode system could help the supermarket manage its stock.

Any three from:

- barcode identifies a (unique) product
- barcode can be used to look up product (in a database)
- data about stock levels can be stored on a system
- stock can be automatically deducted from the system
- can check stock is below a certain level // check stock level
- automatic re-order // Alerts when stock is low
- automatically update new stock level
- to locate if an item of stock is available in another location

[3]

(c) An infrared touch screen is used to view and navigate the supermarket stock system.

Explain how the infrared touch screen detects a user's touch.

Any four from:

- (Infrared) rays are sent across screen (from the edges)
- Has sensors around edge // Sensors capture beams
- (Infrared) rays form a grid across the screen
- (Infrared) ray is broken (by a finger blocking a beam)
- Calculation is made (on where beam is broken) to locate the 'touch' // Co-ordinates are used to locate the touch

IGCSE Exam: Input/output devices: Question with Answer 25

- 9 An advertisement in a magazine displays this barcode:



- (a) Identify this type of barcode.

..... QR/Quick response [1]

- (b) Explain how the data stored in this barcode is read.

Any four from:

- Read/scanned using app (on mobile device)
- It is the camera that is used to scan/capture the image
- The three large squares are used to define the alignment // uses alignment targets/modules
- Black squares reflect less light // white squares reflect more light
- The app/device processes the image
- Each small square/pixel is converted to a binary value

IGCSE Exam: Input/output devices: Question with Answer 26

10 Alexandra has a new mobile device.

It has a touch screen that uses capacitive technology.

(a) Describe how a capacitive touch screen registers Alexandra's touch.

Any four from:

- Conductive layer
- An electrostatic/electric field is created
- **Sensor(s)** (around the screen) monitor the electrostatic field
- When touched (electrostatic) charge is transferred to finger
- Location of touch is calculated // Co-ordinates used to calculate touch

IGCSE Exam: Input/output devices: Question with Answer 27

- 11 A factory uses a security system to control a security light. The system uses a sensor and a microprocessor.

Explain how the security system makes use of the sensor and the microprocessor to control the security light.

Any six from:

- Suitable sensor (motion/infra-red)
- Data converted (from analogue) to digital (using ADC)
- Data sent to microprocessor
- Data is compared to stored value/range ...
- ... if data matches/out of range data security light turned on ..
- ... waits for suitable period/until no motion detected ...
- ... light turned off
- Continuous loop/process

IGCSE Exam: Input/output devices: Question with Answer 28

State **five** sensors that could be used in the following applications.

Give a **different** type of sensor for each application.

Application	Sensor
Weighing a baby in a hospital	Pressure
Turning off a kettle when it boils	Temperature
Controlling an automatic door	Infrared / Light / Pressure
Monitoring the air quality in an aeroplane	Oxygen / Gas / Humidity
Counting cars crossing a bridge	Pressure / Infrared / Magnetic

IGCSE Exam: Input/output devices: Question with Answer 29

- 7 A train station uses large touch screens to allow passengers to search for train information and buy tickets.

(a) State **three** benefits of using a touch screen in the train station.

Benefit 1

Any three from:

- – Does not require peripherals (mouse or keyboard)
- Benefit 2 – Number of possible inputs limited / menu driven interface
- – Less chance of input error
- – Resistant to weather

Benefit 3

[3]

(b) The touch screens at the station use resistive touch technology.

Describe how resistive touch technology works.

- – Uses two/multiple layers
- – When top layer touched / pushed two layers make contact
- – Circuit is completed when layers touch
- – Point of contact is determined/calculated

IGCSE Exam: Input/output devices: Question with Answer 30

(b) Identify the software used to create the computerised designs for 3D printing.

..... Computer Aided Design/CAD [1]

(c) A Digital Light Projector (DLP) is another example of an output device.

Describe how a DLP displays an image.

Three from:

- ☐ Uses a large number of tiny mirrors
- ☐ Mirrors are laid out in a grid/matrix
- ☐ Each mirror creates a pixel in the image
- ☐ Mirrors can tilt toward or away from light source
- ☐ The mirrors reflect light toward a (projection) lens
- ☐ Colour is produced using a colour wheel // Light passes through colour wheel // filters light into red/green/blue
- ☐ Can be used to display an image on a wall/screen

.....

IGCSE Exam: Input/output devices: Question with Answer 31

- 9 A sports stadium uses a pressure sensor and a microprocessor to monitor the number of people entering the sports stadium. For the counter to increment the weight on the pressure sensor must exceed 5 kg.

Explain how the system uses the pressure sensor and the microprocessor to monitor the number of people entering.

Five from:

- ☐ The data is sent to the microprocessor
- ☐ The analogue data is converted to digital (using ADC)
- ☐ The microprocessor compares the data to a stored value of 5 kg .
 - ... If the value is greater than 5 kg
 - ... a counter is added to/incremented
- ☐ The process is continuous

Learning Objectives

3.3 Data storage

Candidates should be able to:

- 1 Understand what is meant by primary storage
- 2 Understand what is meant by secondary storage
- 3 Describe the operation of magnetic, optical and solid-state (flash memory) storage and give examples of each

Notes and guidance

- Primary storage is directly accessed by the CPU
- Including the role of:
 - random access memory (RAM)
 - read only memory (ROM)
- Including why a computer needs both RAM and ROM, and the difference between them
- Secondary storage is not directly accessed by the CPU and is necessary for more permanent storage of data
- Magnetic storage uses platters which are divided into tracks and sectors. Data is read and written using electromagnets
- Optical storage uses lasers to create and read pits and lands
- Solid-state (flash memory) uses NAND or NOR technology. Transistors are used as control gates and floating gates

Learning Objectives

3.3 Data storage

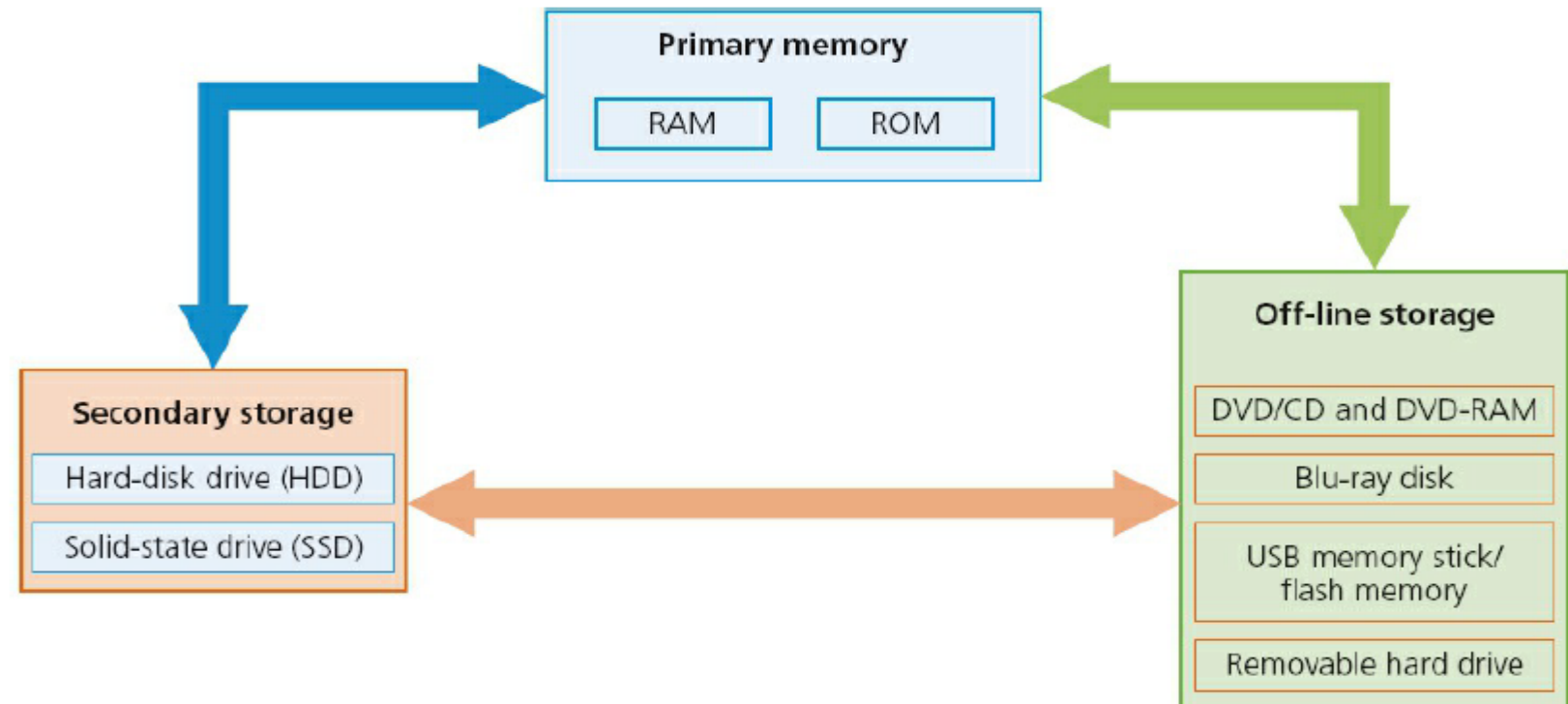
- 4 Describe what is meant by virtual memory, how it is created and used and why it is necessary
- 5 Understand what is meant by cloud storage
 - Pages of data are transferred between RAM and virtual memory when needed
 - Cloud storage can be accessed remotely in comparison to storing data locally
 - Physical servers and storage are needed to store data in cloud storage
- 6 Explain the advantages and disadvantages of storing data on the cloud in comparison to storing it locally

3.3: Data storage

Memory is a hugely important aspect of computing, without memory nothing could be saved and you would have to start every work session from scratch. Even more importantly there would be nowhere to store an operating system thus meaning that computers would not exist as we know them.

Memory and storage devices can be split up into three distinct groups:

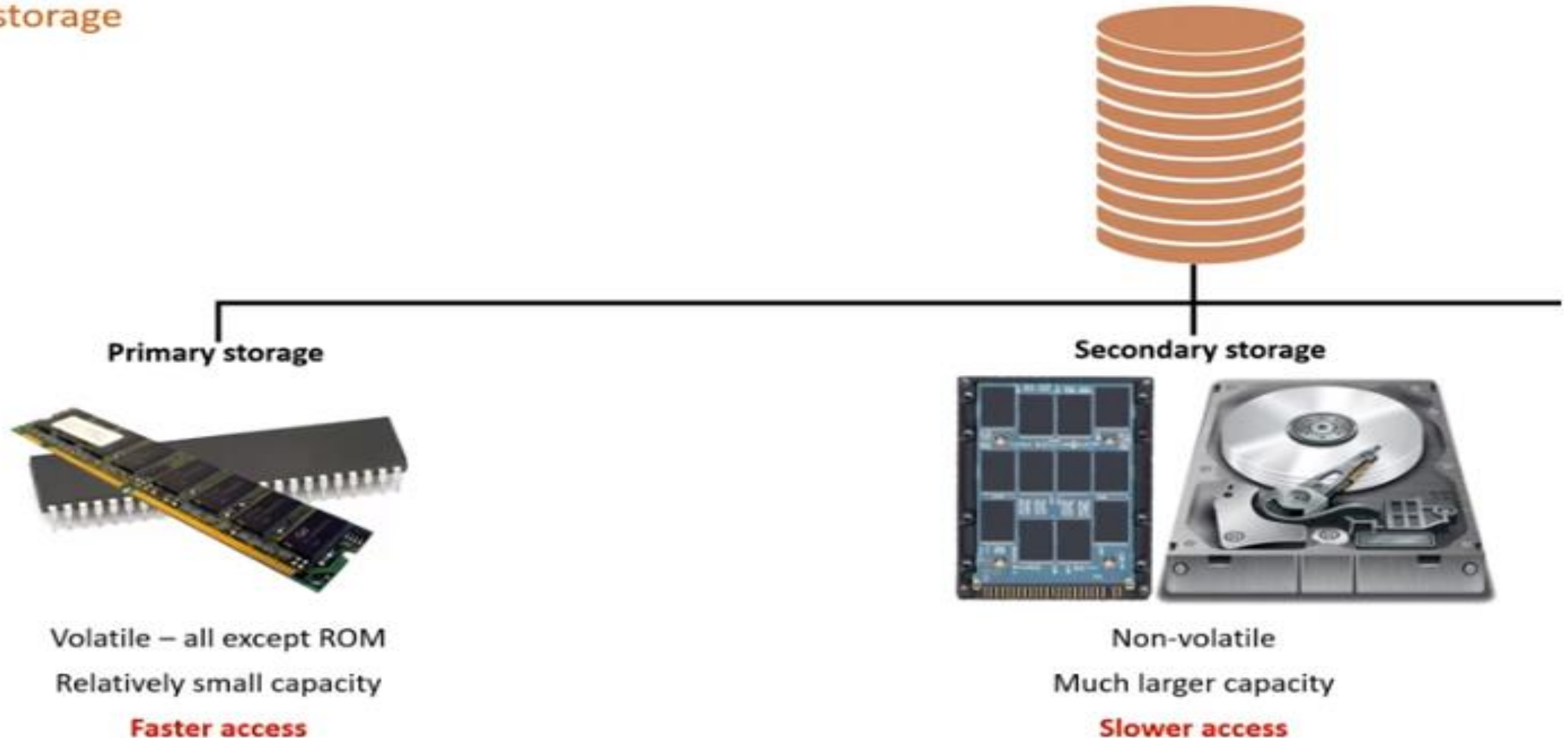
- primary memory
- secondary storage
- off-line storage.



3.3: Data storage

3.3.1: Primary Storage (RAM and ROM)

Types of storage



3.3: Data storage

3.3.1: Primary Storage (RAM and ROM)

<https://www.youtube.com/watch?v=cuqlg31Jlgw>



3.3: Data storage

3.3.1: Primary Storage (RAM and ROM)

<https://www.youtube.com/watch?v=Q2pzT6oYPWg>



3.3: Data storage

3.3.1: Primary Storage (RAM and ROM)

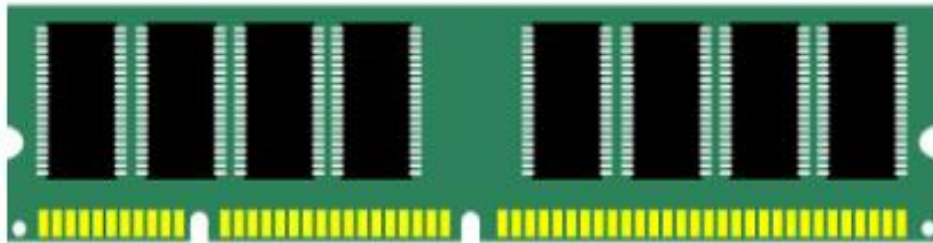
<https://www.youtube.com/watch?v=dhQOkkZXu5w>



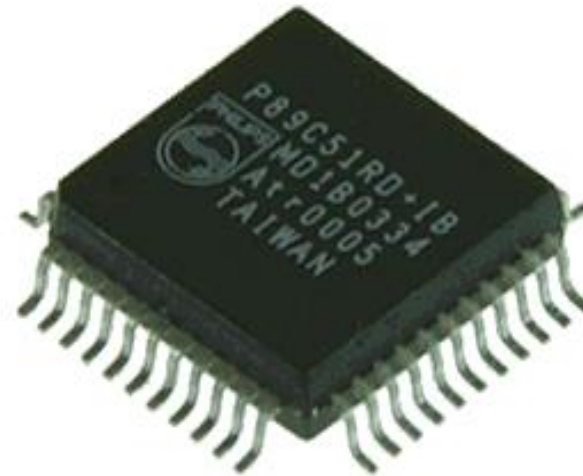
3.3.1: Primary Storage (RAM and ROM)

As a whole, primary memory is the memory that is used by the computer to maintain smooth operation. Primary memory, also known as main memory can be divided into two categories, these are RAM and ROM.

RAM (Random Access Memory)

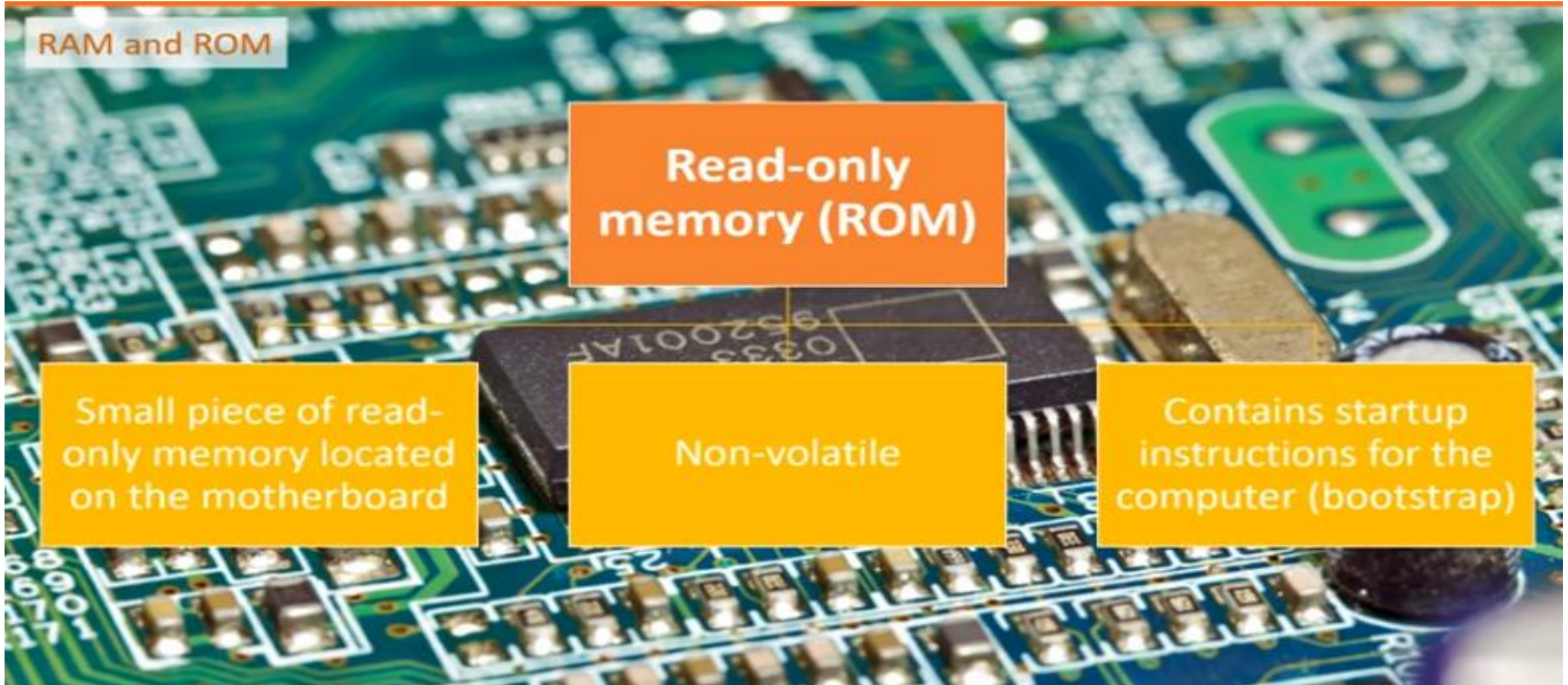


ROM (Read Only Memory)



3.3: Data storage

3.3.1: Primary Storage (ROM)



3.3: Data storage

3.3.1: Primary Storage (ROM)

The main features of READ ONLY MEMORY (ROM): can be summarized as follows:

ROM is different to RAM and has a very specific job. ROM is read only meaning that it cannot be written to, it simply holds data that can be read.

The data that ROM holds is the **instructions that are needed for a computer to start up** once the power is switched on. These instructions are known as the **BIOS (Basic Input Output System)**

Rom is also different to RAM in that it is **Non - Volatile**, This means that it **does not lose its data** when **power is lost**.

3.3: Data storage

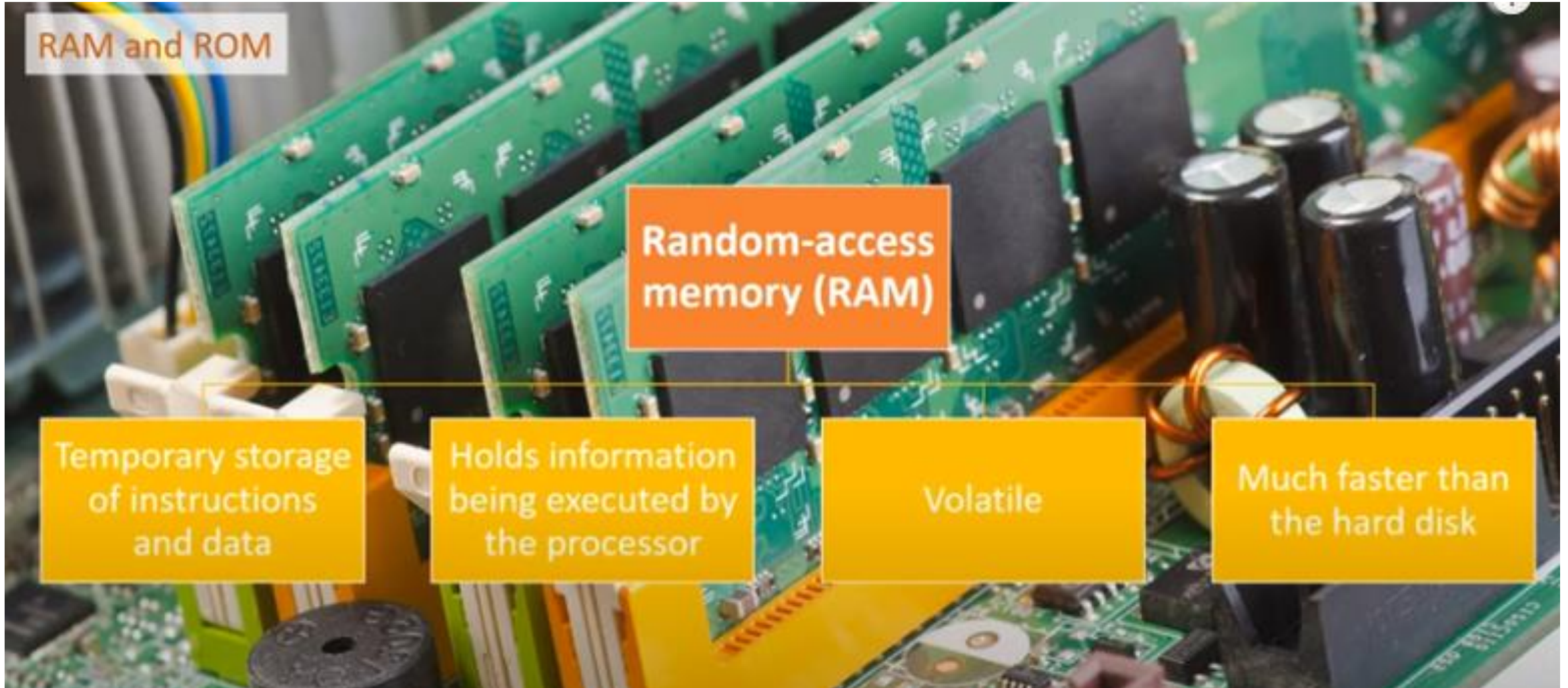
3.3.1: Primary Storage (ROM)

Practical Activities: Try to explore BIOS Command on a MacBook or Window 10

PhoenixBIOS Setup Utility			
Main	Advanced	Security	Boot Exit
System Time: [09:21:30] System Date: [09/02/2016] Legacy Diskette A: [1.44/1.25 MB 3½"] Legacy Diskette B: [Disabled] ▶ Primary Master [None] ▶ Primary Slave [None] ▶ Secondary Master [CD-ROM] ▶ Secondary Slave [None] ▶ Keyboard Features System Memory: 640 KB Extended Memory: 2096128 KB Boot-time Diagnostic Screen: [Enabled]			Item Specific Help <Tab>, <Shift-Tab>, or <Enter> selects field.
F1 Help	↑↓ Select Item	-/+ Change Values	F9 Setup Defaults
Esc Exit	↔ Select Menu	Enter Select ▶ Sub-Menu	F10 Save and Exit

3.3: Data storage

3.3.1: Primary Storage (RAM)



3.3: Data storage

3.3.1: Primary Storage (RAM)

Ram is used to store data / programs that are currently i



The Operating System - e.g. Windows

The software in use - e.g. Photoshop, Excel, Chrome

The data which the software is using - e.g. cookies, graphics

When a program is no longer being used, for example a web browser is closed, it is then removed from RAM to make space for another program.

The features of **RANDOM ACCESS MEMORY (RAM)** are:

- it is volatile/temporary memory (this means that when the power is lost, any data being stored on it is lost.
- it can be written to or read from and the contents of the memory can be changed
- In general, the larger the size of RAM the faster the computer will operate.

3.3: Data storage

3.3.1: Primary Storage (RAM)

DRAMs have a number of advantages over SRAMs:

- they are much less expensive to manufacture than SRAM
- they consume less power than SRAM
- they have a higher storage capacity than SRAM.

Static RAM (SRAM):

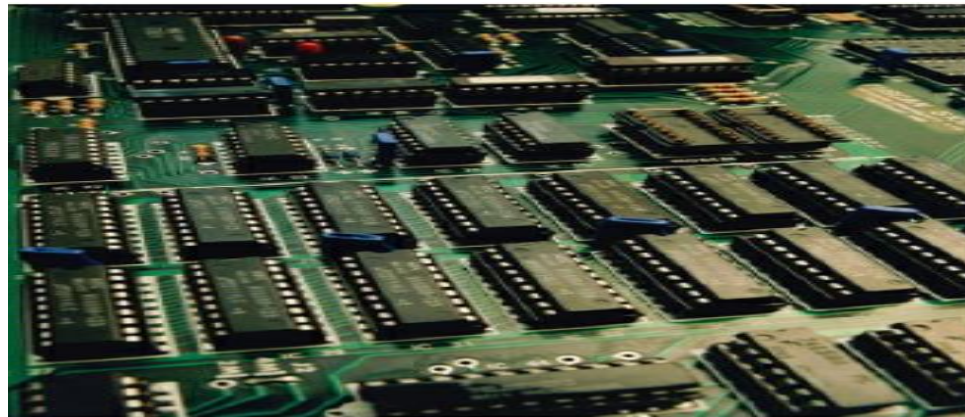
- SRAM is much faster than DRAM when it comes to data access (typically, access time for SRAM is 25 nanoseconds and for DRAM is 60 nanoseconds).
- DRAM is the most common type of RAM used in computers, but where absolute speed is essential, then SRAM is the preferred technology. For example, the processor's **MEMORY CACHE** is the high speed portion of the memory; it is effective because most programs access the same data or instructions many times. By keeping as much of this information as possible in SRAM, the computer avoids having to access the slower DRAM.

3.3: Data storage

3.3.1: Primary Storage (RAM)

Static RAM (SRAM):

- SRAM is much faster than DRAM when it comes to data access (typically, access time for SRAM is 25 nanoseconds and for DRAM is 60 nanoseconds).
- There is also **static RAM (SRAM)**, which does not have to be refreshed. Although faster than DRAM, **SRAM uses** more transistors and is thus more costly; it is **used** primarily for CPU internal registers and cache **memory**.



3.3: Data storage

3.3.1: Primary Storage (RAM)

Difference between SRAM and DRAM

SRAM	DRAM
SRAM stands for Static Random Access Memory	DRAM stands for Dynamic Random Access memory
Make use of 6 transistors for each memory cell	Make use of single transistor for each memory cell
Present on Processors or between main memory and processor	Present on motherboard
Size:1MB to 16MB	Size:1GB to 2GB smart phones and tablets 4GB to 16GB in laptops
Faster	Relatively slow
Do not need to be re-energized	Need to be re-energized constantly
SRAM accesses directly	Accessing speed depends on clock
Expensive	Relatively cheaper

3.3: Data storage

3.3.1: Primary Storage (RAM)

RAM and ROM



RAM



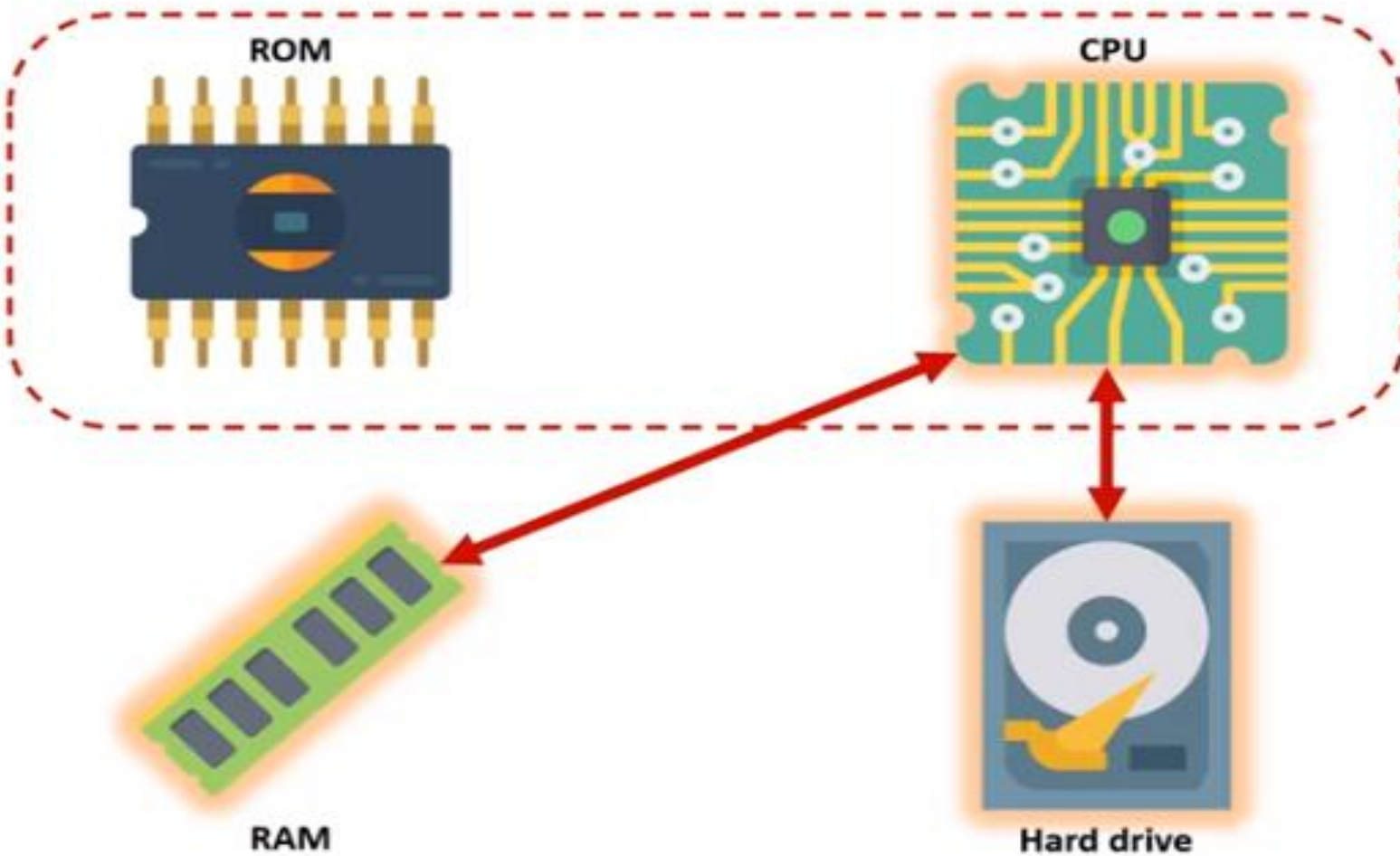
Hard drive

- The solution to this problem is ROM.
- A set of initial startup instructions called the **bootstrap** is placed on the ROM during manufacture.
- ROM is non-volatile, which means that the data it contains is retained when the computer is powered off.
- The first instruction in the bootstrap tells the computer to perform a **power-on self-test (POST)**.
- The POST signals all connected components, wakes them up and makes the CPU aware of them.

3.3: Data storage

3.3.1: Primary Storage (RAM)

RAM and ROM



- As the computer is now aware of the hard drive, it can load the operating system into RAM.
- The process of booting up the operating system can now begin.

3.3: Data storage

3.3.1: Primary Storage (RAM)

Application:

We will now consider an application, other than a computer, where both RAM and ROM chips are used:

A remote-controlled toy car has a circuitry which contains both RAM and ROM chips. The remote control is a hand-held device.

We will consider the function of each type of memory independently:

- ROM

- stores the factory settings such as remote control frequencies
- stores the 'start-up' routines when the toy car is first switched on
- stores the set routines; for example, how the buttons on the hand-held device control turning left, acceleration, stopping, and so on.

- RAM

- the user may wish to program in their own routines; these new instructions would be stored in the RAM chip
- the RAM chip will store the data/instructions received from the remote control

3.3: Data storage

3.3.1: Primary Storage (RAM)

RAM and ROM

Primary storage: Any storage that can be directly accessed by the CPU – this includes:

- Random-access memory (RAM)
- Read-only memory (ROM)
- Registers and cache

This type of storage holds the data and instructions that the CPU needs to access while the computer is running. The CPU can access data from primary storage much faster than secondary storage.

Random-access memory (RAM)

- Holds programs and data currently in use by the CPU.
- Volatile – contents are lost when the computer is powered off.
- Read and write.
- Larger than ROM.

Read-only memory (ROM)

- Holds startup instructions for the computer, known as the bootstrap.
- In embedded systems, programs may be stored in ROM.
- Non-volatile – contents are retained when the computer is powered off.
- Read-only.
- Smaller than RAM.



3.3: Data storage

3.3.2: Secondary storage

<https://www.youtube.com/watch?v=61u6e1onxyE&t=6s>



3.3: Data storage

3.3.2: The need for secondary storage

https://www.youtube.com/watch?v=FNwA-h_tfPo



3.3: Data storage

3.3.2: Secondary storage



Secondary storage

Secondary storage is required because **ROM** is read-only and **RAM** is volatile. Secondary storage is used for:

- Storage of programs and data when the computer is powered off.
- Modifiable, semi-permanent storage of data.
- Backing up or archiving data.

3.3: Data storage

3.3.3: Secondary and off-line storage

Common types of secondary storage

https://www.youtube.com/watch?v=DMq_DV1kWuA



3.3: Data storage

3.3.3: Secondary and off-line storage

Common types of secondary storage

Secondary storage

Optical (CD-R/RW, DVD-R/RW, Blu-ray)

- Low capacity compared to other types of storage.
- Slower access speeds.
- Thin, lightweight and portable.

Magnetic (hard disk drive, tape)

- High storage capacity.
- Faster access speeds.
- Has moving parts that will eventually fail.
- Hard disks perform better if they are defragmented.

Solid-state (SSD, memory stick, flash memory card)

- Medium storage capacity.
- Fastest access speeds.
- No moving parts, so they are very reliable.
- No noise.
- Low power usage.
- Eliminates the need for defragmentation.
- Limited number of write cycles.
- Expensive compared to other types of storage.

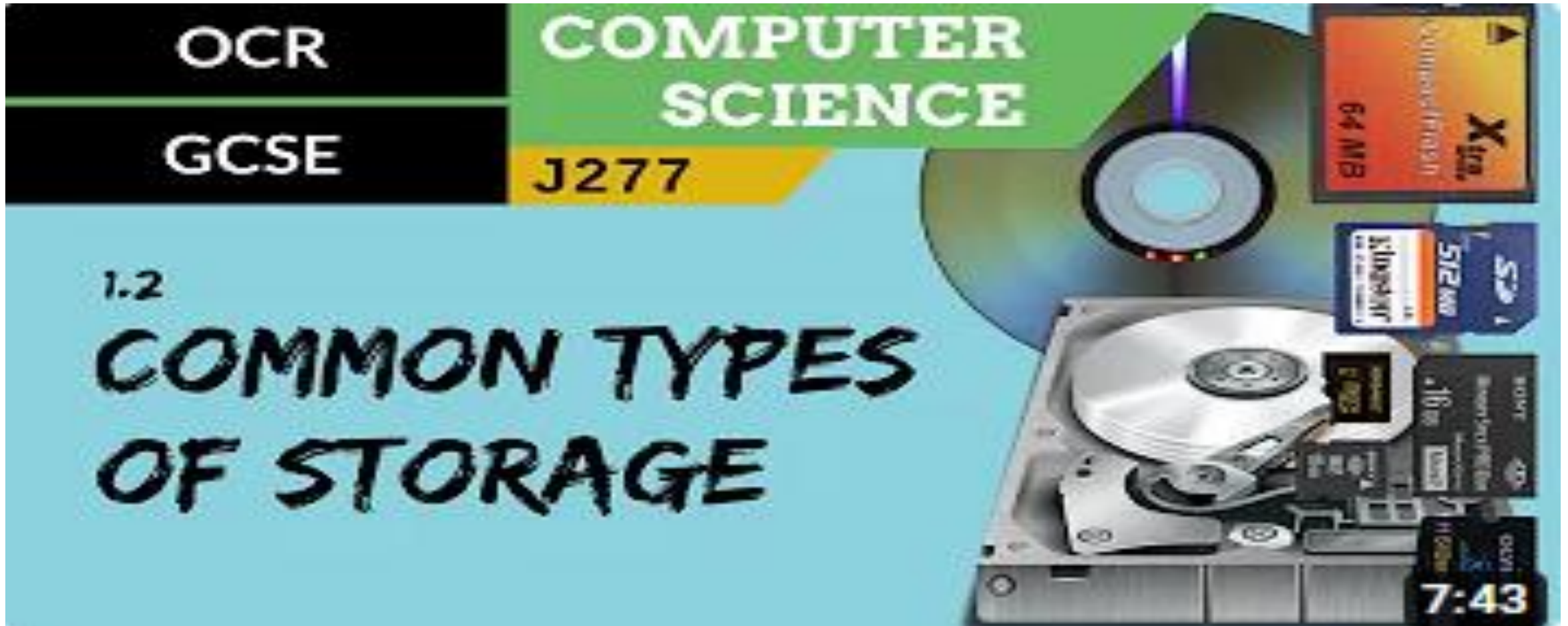


3.3: Data storage

3.3.3: Secondary and off-line storage

Common types of secondary storage

https://www.youtube.com/watch?v=qly_wgo03Oo



3.3: Data storage

3.3.3: Secondary and off-line storage

Suitable storage devices

<https://www.youtube.com/watch?v=xfDwcdap5LA>



3.3: Data storage

3.3.3: Secondary and off-line storage

Secondary memory can be thought of as the computers long term memory, it stores the users files and programs whilst they are not being used. Secondary memory is also **non-volatile** meaning that it will not lose its data when power is lost.

There two main forms of secondary storage that we will look at. These are:

Hard Disk Drive (HDD)



Solid state drive (SSD)



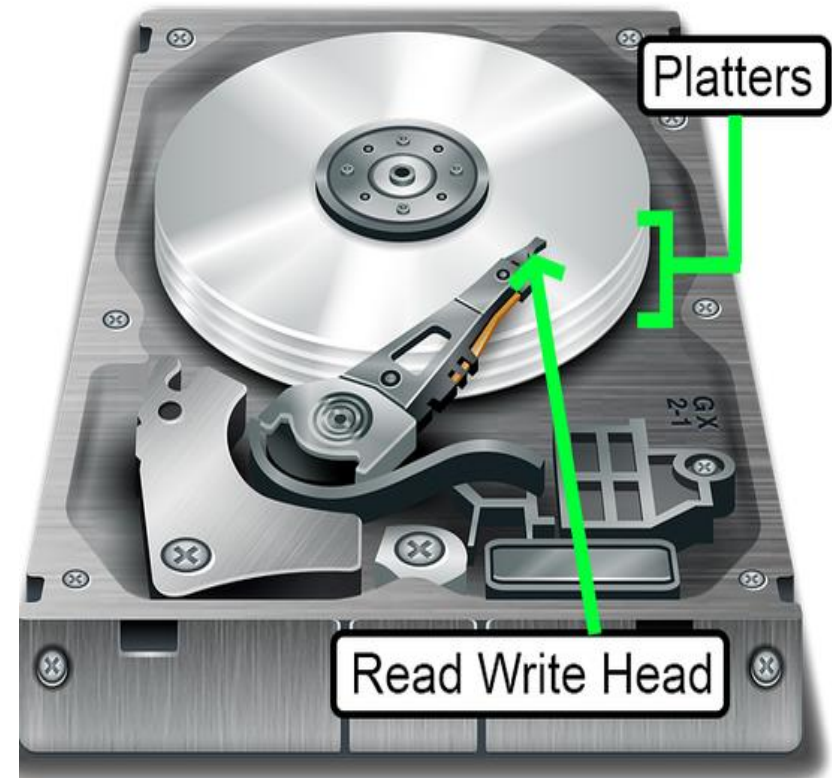
3.3: Data storage

3.3.3: Secondary and off-line storage

Hard Disk Drives (HDD)

Hard disk drives are a form of **magnetic storage** and they are currently the most popular method used to store data on a computer.

Inside the HDD, data is stored on circular, **magnetic surfaces called platters**.



3.3: Data storage

3.3.3: Secondary and off-line storage

Hard Disk Drives (HDD)

Each hard drive can contain multiple platters just like in the image above. These **platters can spin** around at roughly **7200 times a minute**. The HDD also has a **read write head**, this head is responsible for reading data and writing data to the platters. The read write head can move from the center of the platter to the edge at 50 times per second. This gives the read write head access to all parts of the disk.

Spinning 7200 times a minute sounds very fast! but remember, access to data on a HDD is still **slower than RAM** hence the need to store programs currently being used in RAM. The main reason why HDDs are slower than RAM is **latency**. When data is accessed from a hard drive the read write head will constantly look for the correct blocks of data, as the disk is also **constantly spinning** it means that the head will have to be constantly moving and cannot have constant access to the data.

3.3: Data storage

3.3.3: Secondary and off-line storage

Hard Disk Drives (HDD)

This slows down access to the data and is defined as **latency**... latency is the amount of time it takes for a specified block of data to **rotate around to the read write head**, obviously whilst the data is rotating away from the head, it cannot be written or read.

Structure of the Platters

Data on each platter is organised in a way that allows the read write head to easily find the required data. The platters are divided into tracks and sectors.

Track - A single circular area of the disk that works all the way around the surface. Tracks are concentric meaning that they do not spiral, instead they are separate circular entities.

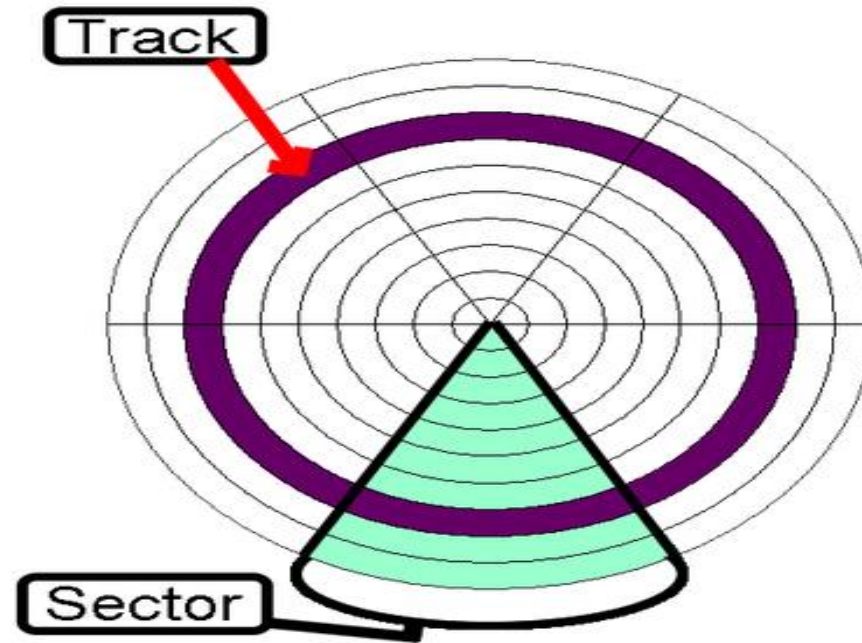
3.3: Data storage

3.3.3: Secondary and off-line storage

Hard Disk Drives (HDD)

Sector - A specific portion of the platter which splits each track

For a better understanding please see the below diagram:



3.3: Data storage

3.3.3: Secondary and off-line storage

Hard Disk Drives (HDD)

Magnetic storage key points:

Advantages

- HDDs are cheap
- HDDs can store large amounts of data
- They have a relatively fast write speed

Disadvantages

- There are a lot of mechanical parts (Which can break)
- Not very portable

Capacity

- Large capacity
- Between 250Gb and 8Tb

Uses

Personal computers
Laptops
Backups

3.3: Data storage

3.3.3: Secondary and off-line storage

Solid State Drives

Solid state drives have the same purpose as HDDs in that they provide long term storage of a user's files and programs even when power is lost - They are also nonvolatile.

The main difference between SSD and HDD is that SSDs have no moving / mechanical parts as a result, the read / write process is a lot faster than with HDDs.



3.3: Data storage

3.3.3: Secondary and off-line storage

Solid State Drives (SSD)

- Latency is an issue in HDDs as described earlier. Solid state drives (SSD) remove this issue considerably since they have no moving parts and all data is retrieved at the same rate. They don't rely on magnetic properties; the most common type of solid state storage devices store data by controlling the movement of electrons within NAND or NOR chips. The data is stored as 0s and 1s in millions of tiny transistors (at each junction one transistor is called a floating gate and the other is called a control gate within the chip. This effectively produces a nonvolatile rewritable memory.

3.3: Data storage

3.3.3: Secondary and off-line storage

So what are the main benefits of using SSD rather than HDD? The main benefits of SSDs are summarised below:

- they are more reliable (no moving parts to go wrong)
- they are considerably lighter (which makes them suitable for laptops)
- they don't have to 'get up to speed' before they work properly
- they have a lower power consumption
- they run much cooler than HDDs (these last two points again make them very suitable for laptop computers)
- because they have no moving parts, they are very thin
- data access is considerably faster than HDD.

3.3: Data storage

3.3.3: Secondary and off-line storage

Solid State Drives

The main drawback of SSD is still the longevity of the technology (although this is becoming less of an issue). Most solid state storage devices are conservatively rated at only 20 GB of write operations per day over a three year period – this is known as **SSD endurance**. For this reason, SSD technology is still not used in all servers, for example, where a huge number of write operations take place every day. However, the durability of these solid state systems is being improved by a number of manufacturers and they are rapidly becoming more common in applications such as servers and **cloud storage** devices.

3.3: Data storage

3.3.3: Secondary and off-line storage

Solid State Drives

SSD Key points

Advantages

- More durable (no moving parts)
- Very fast (no moving parts)
- Faster start up time (No Latency)
- Less noise
- Smaller and less portable

Disadvantages

- More expensive
- Less storage capacity

Capacity

50Gb to 1Tb

Uses

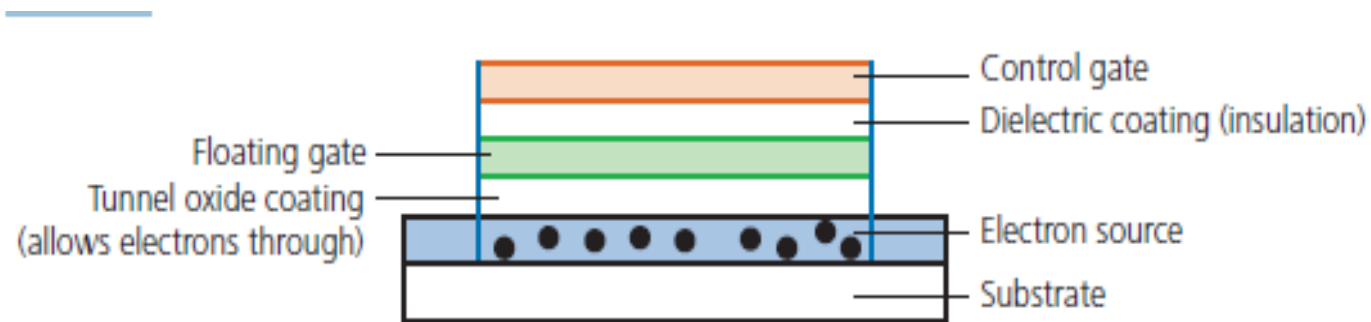
- Mobile devices
- Additional PC storage

3.3: Data storage

3.3.3: Secondary and off-line storage

Floating gate and control gate transistors

Floating gate and control gate transistors use CMOS (complementary metal oxide semi-conductor) NAND technology. Flash memories make use of a matrix; at each intersection on the matrix there is a floating gate and a control gate arranged as follows:



3.3: Data storage

3.3.3: Secondary and off-line storage

Floating gate and control gate transistors

A dielectric coating separates the two transistors, which allows the floating gate transistor to retain its charge (which is why the memory is non-volatile). The floating gate transistor has a value of 1 when it is charged and a value of 0 when it isn't. To program one of these 'intersection cells' a voltage is applied to the control gate and electrons from the electron source are attracted to it. But due to the dielectric coating, the electrons become trapped in the floating gate. Hence, we have control over the bit value stored at each intersection. (**Note:** After about 12 months, this charge can leak away, which is why a solid state device should be used at least once a year to be certain it will retain its memory.)

3.3: Data storage

3.3.3: Secondary and off-line storage

Offline storage is any data storage device that can hold data and is not permanently connected to a computer.

Offline storage is great for backing up data as it can be taken away and kept in a secure location, it is also good for transporting data from one device to another.

Examples of offline storage are:

USB Flash Drives

CDs / DVDs / BlueRay

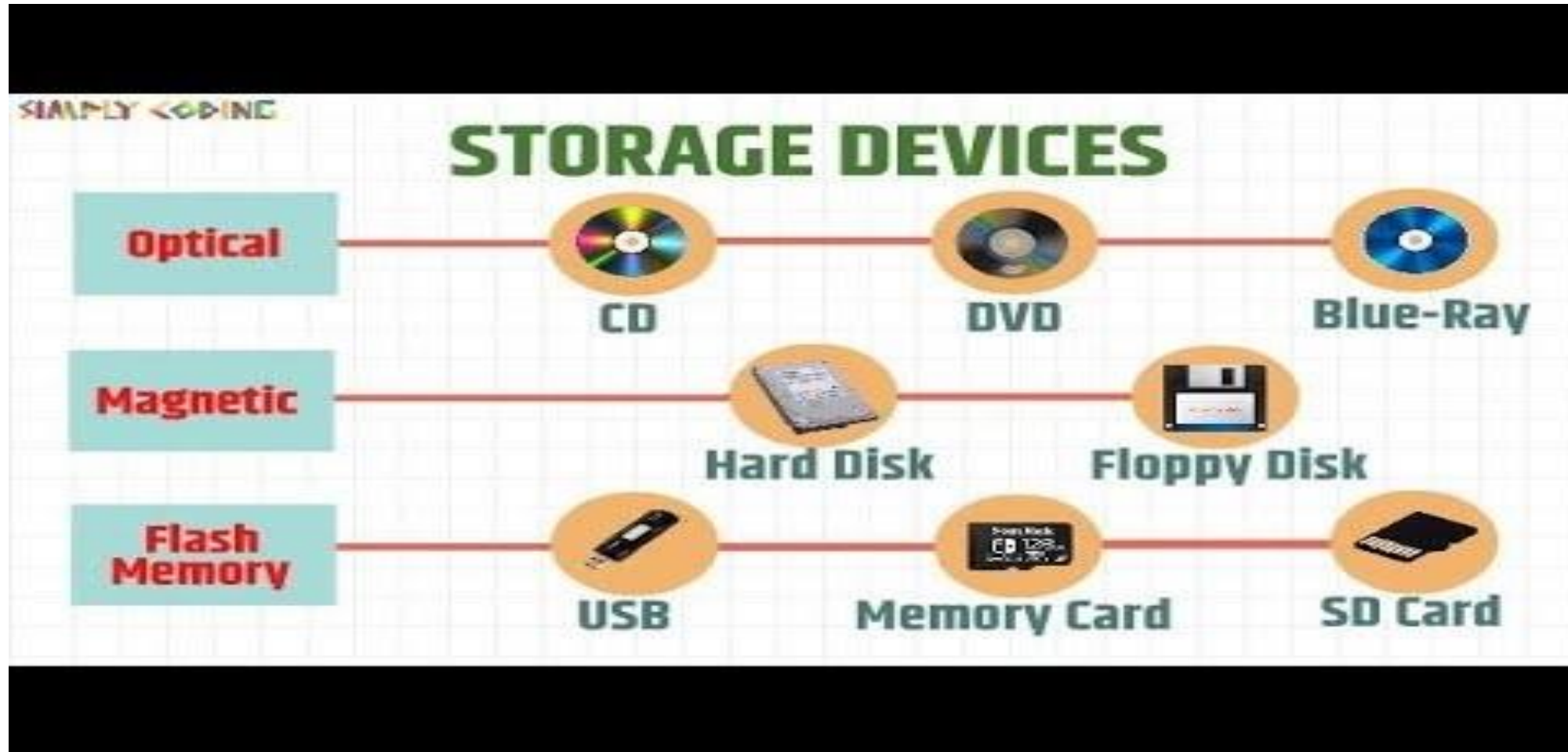
Magnetic Tapes

Portable HDDs

3.3: Data storage

3.3.3: Secondary and off-line storage

<https://www.youtube.com/watch?v=OsEDJM9NuGA>

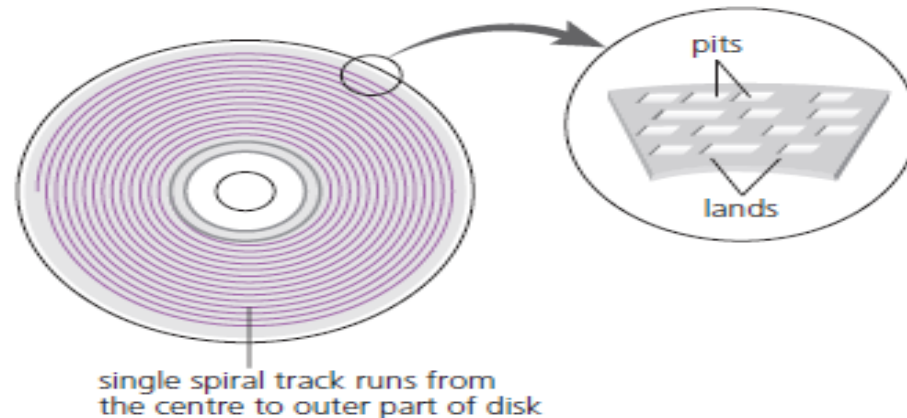
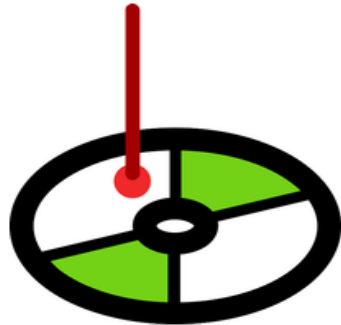


3.3: Data storage

3.3.3: Secondary and off-line storage

Optical Storage - CD / DVD disks:

CDS and **DVDS** are described as **OPTICAL STORAGE DEVICES**. Laser light is used to read data and to write data in the surface of the disk. Both CDs and DVDs use a thin layer of metal alloy or light-sensitive organic dye to store the data.



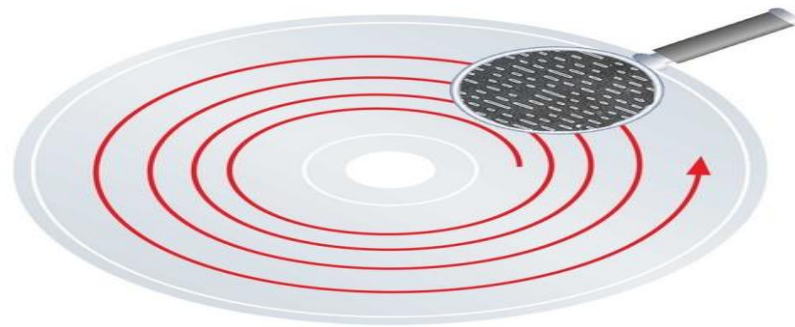
On these disks **data is stored in pits and lands or cracks** that are **burned** into a **spiral track** that circles outwards from the center of the disk.

3.3: Data storage

3.3.3: Secondary and off-line storage

Optical Storage - CD / DVD disks:

In the diagram below you can see that the **spiral track** is illustrated with a red line. If you were to look at this disk through a magnifying disk you would see what is visible in the magnifying glass. Here you can see the **actual pits and lands (cracks)** that are **burned on to the disk with light**.



DVD technology is slightly different to that used in CDs. One of the main differences is the use of **DUAL-LAYERING** which considerably increases the storage capacity. Basically, this means that there are two individual recording layers.

3.3: Data storage

3.3.3: Secondary and off-line storage

Optical Storage - CD / DVD disks :

Reading and writing to a disk

It has already been mentioned that data is written to an optical disk with a laser which leaves pits and lands in the disk.

To read from the disk a **laser is shone (shine) at the tracks of the disk**. The laser will **reflect** from the track and direct the **reflected light into a sensor**. The **intensity** of the reflected light will determine whether it is **interpreted as a 1 or a 0**. Technically speaking, if the **laser hits a pit**, the light does not reach the sensor and is **recorded as a 1**, if the light **hits a land**, it will reflect into the sensor and is **recorded as a 0**.

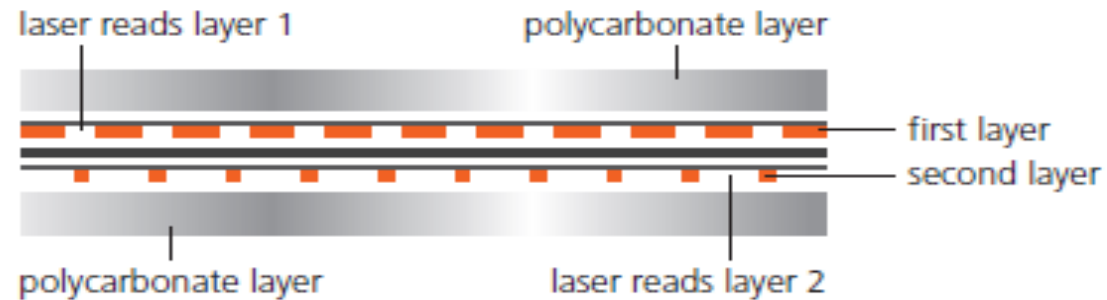
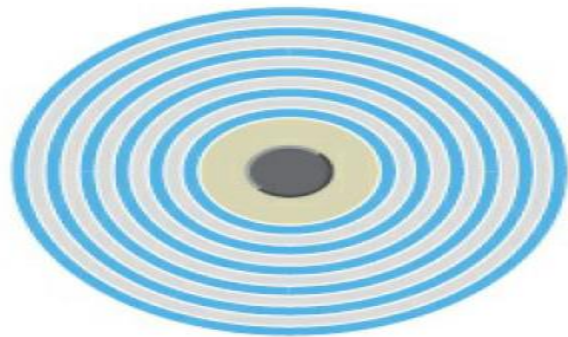
3.3: Data storage

3.3.3: Secondary and off-line storage

Optical Storage - DVD-RAM:

DVD-RAM uses a very different technology to CDs and DVDs. They have the following features:

- Instead of a single, spiral track, they use a number of concentric tracks see figure below:



- Use of concentric tracks allows simultaneous read and write operations to take place.
- They allow numerous read and write operations (up to 100 000 times) and have great longevity (over 30 years) which makes them ideal for archiving.

3.3: Data storage

3.3.3: Secondary and off-line storage

Optical Storage - Blue-ray disks:

BLUE-RAY DISKS are another example of optical storage media. However, they are fundamentally different to DVDs in their construction and in the way they carry out read–write operations.



3.3: Data storage

3.3.3: Secondary and off-line storage

Optical Storage - Blue-ray disks:

The main differences are:

- a blue laser, rather than a red laser, is used to carry out read and write operations; the wavelength of blue light is only 405 nanometres (compared to 650 nm for red light)
- using blue laser light means that the 'pits' and 'bumps' can be much smaller; consequently, Blu-ray can store up to five times more data than normal DVD
- Blu-ray uses a single 1.1 mm thick polycarbonate disk; normal DVDs use a sandwich of two 0.6 mm thick disks.
- Blu-ray disks automatically come with a secure encryption system which helps to prevent piracy and copyright infringement.

3.3: Data storage

3.3.3: Secondary and off-line storage

Summarises the main differences between CDs, DVDs and Blu-ray.

	DVD	HD-DVD	Blu-ray
Storage (single/dual layer)	4.7GB/8.5GB	15GB/30GB	25GB/50GB
Laser Type	650nm (red laser)	405 nm (blue-violet laser)	405 nm (blue-violet laser)
Disc Diameter	120 mm	120 mm	120 mm
Disc Thickness	1.2 mm	1.2 mm	1.2 mm
Video Resolution (maximum)	720 × 480 / 720 × 576 (480i/576i)	1920 × 1080 (1080p)	1920 × 1080 (1080p)
Video Codecs	MPEG-1 MPEG-2	MPEG-2 MPEG-4 AVC SMPTE VC-1	MPEG-2 MPEG-4 AVC SMPTE VC-1
Audio Codecs	Linear PCM Dolby Digital DTS Digital Surround	Linear PCM Dolby Digital Dolby Digital Plus Dolby True HD DTS Digital Surround DTS-H	Linear PCM Dolby Digital Dolby Digital Plus Dolby True HD DTS Digital Surround DTS-H
Equipment Needed to Burn Discs	DVD burner like DVD Creator Blank DVD-R or DVD-RW discs	DVD burner Blank HD-DVD-R or HD-DVD-RW discs	Blu-ray burner Blank BDR or BDRE discs
Equipment Needed to View Discs	DVD or Blu-ray player	Blu-ray player	Blu-ray player only



(Note: nm = 10^{-9} metres and μm = 10^{-6} metres.)

3.3: Data storage

3.3.3: Secondary and off-line storage

Comparison of CD, DVD and Blu-ray (Note: nm = 10^{-9} metres and μm = 10^{-6} metres)

Disk type	Laser colour	Wavelength of laser light	Disk construction	Track pitch (distance between tracks)
CD	Red	780 nm	single 1.2 mm polycarbonate layer	1.60 μm
DVD (dual-layer)	Red	650 nm	two 0.6 mm polycarbonate layers	0.74 μm
Blu-ray (single layer)	Blue	405 nm	single 1.2 mm polycarbonate layer	0.30 μm
Blu-ray (dual-layer)	Blue	405 nm	two 0.6 mm polycarbonate layers	0.30 μm

(Note: nm = 10^{-9} metres and μm = 10^{-6} metres.)

3.3: Data storage

3.3.3: Secondary and off-line storage

USB flash memories:

MEMORY STICKS/FLASH MEMORIES (also known as pen drives) use solid-state technology. USB usually connect to a computer through the USB port.

- Main advantage is that they are very small, lightweight devices which make them suitable as a method for transferring files between computers.
- Digital cameras use a slightly different form of solid-state memory, known as XD (eXtreme Digital) or SD (Secure Digital) cards).



3.3: Data storage

3.3.3: Secondary and off-line storage

USB flash memories:

- All solid-state memories need to be treated with some care. If they are removed from a device before being made safe (for example, while still in the middle of a read–write operation), data on the memory card or stick can be corrupted. In extreme cases, the memory card or stick become totally unusable. It is essential to wait for the message: it is now safe to remove your device if plugged into a computer.



3.5.4: Off-line storage

Removable hard disk drives :

- **REMOVABLE HARD DISK DRIVES** are essentially HDD but can be connected to the computer using one of the USB ports. In this way, they can be used as a back-up device or as another way of transferring files between computers.



IGCSE Question without answers therefore you need
to find all the answers below using the past exam
papers

IGCSE Question 1

(c) Explain **three** ways that RAM is different to ROM.

1

.....

.....

2

.....

.....

3

.....

.....

[3]

IGCSE Exam Question 2

A computer uses RAM and ROM to store data.

(a) The table contains three statements about RAM or ROM.

Tick (✓) to show whether each statement describes **RAM** or **ROM**.

Statement	RAM (✓)	ROM (✓)
Stores the programs and data that are currently in use		
Used to boot up the computer when power is turned on		
Contents are retained when power is turned off		

[3]

IGCSE Question 3

(b) A microprocessor uses ROM.

Explain what is meant by ROM.

.....

.....

.....

.....

.....

..... [3]

IGCSE Exam Question 4

2 A computer uses RAM and ROM to store data.

(a) The table contains three statements about RAM or ROM.

Tick (✓) to show whether each statement describes **RAM** or **ROM**.

Statement	RAM (✓)	ROM (✓)
Stores the programs and data that are currently in use		
Used to boot up the computer when power is turned on		
Contents are retained when power is turned off		

[3]

IGCSE Exam Question 5

- 9 Describe **two** differences between Read Only Memory (ROM) and Random Access Memory (RAM).

Difference 1

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.....

.....

Difference 2

.....

.....

.....

[4]

IGCSE EXAM: Question 6

Identify **three** similarities between CDs and DVDs.

1

.....

2

.....

3

.....

[3]

IGCSE EXAM: Question 7

- 9 (a) Optical storage media can be used to store data.

Describe how the data is read from a Compact Disc (CD).

.....

.....

.....

.....

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.....

.....

.....[4]

IGCSE EXAM: Question 8

Primary, secondary and off-line are types of storage.

Give an example of each type of storage.

For each example state how it is used.

Primary storage

Example

Use

.....

Secondary storage

Example

Use

.....

Off-line storage

Example

Use

.....

IGCSE EXAM: Question 9

The supermarket uses secondary storage and off-line storage to store data about its stock.

Explain what is meant by secondary storage and off-line storage.

Secondary storage

.....

.....

.....

Off-line storage

.....

.....

.....

[4]

IGCSE EXAM: Question 10

- 5 (a) **Five** storage devices or media are listed in the table.

Tick (✓) to show whether each storage device or media is an example of **primary**, **secondary** or **off-line** storage.

Storage device or media	Primary (✓)	Secondary (✓)	Off-line (✓)
External HDD			
RAM			
Internal SSD			
ROM			
DVD			

[5]

IGCSE EXAM: Question 11

(b) Users can store their data on optical storage media.

Explain how data is written to optical storage media.

.....

.....

.....

.....

.....

.....

.....

..... [4]

IGCSE EXAM: Question 12

(b) Circle the storage category that includes both RAM and ROM.

Primary

Secondary

Off-line

[1]

(c) Explain what is meant by off-line storage.

.....

.....

.....

.....[2]

IGCSE EXAM: Question 13

(c) Describe the differences between primary and secondary storage.

.....

.....

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.....

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.....[4]

IGCSE EXAM: Question 14

- 2 Storage devices and storage media can be categorised as primary, secondary or off-line.

Write **primary**, **secondary** or **off-line** next to each storage device or medium to indicate its most suitable category.

HDD

RAM

ROM

CD-ROM

SSD

DVD-RAM

[6]

IGCSE EXAM: Question 15

8 Complete the paragraph by choosing **six** correct terms from the list.

- Optical
- On-line
- RAM
- HDD
- Primary
- SSD
- Secondary
- ROM
- Off-line

A computer has two different types of memory. memory is not directly accessed by the CPU, but it allows a user to store data that can easily be accessed by applications.

Two examples of this type of memory are and The second type of memory is memory. This memory is directly accessed by the CPU. It allows the processor to access data and instructions that are stored in this memory.

Two examples of this memory are and

[6]

IGCSE EXAM: Question 16

- 13 (a) Explain what is meant by primary, secondary and off-line storage. Give an example of each.

Primary storage

.....

.....

Example

Secondary storage

.....

.....

Example

Off-line storage

.....

.....

Example

IGCSE EXAM: Question 18

5 Five storage devices are described in the table below.

In column 2, name the storage device being described.

In columns 3, 4, or 5, tick (✓) to show the appropriate category of storage.

1	2	3	4	5
Description of storage device	Name of storage device	Category of storage		
		Primary	Secondary	Off-line
optical media which use one spiral track; red lasers are used to read and write data on the media surface; makes use of dual-layering technology to increase the storage capacity				
non-volatile memory chip; contents of the chip cannot be altered; it is often used to store the start up routines in a computer (e.g. the BIOS)				
optical media which use concentric tracks to store the data; this allows read and write operations to be carried out at the same time				
non-volatile memory device which uses NAND flash memories (which consist of millions of transistors wired in series on single circuit boards)				
optical media which use blue laser technology to read and write data on the media surface; it uses a single 1.1 mm polycarbonate disc				

IGCSE EXAM: Question 19

- (c) A sports events company uses a digital camera attached to a drone (small flying system), to video their events from the sky.

The video is stored as it is captured, on a device that is attached to the drone.

- (i) Circle the most suitable type of storage to store the video.

Optical

Magnetic

Solid state

[1]

- (ii) Explain the reasons for your choice in **part (c)(i)**.

.....

.....

.....

..... [2]

IGCSE EXAM: Question 20 with Answer

The law company also uses optical storage.

Give **three** different examples of optical storage.

- 1
 - DVD
- 2
 - CD
- 3
 - Blu-ray

IGCSE EXAM: Question 21 with Answer

(a) Marley wants to store a video he has created for his school project.

He considers using a DVD or a Blu-ray to store the video.

Explain **two** differences between a DVD and a Blu-ray.

1 Two from e.g.:

- DVD uses red laser/light whereas Blu-ray uses blue/violet laser/light
- - DVD has a **smaller** (storage) capacity // Blu-ray has a **larger** (storage) capacity
- DVD has **two layers** (of polycarbonate) whereas Blu-ray disks have a **single layer** (of polycarbonate)
- - DVD has a **slower transfer rate** (of approximately 10 mbps) // Blu-ray has a **faster transfer rate** (of approximately 36 mbps)

3.3: Data storage

3.3.4: Virtual memory

<https://www.youtube.com/watch?v=M31SS70Od08>



3.3: Data storage

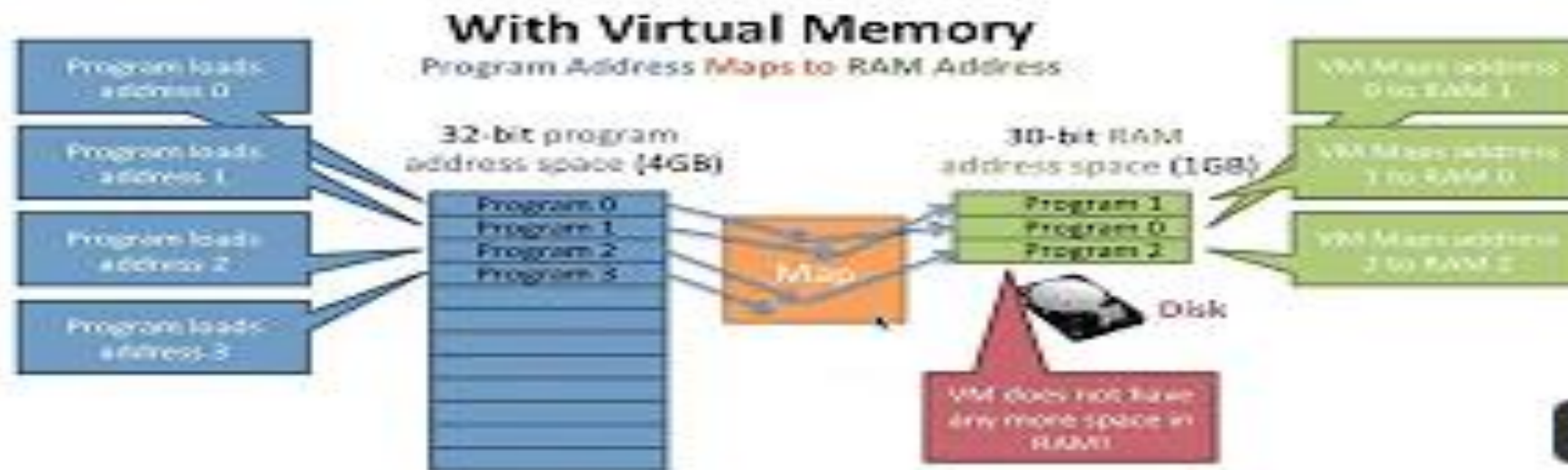
3.3.4: Virtual memory

What is Virtual Memory?

<https://www.youtube.com/watch?v=qlH4-oHnBb8>

Solving the problems: #1 not enough memory

- **Map** some of the program's address space to the **disk**
- When we need it, we bring it into memory



8:47

3.3: Data storage

3.3.4: Virtual memory

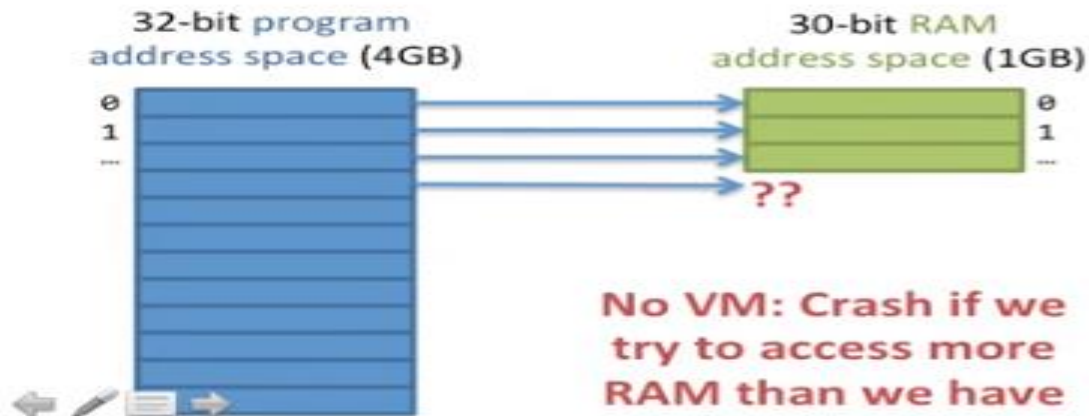
One of the problems associated with memory management is the case when processes run out of RAM. If the amount of available RAM is exceeded due to multiple programs running, it is likely to cause a system crash. This can be solved by utilising the hard disk drive (or SSD) if we need more memory. This is the basis behind **virtual memory**. Essentially RAM is the **physical memory**, while **virtual memory** is RAM + swap space on the hard disk or SSD. To execute a program, data is loaded into memory from HDD (or SSD) whenever required. It is possible to show the differences between using normal memory management and virtual memory management in two simple diagrams below.

“Any problem in computer science can be solved by adding indirection.”

Virtual memory takes **program addresses** and **maps** them to **RAM addresses**

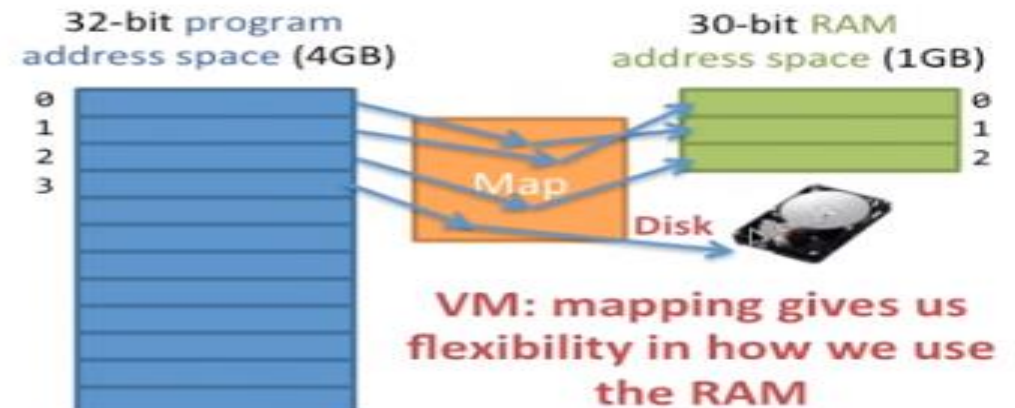
Without Virtual Memory

Program Address = RAM Address



With Virtual Memory

Program Address Maps to RAM Address



3.3: Data storage

3.3.4: Virtual memory



- Virtual memory is needed when there is not enough physical RAM to store the open programs.
- Virtual memory is held on the hard disk.
- Programs are transferred out to virtual memory from the RAM when they are not currently being executed.
- Programs are transferred back to RAM from virtual memory when they are needed.

3.3: Data storage

3.3.5:cloud storage

<https://www.youtube.com/watch?v=q4j8zE5LTGw>



By: Craig'n'Dave

3.3: Data storage

3.3.5:cloud storage



3.3: Data storage

3.3.5:cloud storage



3.3: Data storage

3.3.5:cloud storage

Benefits and drawbacks of cloud storage

Benefits of using cloud storage	Drawbacks of using cloud storage
customer/client files stored on the cloud can be accessed at any time from any device anywhere in the world provided internet access is available	if the customer/client has a slow or unstable internet connection, they would have many problems accessing or downloading their data/files
there is no need for a customer/client to carry an external storage device with them, or even use the same computer to store and retrieve information	costs can be high if large storage capacity is required; it can also be expensive to pay for high download/upload data transfer limits with the customer/client internet service provider (ISP)
the cloud provides the user with remote back-up of data with obvious benefits to alleviate data loss/disaster recovery	the potential failure of the cloud storage company is always possible – this poses a risk of loss of all back-up data
if a customer/client has a failure of their hard disk or back-up device, cloud storage will allow recovery of their data	
the cloud system offers almost unlimited storage capacity	

3.3: Data storage

3.3.5:cloud storage



3.3: Data storage

3.3.5:cloud storage



Cloud storage

Cloud storage: The concept of storing and retrieving data virtually via the internet instead of a local storage device.

Advantages

- Data can be accessed at any time from any device, providing it has internet access.
- Data can easily be shared without physical transfer (e.g., USB drive).
- Easy collaboration with shared documents
- Storage considered to be limitless from the user's perspective.

Disadvantages

- Can be expensive.
- Slow to access if connectivity is poor.
- No internet connection means no access.

All files kept in **cloud storage** are still stored on physical storage media in a remote location.

Cloud storage warehouses often have thousands of servers containing hard drives.

Learning Objectives

3.4 Network hardware

Candidates should be able to:

- 1 Understand that a computer needs a network interface card (NIC) to access a network
- 2 Understand what is meant by and the purpose of a media access control (MAC) address, including its structure
- 3 (a) Understand what is meant by and the purpose of an internet protocol (IP) address
(b) Understand that there are different types of IP address
- 4 Describe the role of a router in a network

Notes and guidance

- A network interface card is given a MAC address at the point of manufacture
- MAC addresses are usually written as hexadecimal
- MAC addresses are created using the manufacturer code and the serial code
- An IP address is allocated by the network and they can be static or dynamic
- Including the characteristics of and differences between IPv4 and IPv6
- A router sends data to a specific destination on a network
- A router can assign IP addresses
- A router can connect a local network to the internet

3.4 Network hardware

3.4.1 Network interface card (NIC)

https://www.youtube.com/watch?v=ebgld_QwrJw



3.4 Network hardware

3.4.1 Network interface card (NIC)

Standalone vs networked



A computing device that is physically unable to communicate with any other device is known as *standalone*.

3.4 Network hardware

3.4.1 Network interface card (NIC)

Standalone vs networked



As soon as we provide some mechanism for the device to communicate with another device, whether that be via a wired or wireless solution, we can say the device is *networked*.

3.4 Network hardware

3.4.1 Network interface card (NIC)

Network interface card (NIC)



Every device connecting to a network needs a **network interface card (NIC)**.

NICs are often also referred to as:

- Network interface controller
- Network adapter

A NIC uses a set of rules to determine how the connection should work

A NIC allows a device to connect to either a wired or wireless network

3.4 Network hardware

3.4.1 Network interface card (NIC)

Network interface card (NIC)



Early **NICs** were commonly implemented as expansion cards plugged into a slot in a computer's motherboard.

Today, the low cost and universal nature of the Ethernet standard means **NICs** are now built directly into the motherboard.

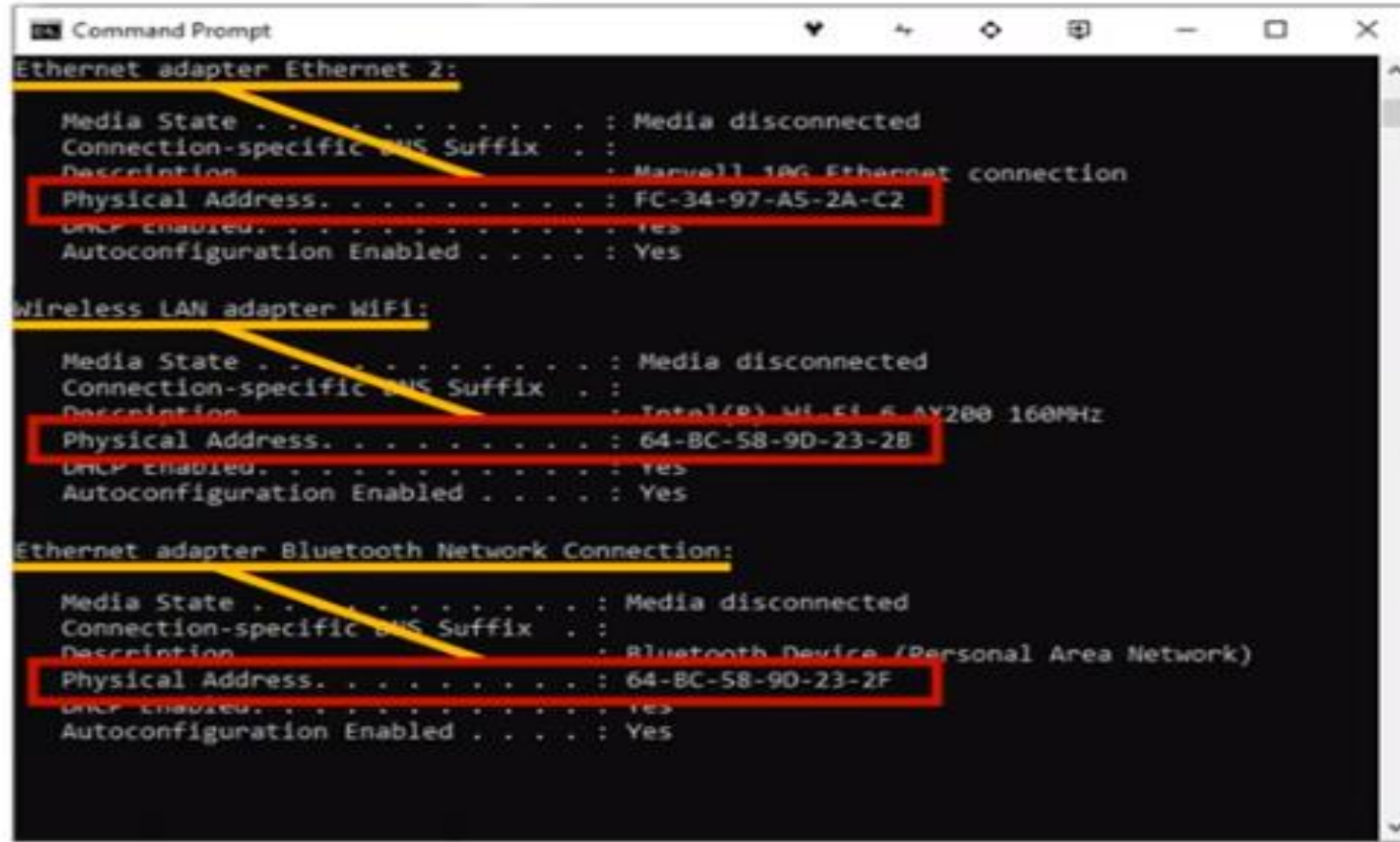
Wireless **NICs** (WNICs) allows a device to connect without cables – they can be:

- integrated into the device; or
- plugged in – e.g., a USB dongle.

3.4 Network hardware

3.4.1 Network interface card (NIC)

Network interface card (NIC)



```
Command Prompt
Ethernet adapter Ethernet 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . : 
Description . . . . . : Marvell 10G Ethernet connection
Physical Address. . . . . : FC-34-97-A5-2A-C2
Dhcp Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter WiFi:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . : 
Description . . . . . : Intel(R) Wi-Fi 6 AX200 160MHz
Physical Address. . . . . : 64-BC-58-9D-23-2B
Dhcp Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 64-BC-58-9D-23-2F
Dhcp Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

A NIC contains a **media access control (MAC)** address, generated at the manufacturing stage.

It is possible for a device to have multiple separate connections to a network via **NICs**.

Each connection is assigned its own unique **MAC address**.

3.4 Network hardware

3.4.1 Network interface card (NIC)



Network interface card (NIC)

- A **network interface card (NIC)** connects a device to a wired or wireless network.
- A **NIC** uses a protocol to ensure successful communication.
- A **NIC** contains a **media access control (MAC)** address, generated at the manufacturing stage.

3.4 Network hardware

3.4.2 Media Access Control (MAC)

<https://www.youtube.com/watch?v=ym7c3oAB1U4>

CAMBRIDGE
IGCSE

**COMPUTER
SCIENCE**
0478 | 0984

3.4
MAC
ADDRESS

The screenshot shows a Windows command prompt window titled "C:\WINDOWS\system32\cmd.exe". The user has entered the command "ipconfig /all". The output displays network configuration for the Ethernet adapter Ethernet0. The Physical Address (MAC) is highlighted with a red box and is 94-BE-6B-77-8D-4C. Other visible details include the IP Address 192.168.1.102, Subnet Mask 255.255.255.0, and Default Gateway 192.168.1.1.

```
C:\Users\Noureddine>ipconfig /all

Windows IP Configuration

Ethernet adapter Ethernet0:

   Connection-specific IP Settings: . . . . . : 
   Physical Address. . . . . : 94-BE-6B-77-8D-4C
   Internet Protocol Settings: . . . . . : 
   Link-local IPv6 Address . . . . . : fe80::94be6b778d4c%1
   Subnet Mask . . . . . : 255.255.255.0
   Lease Obtained. . . . . : 20/06/2015 23:08:53
   Lease Expires . . . . . : 20/06/2015 23:09:12
   IPv4 Address. . . . . : 192.168.1.102
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 192.168.1.1
   DHCP Server . . . . . : 192.168.1.1
   DHCPv6 IAID. . . . . : 261414128
   DHCPv6 Client ID. . . . . : 94-BE-6B-77-8D-4C
   DNS Servers . . . . . : 192.168.1.1
   NetBIOS over Tcpip. . . . . : Enabled
```

2:06

3.4 Network hardware

3.4.2 Media Access Control (MAC)

<https://www.youtube.com/watch?v=TliQiw7fpsU>

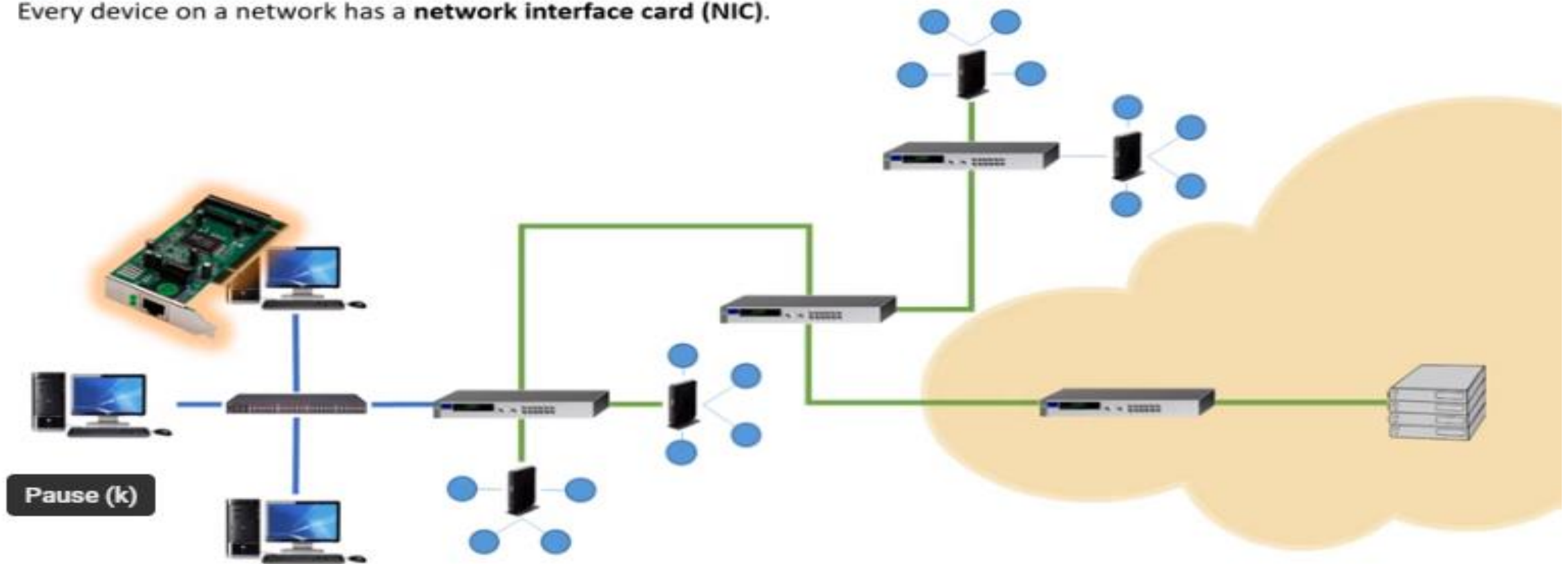


3.4 Network hardware

3.4.2 Media Access Control (MAC)

MAC addressing

Every device on a network has a **network interface card (NIC)**.



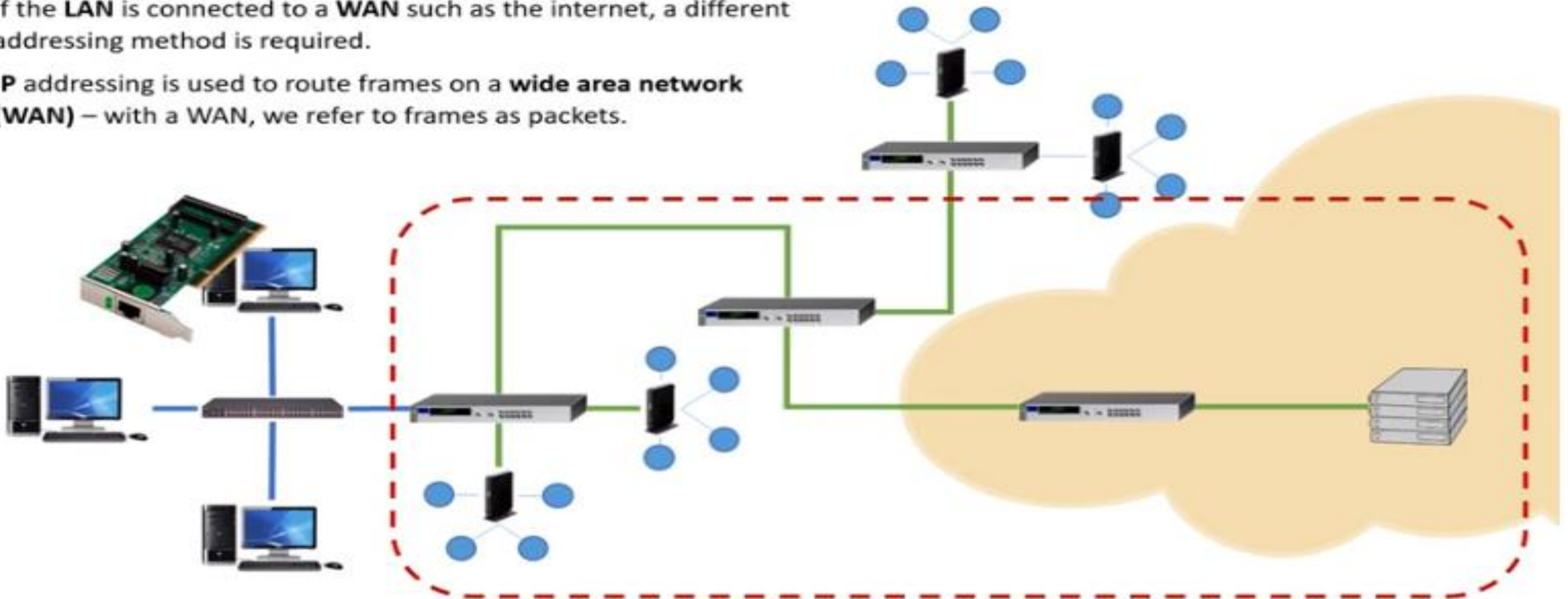
3.4 Network hardware

3.4.2 Media Access Control (MAC)

MAC addressing

If the **LAN** is connected to a **WAN** such as the internet, a different addressing method is required.

IP addressing is used to route frames on a **wide area network (WAN)** – with a WAN, we refer to frames as packets.



3.4 Network hardware

3.4.2 Media Access Control (MAC)

MAC addressing

While the **IP address** assigned to a device can change, the **MAC address** assigned to a **NIC** is:

- Unique
- Static
- Set by the manufacturer

If the **NIC** is replaced, the **MAC address** will also change.

2A – B3 – 19 – F6 – 82 – CD

The diagram shows a MAC address in red text: 2A – B3 – 19 – F6 – 82 – CD. An orange bracket is positioned under the first three pairs of hex digits (2A, B3, 19). A yellow bracket is positioned under the last three pairs of hex digits (F6, 82, CD).

The first six hex digits
(3 bytes) represent a
unique manufacturer code

The second six hex digits
(3 bytes) represent the unique
serial code for the device

3.4 Network hardware

3.4.2 Media Access Control (MAC)

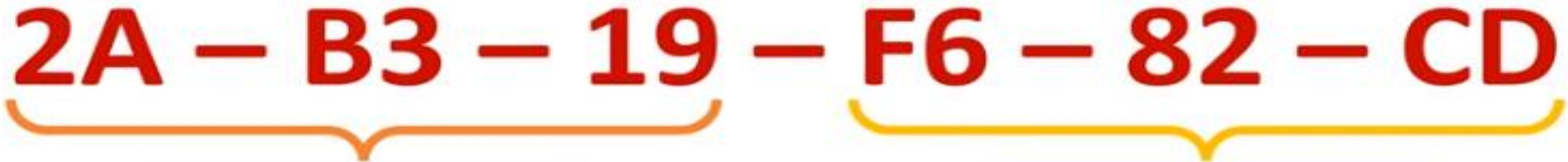
MAC addressing

While the **IP address** assigned to a device can change, the **MAC address** assigned to a **NIC** is:

- Unique
- Static
- Set by the manufacturer

If the **NIC** is replaced, the **MAC address** will also change.

2A – B3 – 19 – F6 – 82 – CD

The diagram shows a MAC address in red text: 2A – B3 – 19 – F6 – 82 – CD. An orange bracket is positioned under the first three pairs of hex digits (2A, B3, 19). A yellow bracket is positioned under the last three pairs of hex digits (F6, 82, CD).

The first six hex digits
(3 bytes) represent a
unique manufacturer code

The second six hex digits
(3 bytes) represent the unique
serial code for the device

3.4 Network hardware

3.4.2 Media Access Control (MAC)



Media access control (MAC) address

- **MAC** addressing is used to route frames on a **local area network (LAN)**.
- Each **MAC address** is unique to every **network interface card (NIC)**.
- **MAC addresses** are static and assigned by the manufacturer.
- A **MAC address** is 48 bits in length.
- **MAC addresses** are represented as six groups of hexadecimal digits (e.g., 00:0a:95:9d:68:16).
 - The first six hex digits (3 bytes) represent a unique manufacturer code.
 - The second six hex digits (3 bytes) represent the unique serial code for the device

3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv4 and IPv6

<https://www.youtube.com/watch?v=-Zhe2-f7otc>

The image is a YouTube video thumbnail with an orange background. At the top left, there is a black box with 'CAMBRIDGE' in white, and below it, another black box with 'IGCSE' in white. To the right of these, the text 'COMPUTER SCIENCE' is written in white, with '0478 | 0984' below it. The main title '3.4 IP ADDRESS IPV4 & IPV6' is in large, bold, black letters. To the right of the title, the IP address '193 . 74 . 211 . 63' is displayed in large red numbers. Below each number is its 8-bit binary representation: '11000001' for 193, '01001010' for 74, '11010011' for 211, and '00111111' for 63. A bracket underneath these four binary strings is labeled '32 bits (4 bytes)'. In the bottom right corner, there is a black box with the text '7:21' in white.

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3.4
IP ADDRESS
IPV4 & IPV6

193 . 74 . 211 . 63
11000001 01001010 11010011 00111111

32 bits (4 bytes)

7:21

3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv4 and IPv6

<https://www.youtube.com/watch?v=ThdO9beHhpA>



3.4 Network hardware

3.4.3: Public vs Private IP Address

<https://www.youtube.com/watch?v=po8ZFG0Xc4Q>



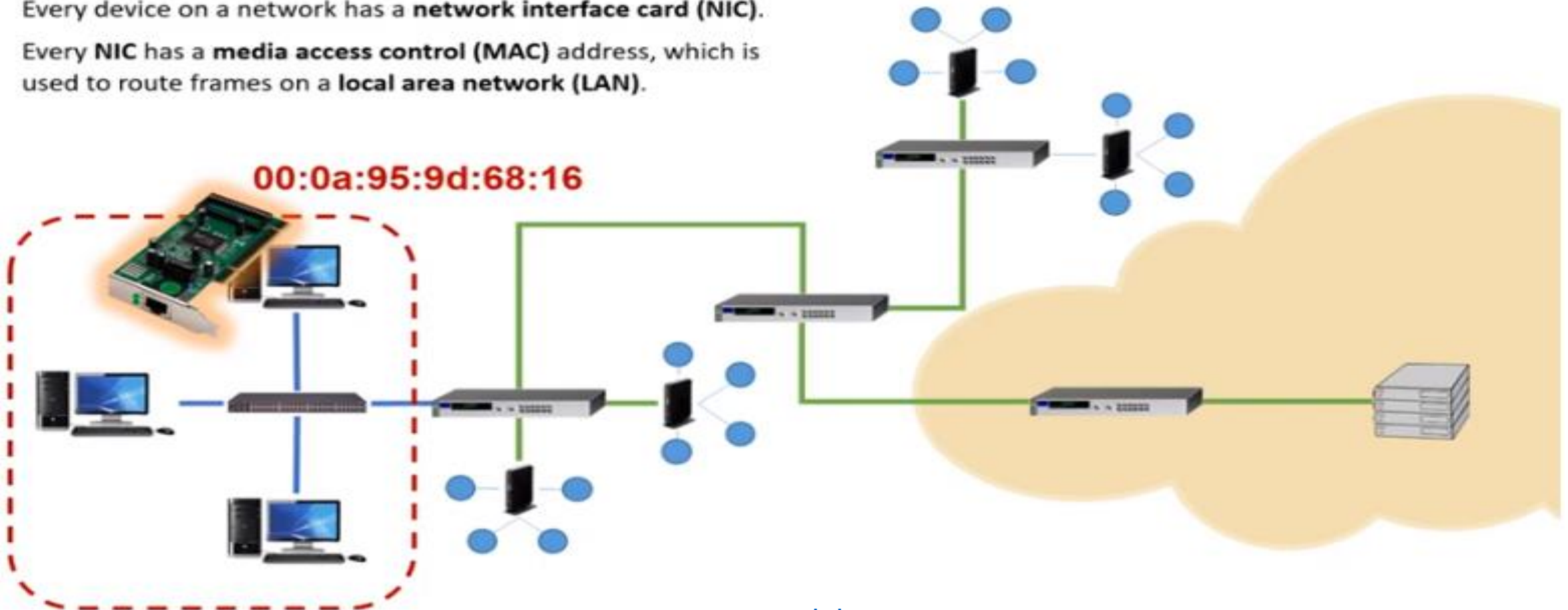
3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv4 and IPv6

IP addressing

Every device on a network has a **network interface card (NIC)**.

Every **NIC** has a **media access control (MAC)** address, which is used to route frames on a **local area network (LAN)**.



By: [Craig'n'Dave](#)

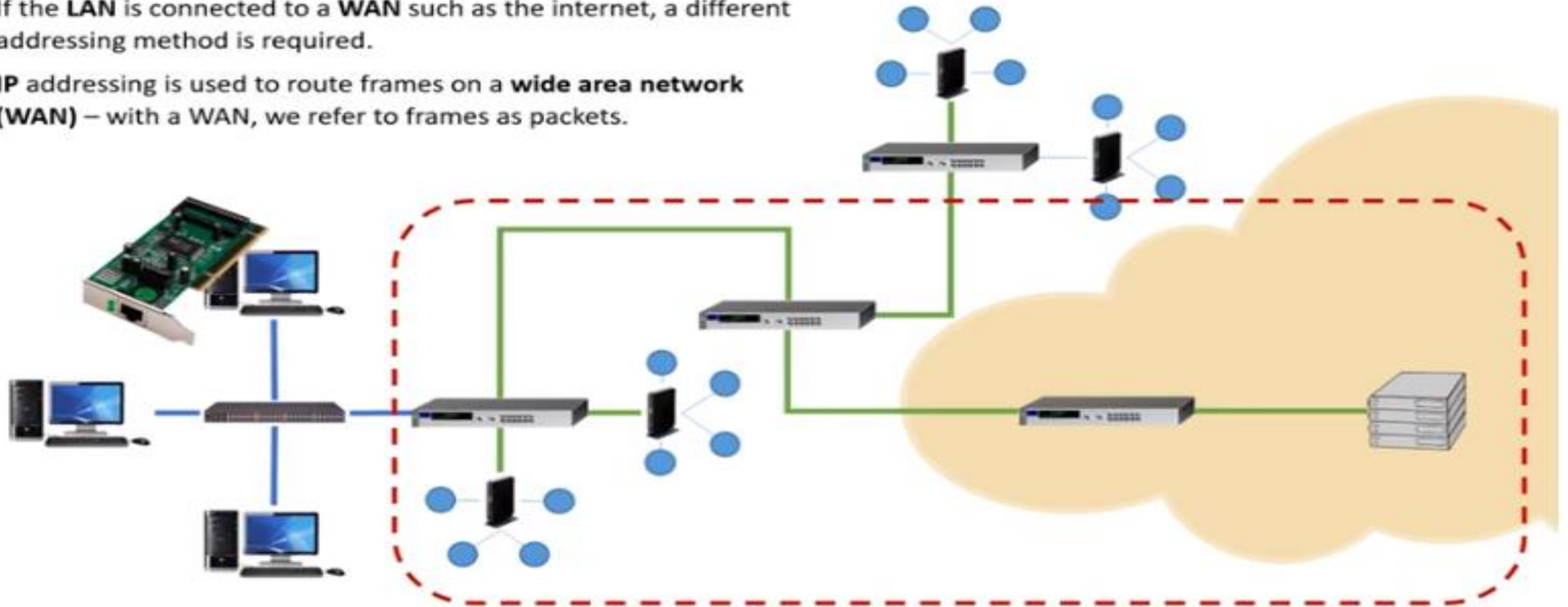
3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv4 and IPv6

IP addressing

If the **LAN** is connected to a **WAN** such as the internet, a different addressing method is required.

IP addressing is used to route frames on a **wide area network (WAN)** – with a WAN, we refer to frames as packets.

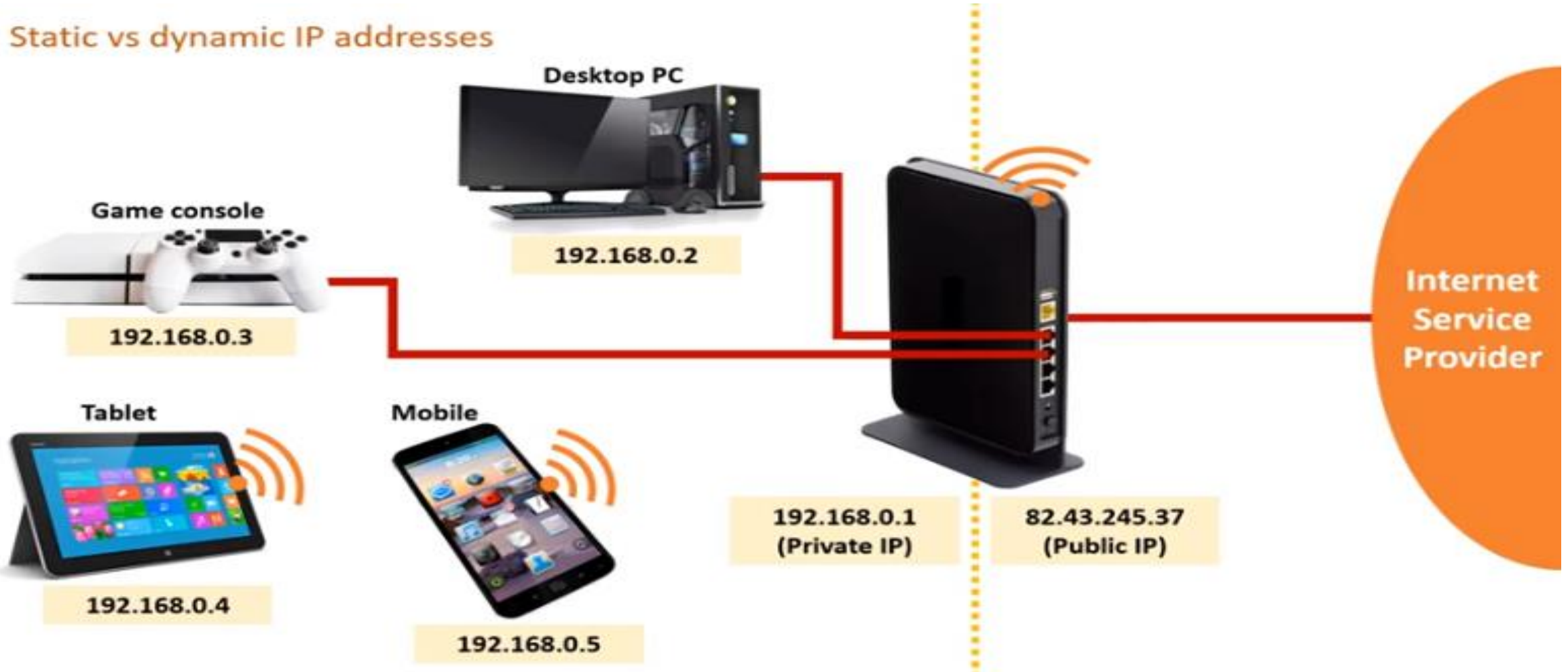


By: [Craig'n'Dave](#)

3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv4 and IPv6

Static vs dynamic IP addresses



By: [Craig'n'Dave](#)

3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv4 and IPv6

Each device on the internet is given a unique address known as the **INTERNET PROTOCOL (IP) ADDRESS**. This is a 32-bit number which is usually written in the

- form: 192.168.1.1
- For window user: click command prompt – type: ipconfig/all

Activity: Find your IP address on your laptop or MacBook ?

3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv4

IPv4

An **IP (Internet Protocol)** address is a unique number used to identify a host computer or node that communicates over the internet. There are currently two versions of **IP** in use – IPv4 and IPv6.

193 . 74 . 211 . 63
11000001 01001010 11010011 00111111



32 bits (4 bytes) in size

IPv4 format: Four numbers between 0 and 255 separated by periods.

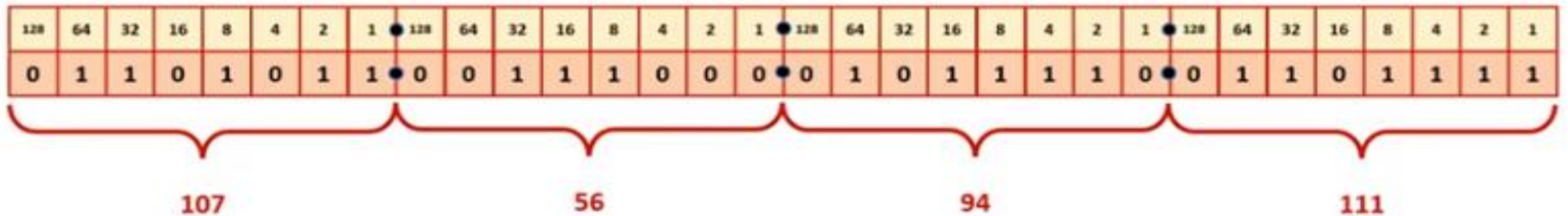
[By: Craig'n'Dave](#)

3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv4

IPv4

107.56.94.111



IPv4 provides us with approximately 4 billion possible addresses from 000.000.000.000 to 255.255.255.255.

The actual number is slightly lower, as some addresses are reserved for special uses.

By: Craig'n'Dave

3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv4

IPv4

107.56.94.111

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
0	1	1	0	1	0	1	1	0	0	1	1	1	0	0	0	0	1	0	1	1	1	1	0	0	1	1	0	1	1	1	

Network identifier (3 bytes): 107.56.94

Host identifier (1 byte): 111

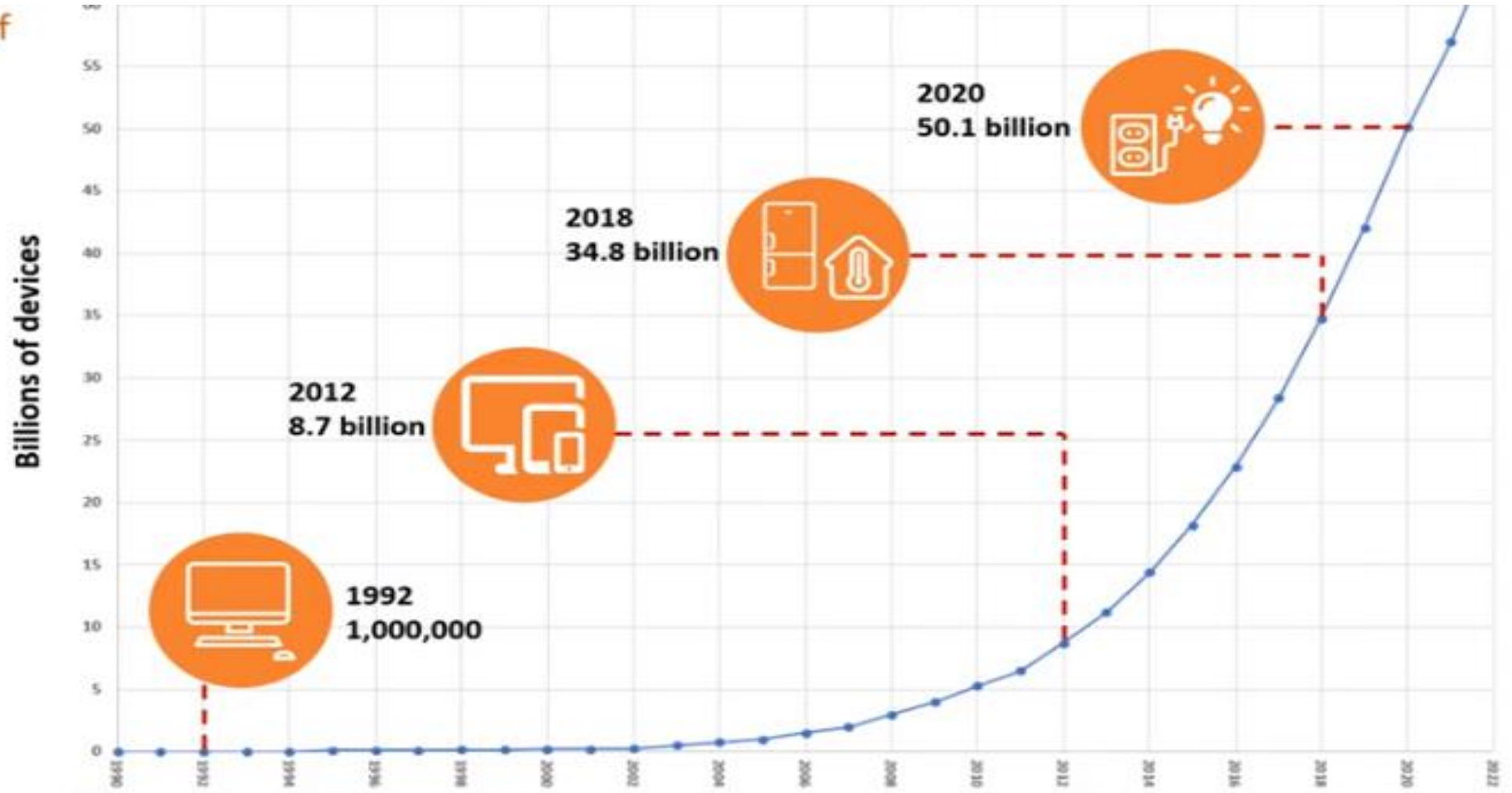
An IPv4 address has two parts:

- Network identifier
- Host identifier

3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv4

The Internet of Things (IoT)



By: [Craig'n'Dave](#)

3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv6

IPv6

49cd:c92d:dabb:469f:003c:4877:f44b:5441

0100100111001101 1100100100101101 1101101010111011 0100011010011111 0000000000111100 0100100001110111 1111010001001011 0101010001000001

128 bits (16 bytes) in size.
Approximately 340 trillion-trillion-trillion combinations

IPv6 format: Eight groups of four 16-bit hex values separated by colons.

[By: Craig'n'Dave](#)

3.4 Network hardware

3.4.3: Internet Protocol (IP) address - IPv4 and IPv6



Internet Protocol (IP) address

IP addressing is used to route packets on a **wide area network (WAN)**.

There are two versions of IP addresses – IPv4 and IPv6.

IP addresses can be static, meaning they do not change, or dynamic, meaning they *can* change.

IPv4

- 32 bits in size.
- Four numbers between 0 and 255 separated by periods (e.g., 69.89.31.212).
- A router has a unique, **WAN-facing IP address** and a LAN-facing **IP address** – this enables a **LAN** device to have the same **IP address** as another device on a separate **LAN**.
- **IPv4** is being replaced by **IPv6** because unique static addresses are running out.

IPv6

- 128 bits in size.
- Eight groups of four 16-bit hex values separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).

3.4 Network hardware

3.4.3: The use of IP and MAC addressing

<https://www.youtube.com/watch?v=p9D0Ca3VNpM>

OCR

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J277

1.3

IP AND MAC
ADDRESSING

```
C:\WINDOWS\system32\cmd.exe
C:\Users\Noureddine>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Noureddine-PC
Primary Dns Suffix . . . . . : 
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet0:

Connection-specific DNS Suffix . : 
Physical Address. . . . . : 84-BE-0A-37-89-40
Link-local IPv6 Address . . . . . : fe80::814d:f463:a8a7:377f%1
IPv4 Address. . . . . : 192.168.0.25 Preferred
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 22 Jul 2015 22:00:53
Lease Expires . . . . . : 24 Jul 2015 15:00:33
Default Gateway . . . . . : 192.168.0.1
DHCP Server . . . . . : 192.168.0.1
DHCPv6 IAD . . . . . : No
DHCPv6 Client ID . . . . . : 80-81-80-0

DHCP Servers . . . . . : 192.168.0.1
NetBIOS over Tcpip. . . . . : Enabled
```


3.4 Network hardware

3.4.3: The use of IP and MAC addressing



- MAC addressing is used to route frames on a local area network (LAN):
 - Example MAC address: 00:0a:95:9d:68:16
 - Each MAC address is unique to every network interface card
- IP addressing is used to route packets on a wide area network (WAN):
 - There are two versions of IP addresses IPv4 and IPv6
 - IPv4 address:
 - 32-bits in size
 - Written as four numbers separated by periods (.), each number in the range 0-255.
 - Example: 69.89.31.212
 - A router will have a unique WAN facing IP address and a LAN facing IP address
 - This enabled a LAN device to have the same IP address as another device on another LAN
 - IPv4 is being replaced by IPv6 because unique static addresses are running out
 - IPv6 address:
 - 128-bits in size
 - Written as eight groups separated by colons (:), each group made up of four hex values representing 16-bits
 - Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

3.4 Network hardware

MAC addresses and IP addresses

IP addresses and MAC addresses:

So what is the difference between an IP address and a MEDIA ACCESS CONTROL (MAC) address?

- The IP address gives the location of a device on the internet, whereas the MAC address identifies the device connected to the internet.
- You can think of the IP address as the address of the house you live in (it will have some unique way of identifying it, such as a post or zip code). Using this example, the MAC address can be thought of as a way of uniquely identifying each person living in that house. It is possible to move house (so your IP address will change) but the same people will be living in the new house (so their MAC addresses will remain unchanged).

3.4 Network hardware

MAC addresses and IP addresses

MAC addresses	IP addresses
identifies the physical address of a device on the network	identifies the global address on the internet
unique for device on the network	may not necessarily be unique
assigned by the manufacturer of the device and is part of the NIC	dynamic IP addresses are assigned by ISP using DHCP each time the device connects to the internet (see later)
they can be universal or local	dynamic IP addresses change every time a device connects to the internet; static IP addresses don't change
when a packet of data is sent and received, the MAC address is used to identify the sender's and recipient's devices	used in routing operations as they specifically identify where the device is connected to the internet
use 48 bits	use either 32 bits (IPv4) or 128 bits (IPv6)
can be UAA or LAA	can be static or dynamic

3.4 Network hardware

3.4.4: The role of a router

<https://www.youtube.com/watch?v=Ztw50p0soeA>



3.4 Network hardware

3.4.4: The role of a router

https://www.youtube.com/watch?v=1z0ULvg_pW8



3.4 Network hardware

3.4.4: The role of a router

<https://www.youtube.com/watch?v=Mad4kQ5835Y>



3.4 Network hardware

3.4.4: The role of a router

Routers

Routers come in many different shapes and sizes, but they all essentially do the same thing.



By: Craig'n'Dave

3.4 Network hardware

3.4.4: The role of a router

Routers

Routers come in many different shapes and sizes, but they all essentially do the same thing.

Join local area networks (LANs) to the internet with a public IP address

Direct traffic (packets) from one router to another using IP addresses

Pass local data on to an external network (internet)

Receive inbound data and direct it to a specific device on a network (LAN)



By: [Craig'n'Dave](#)

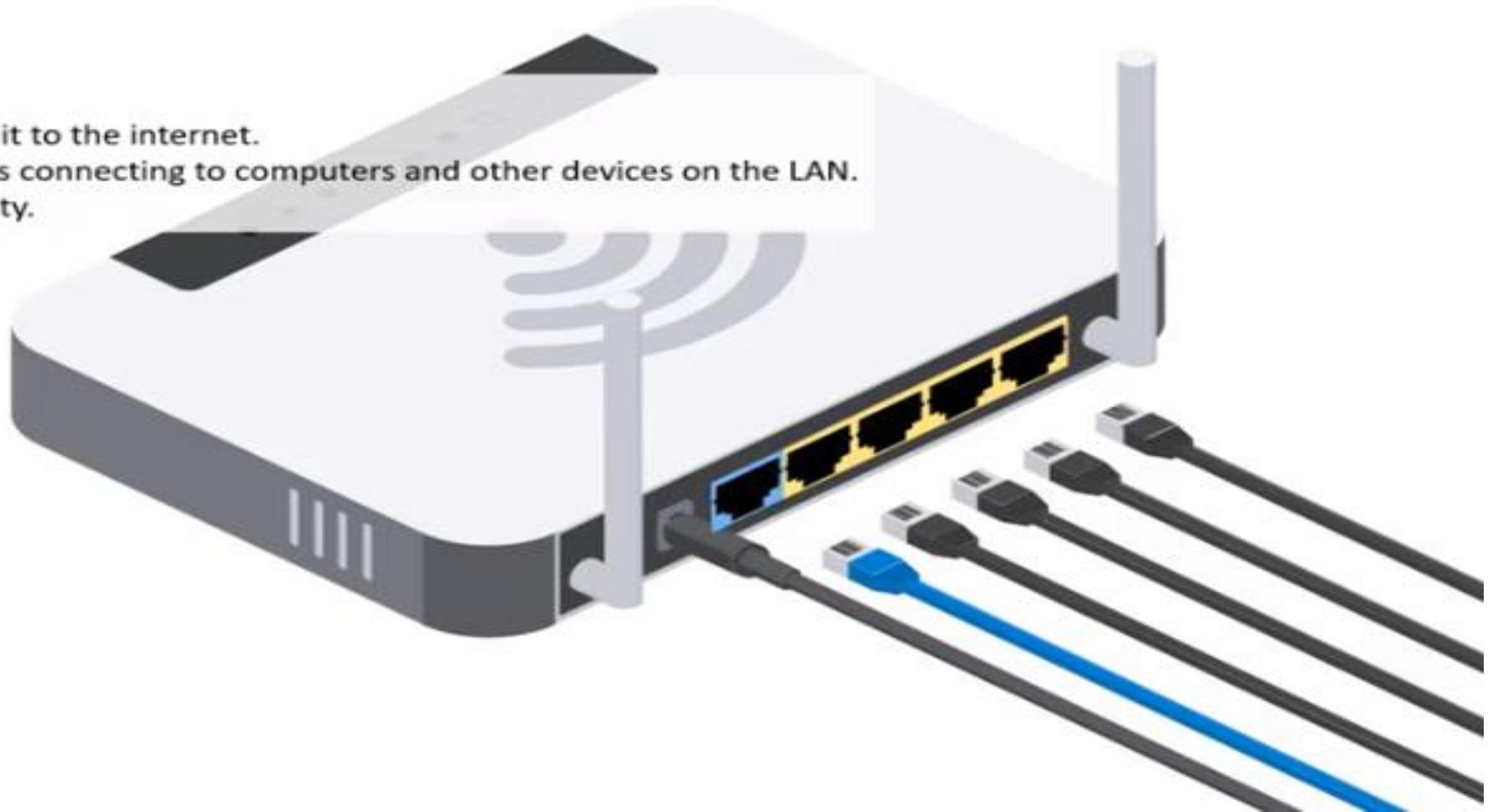
3.4 Network hardware

3.4.4: The role of a router

Routers

A typical **router** will have:

- A cable connecting it to the internet.
- Several other cables connecting to computers and other devices on the LAN.
- Wireless connectivity.

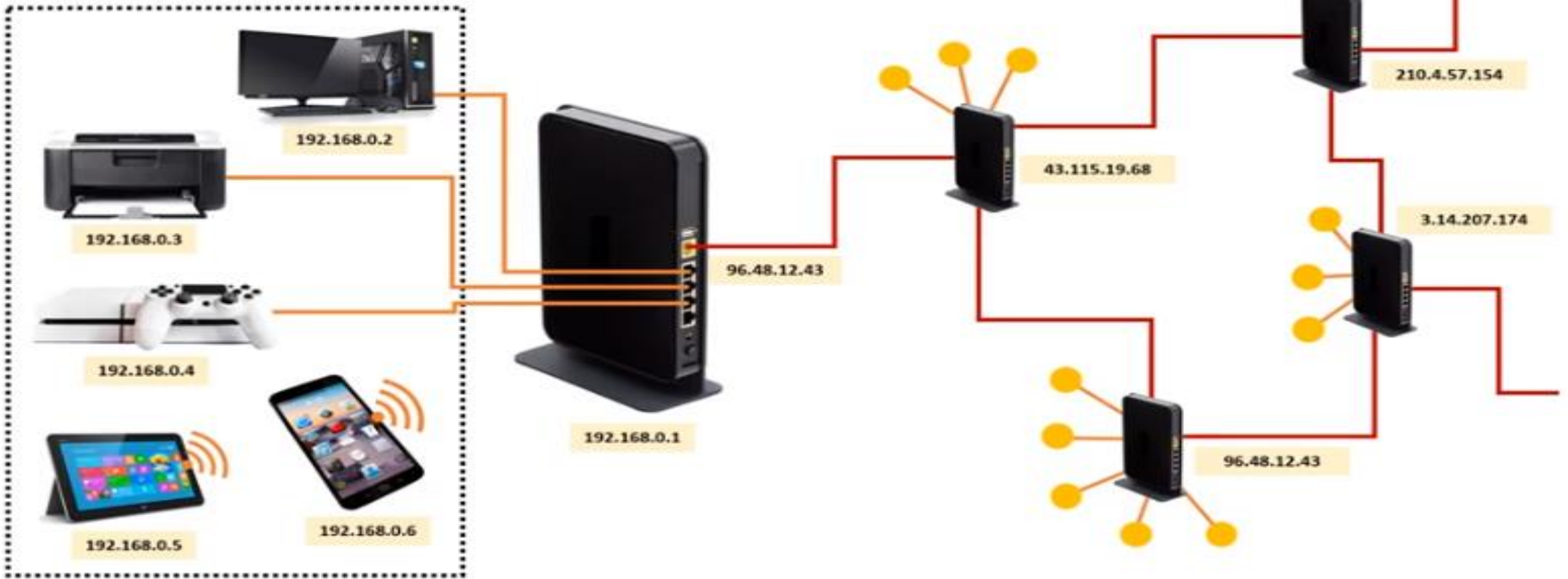


By: Craig'n'Dave

3.4 Network hardware

3.4.4: The role of a router

Routers



By: [Craig'n'Dave](#)

3.4 Network hardware

3.4.4: The role of a router



Routers

A **router** sends data between networks – it is required to connect a **local area network** to a **wide area network**.

Routers use devices' **IP address** to route traffic to other **routers**.

Routers can assign IP addresses to devices on a local network.

3.4 Network hardware

IGCSE Past Paper Exam Question and Answer 1

(a) A computer can have both a MAC address and an IP address.

Four statements are given about MAC addresses and IP addresses.

Tick (✓) to show whether each statement is **True** or **False**.

Statement	True (✓)	False (✓)
A MAC address is unique to a computer on a network	✓	
Once an IP address has been set it cannot be changed		✓
A MAC address is made up of the computer's serial number and the IP address		✓
If a computer does not have an IP address it cannot communicate with another device using the Internet	✓	

3.4 Network hardware

IGCSE Past Paper Exam Question and Answer 2

1 A school network has several computers.

Each computer in the network has a media access control (MAC) address.

Hexadecimal is used for MAC addresses.

Part of a MAC address is given.

97–5C–E1

Each pair of digits is stored as binary in an 8-bit register.

(a) Complete the binary register for these two pairs of digits.

97	<table><tbody><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr></tbody></table>	1	0	0	1	0	1	1	1			
1	0	0	1	0	1	1	1					
5C	<table><tbody><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></tbody></table>	0	1	0	1	1	1	0	0			
0	1	0	1	1	1	0	0					

[4]

(b) Describe what is meant by a MAC address.

Any four from:

- used to identify a device
- it is a unique (address)
- it is a static address // it does not change
- it is set by the manufacturer
- the first part is the manufacturer ID / number / identifies the manufacturer
- the second part is the serial number / ID.

3.4 Network hardware

IGCSE Past Paper Exam Question and Answer 3

The table shows **four** definitions.

Complete the table giving the missing **Term** for each definition.

Term	Definition
Serial (ignore any ref to simplex etc.)	A data transmission method that sends data one bit at a time, down a single wire
IP (address)	An address given to a device on a network. The address is assigned by the network
Browser	The software used to render HTML and display a web page
MAC (address)	An address given to a device at the manufacturing stage that can be used to identify the device on a network

3.4 Network hardware

IGCSE Past Paper Exam Question and Answer 4

(b) Explain what is meant by a MAC address.

Four from:

- Media Access Control (address)
 - Used to identify a device
 - It is a **unique** (address)
 - It is a static address // It does not change
 - It is set by the manufacturer
 - The first part is the manufacturer ID/number/identifies the manufacturer
 - The second part is the serial number/ID
-

3.4 Network hardware

IGCSE Past Paper Exam Question and Answer 5

(c) Give **two** other examples where hexadecimal can be used.

Two from e.g.:

Example 1

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Example 2

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- Colour codes // Colour in HTML / CSS
- Error messages
- Locations in memory
- Memory dump // debugging
- IP address
- ASCII // Unicode
- Assembly language
- URL

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3.4 Network hardware

IGCSE Past Paper Exam Question and Answer 6

13 Describe the role of an Internet Service Provider (ISP).

.... Any six from e.g.:

- – Provide access to the internet / dial up / broadband
- – Usually charge a monthly fee
- – Monitor usage
- – Give users an IP address
- – Determine bandwidth
- – Supports domain names
- – Provide security services
- – Provide web hosting facilities
- – Provide access to Email / Mailbox
- – Provides online data storage

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3.4 Network hardware

IGCSE Past Paper Exam Question and Answer 7

8 Kamil correctly answers an examination question about a number of internet terms.

Six different terms have been removed from Kamil's answer.

Complete the sentences in Kamil's answer, using the list given. Not all terms in the list need to be used.

- browser
- connection
- domain name server (DNS)
- Internet
- Internet Service Provider (ISP)
- IP address
- MAC address
- network
- protocol
- uniform resource locator (URL)
- webpages
- hypertext mark-up language (HTML)

A **browser** is a program that allows a user to view **webpages**

An **Internet Service Provider (ISP)** is a company that provides a connection to access the **Internet**

The main **protocol** that governs the transmission of data using the Internet is http.

The **IP address** is provided by the network, and given to each device on the network.